Application Note:

LR2021

FLRC Regulatory Measurement Reports

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1 Introduction

The use of FLRC in the sub-GHz bands gives the significant advantages of both long range and high data rate up to 2.6 Mbps. With these higher data rates come new regulatory considerations when certifying a device using FLRC modulation.

This document is broken down into 2 main sections: one dealing with FLRC in the EU sub-bands and ETSI certification, the second considers FCC compliance of FLRC in the US ISM bands.

This Application Note is designed to be used in conjunction with Semtech Application Notes AN1200.108 [5], that covers general regulatory compliance of the LR2021, as well as AN1200.101 [6], describing the FLRC Improvements brought by the LR2021.

2 Part I: European Regulatory Compliance

2.1 ERC REC 70-03 Applicable Bands

The sub-GHz ISM bands are governed by the CEPT ERC REC 70-03 [1] which details the bands of operation and the applicable power limits and the normalized ETSI standard EN 300 220 that details the test methods and all other regulatory limits imposed on devices operating the ISM band.

The LR2021 is well suited to operation in the 868 MHz European ISM bands identified in [1]. However, because LR2021 offers the ability to set FLRC for large bandwidths, not all 868 MHz sub-bands will be applicable. For example, Annexes 2 (Tracking systems) and 7 (Social Alarms), do not allow adequate modulation bandwidth for FLRC operation. Those bands suitable for European operation are reproduced from [1] below:

Table 1. Annex 1 SRDs

	Frequency Band	Power / Magnetic Field	Spectrum access and mitigation requirements	Modulation / maximum occupied bandwidth	ECC/ERC Deliverable	Notes
h0	862-863 MHz	25 mW e.r.p.	≤ 0.1% duty cycle	≤ 350 kHz		
h1.0	863-870 MHz (note 2)	25 mW e.r.p.	≤ 0.1% duty cycle (note 1)	≤ 100 kHz for 47 or more hop channels		FHSS. Parts of the frequency band are also identified in Annexes 2, 3, 10 and 11
h1.1	865-868 MHz	25 mW e.r.p.	≤ 1% duty cycle (note 1)	≤ 50 kHz for 58 or more hop channels		FHSS. The frequency band is also identified in Annexes 2, 3 and 11
h1.2	863-870 MHz (note 2)	25 mW e.r.p. -4.5 dBm/100 kHz	≤ 0.1% duty cycle or LBT+AFA	Not specified		Non-FHSS. Parts of the frequency band are also identified in Annexes 2, 3, 10 and 11
h1.3	863-865 MHz	25 mW e.r.p.	≤ 0.1% duty cycle or LBT+AFA	Not specified		The frequency band is also identified in Annexes 3 and 10
h1.4	865-868 MHz	25 mW e.r.p.	≤ 1% duty cycle or LBT +AFA	Not specified		The frequency band is also identified in Annexes 2, 3 and 11
h1.5	868-868.6 MHz	25 mW e.r.p.	≤ 1% duty cycle or LBT +AFA	Not specified		
h1.6	868.7-869.2 MHz	25 mW e.r.p.	≤ 0.1% duty cycle or LBT+AFA	Not specified		
h1.7	869.4-869.65 MHz	500 mW e.r.p.	≤ 10% duty cycle or LBT+AFA	Not specified		
h1.8	869.7-870 MHz	5 mW e.r.p.	No requirement (note 3)	Not specified		
h1.9	869.7-870 MHz	25 mW e.r.p.	≤ 1% duty cycle or LBT +AFA	Not specified		

Table 2. Annex 10 Audio Applications

	Frequency Band	Power / Magnetic Field	Spectrum access and mitigation requirements	Modulation / maximum occupied bandwidth	ECC/ERC Deliverable	Notes
g	863-865 MHz	10 mW e.r.p.	No requirement	Not specified		Radio microphones, wireless audio and multimedia streaming devices. The frequency band is also identified in Annex 1

2.2 EN 300 220 Applicable Regulatory Tests

The impact of FLRC is principally on the transmitter regulatory compliance of the radio.

All the transmitter testing sections from the EN 300 220-1 are outlined below, together with:

- the parts that apply to a design using the LR2021 in FLRC mode
- tests whose results would be directly impacted by the use of FLRC modulation.

Table 3. EU Regulatory Test Specific to FLRC

Ref	Description	Applies?	FLRC Specific?
5.1	Operating frequency	Y	N
5.2	Effective Radiated Power	Y	Y
5.3	Maximum Effective Radiated PSD	N	N
5.4	Duty Cycle	Y	N
5.5	Duty Cycle Template	Y	N
5.6	Occupied Bandwidth	Y	Y
5.7	Frequency error	Y	N
5.8	Tx Out of Band Emissions	Y	Y
5.9	Unwanted Emissions in the Spurious Domain	Y	Y
5.10	Transient Power	Y	N
5.11	Adjacent Channel Power	N	N
5.12	TX behaviour under Low Voltage Conditions	Y	N
5.13	Adaptive Power Control	N	N

The regulatory testing performed in this Part, therefore concentrates on the four test areas identified above, specific to FLRC.

2.3 Output Power Regulations

Section 5.2 of the EN 300 220 gives the procedure for measuring the effective radiated power (ERP). Although this step is not strictly dependent on the modulation selected, the configured output power will influence all the other regulatory behaviour. As seen in the ERC REC 70-03, powers of +10 dBm (Annex 10) and +14 dBm (all other bands) are of interest.

Following the method outlined in the EN 300 220, a CW tone can be used for simple measurement of the output power. In a real regulatory scan this would ideally be done by radiated measurement. Here we perform the equivalent conducted measurement and assume the use of a 0 dBd antenna.

2.4 Output Power Measurements

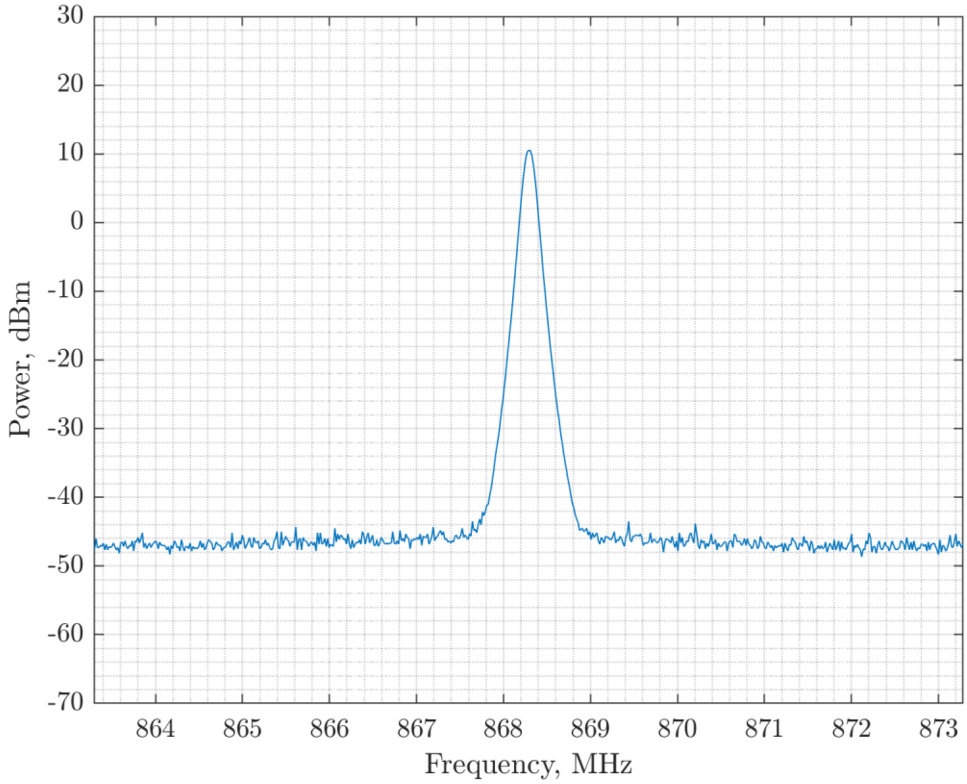


Figure 1. ETSI Conducted Output Power Measured from the LR2021 at +10.5 dBm

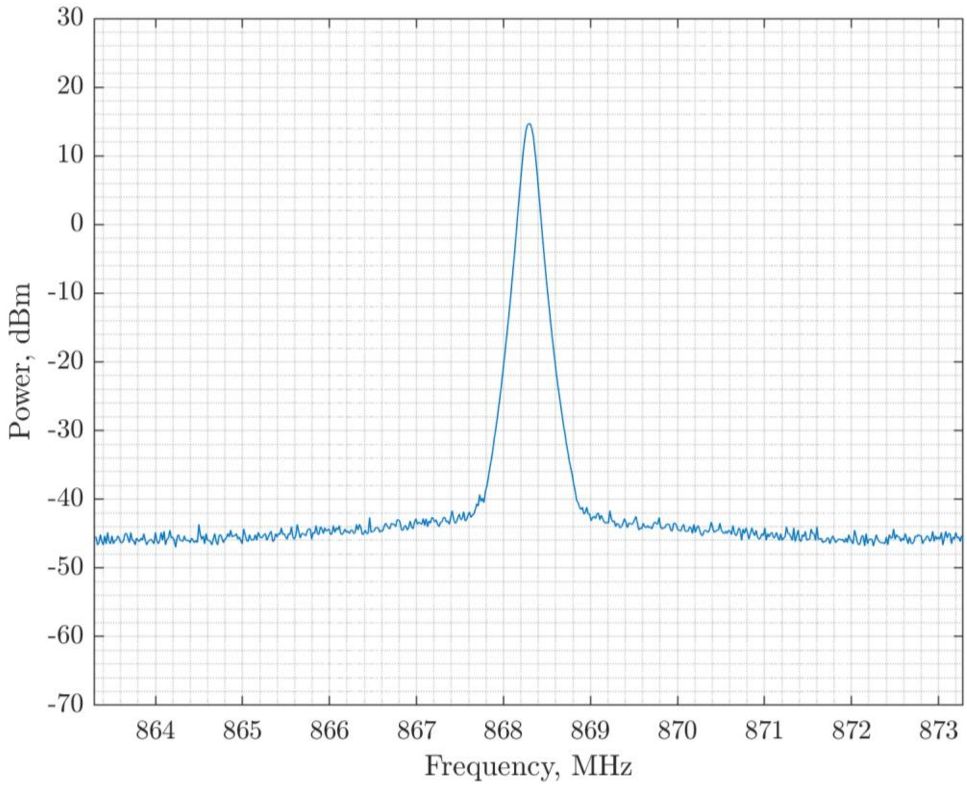


Figure 2. ETSI Conducted Output Power Measured from the LR2021 at +14.6 dBm

For all remaining measurements in this report the more conservative, +14 dBm output power setting is employed.

2.5 Occupied Bandwidth Regulations

Section 5.6 of the EN 300 220 details the procedure for measurement of the occupied bandwidth (OBW). Some sub-bands impose bandwidth limitations, to which this measurement is applied. It is mandatory to measure and record this information. OBW is also related to the occupied channel width (OCW) declared by the manufacturer. For each data rate and bandwidth combination of the LR2021, the declared OCW is listed below, together with the resolution bandwidth (RBW) used for the OBW measurement.

Table 4. FLRC Data Rates, Bandwidth and OCW

Data Rate (kbps)	BW (MHz)	OCW	RBW
2600	2.666	2.8 MHz	30 kHz
2080	2.222	2.4 MHz	30 kHz
1300	1.333	1.4 MHz	30 kHz
1040	1.333	1.4 MHz	30 kHz
650	0.740	0.8 MHz	10 kHz
520	0.571	0.7 MHz	10 kHz
325	0.357	0.4 MHz	10 kHz
260	0.307	0.4 MHz	10 kHz

Following the measurement procedure detailed in [2] the occupied channel width is measured at a centre frequency of 868.3 MHz for all combinations of FLRC data rate and Gaussian filtering with an output power of +14 dBm (unless another power is stated).

The Gaussian filtering of the FLRC modulation is controlled according to it’s BT (Bandwidth-Time product) which is a filter coefficient that defines the relationship between the filter's bandwidth and the symbol time. The smaller the BT value, the greater the filters rejection of the out-of-channel emissions, but the greater the potential for Inter Symbol Interference (ISI), which can lead to a marginal reduction in link budget.

2.6 Occupied Bandwidth Measurements

The occupied bandwidth measurements are presented here, please note that non-compliant results are omitted for clarity. For measurement bandwidths refer to previous table.

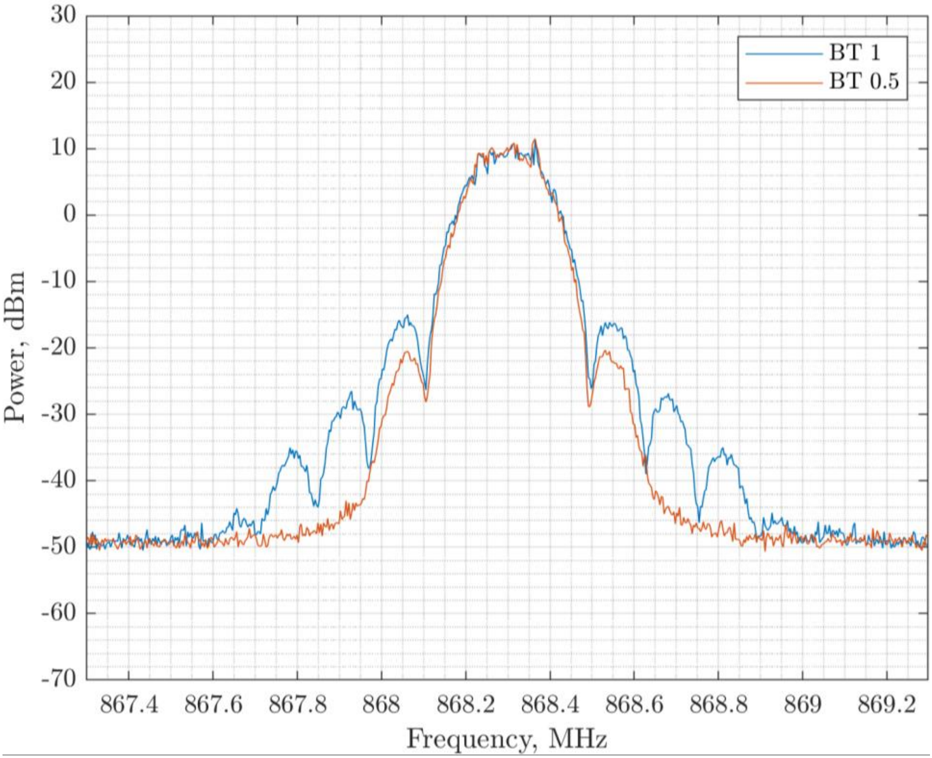


Figure 3. 260 kbps: 326 kHz OBW with BT = 1 and 306 kHz OBW with BT = 0.5

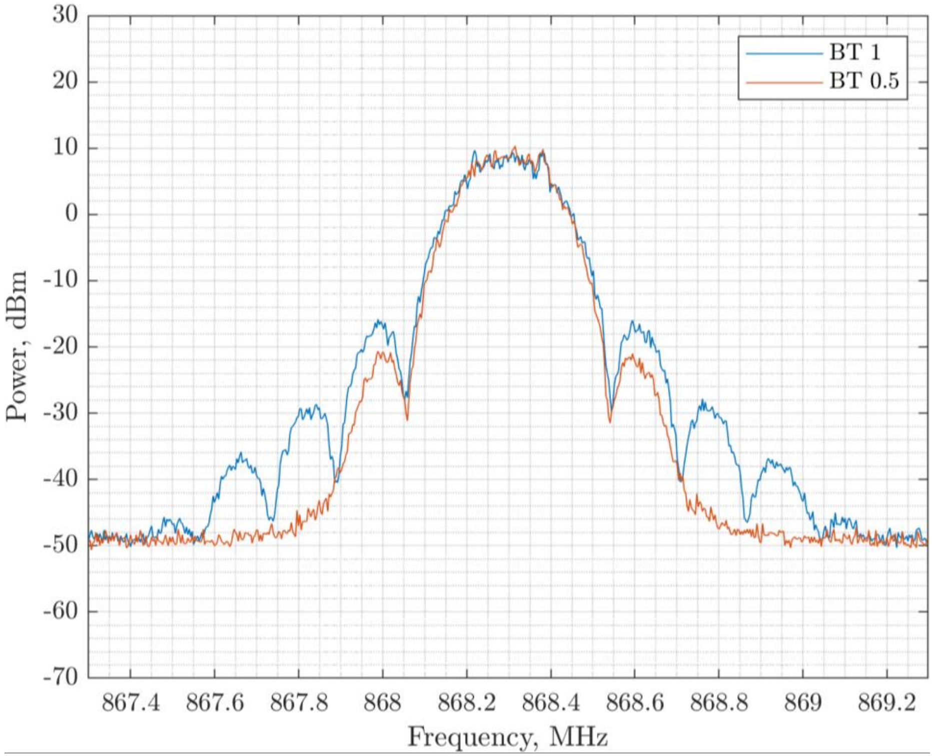


Figure 4. 325 kbps: 402 kHz OBW with BT = 1 and 382 kHz OBW with BT = 0.5

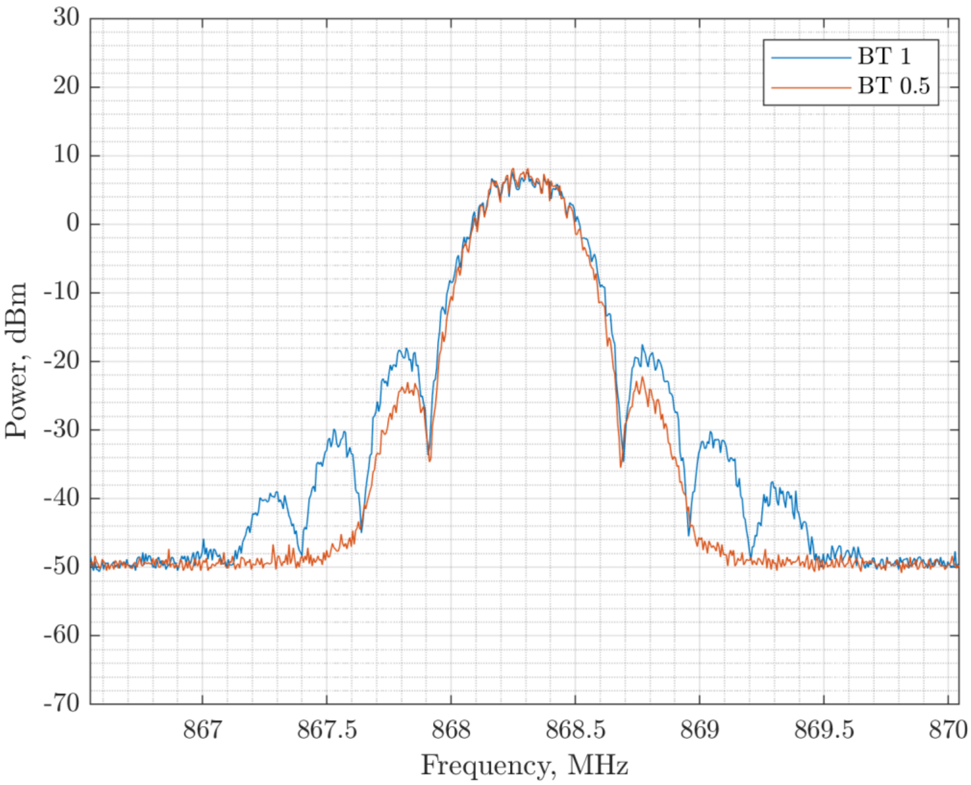


Figure 5. 520 kbps: 663 kHz OBW with BT = 1 and 617 kHz OBW with BT = 0.5

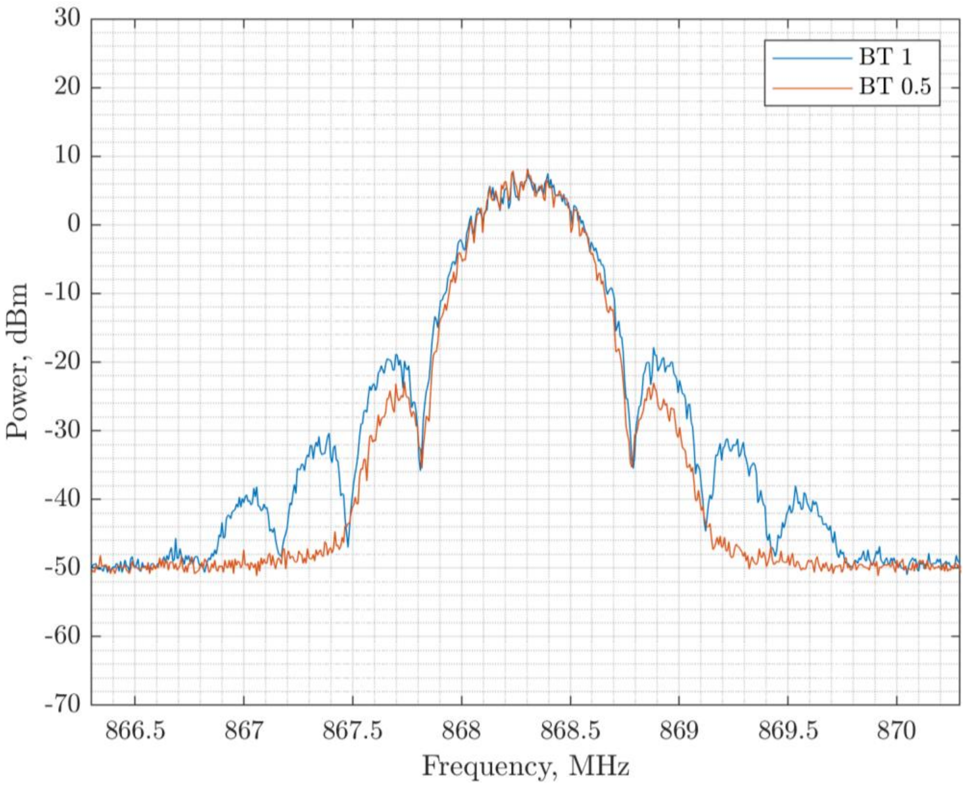


Figure 6. 650 kbps: 825 kHz OBW with BT = 1 and 758 kHz OBW with BT = 0.5

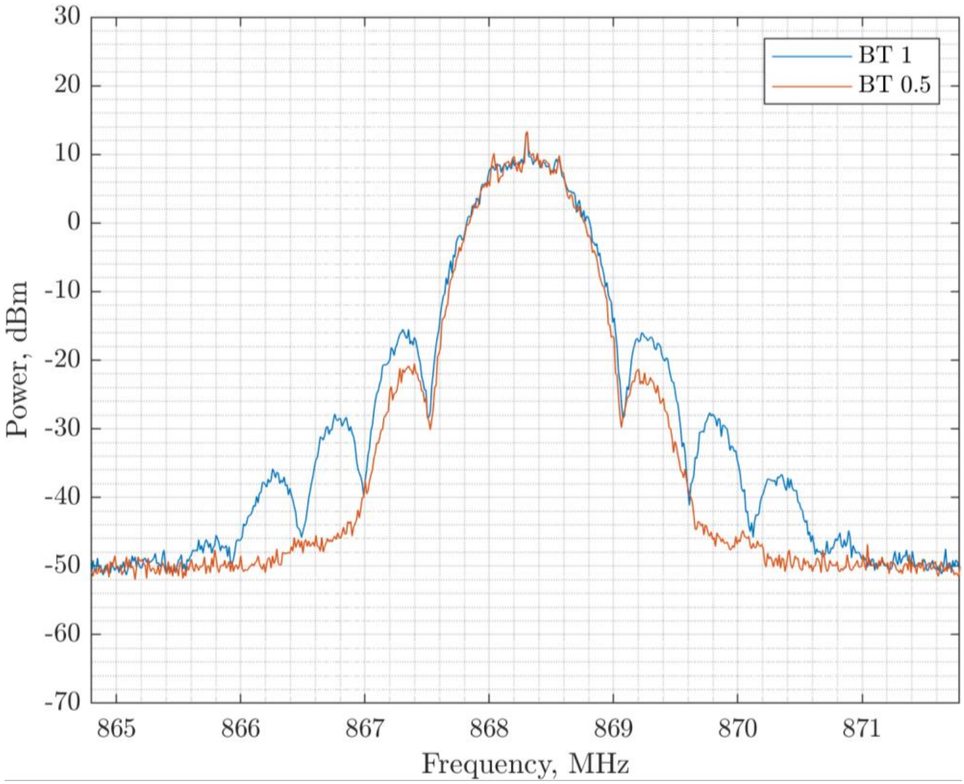


Figure 7. 1.040 Mbps: 1234 kHz OBW with BT = 1 and 1153 kHz OBW with BT = 0.5

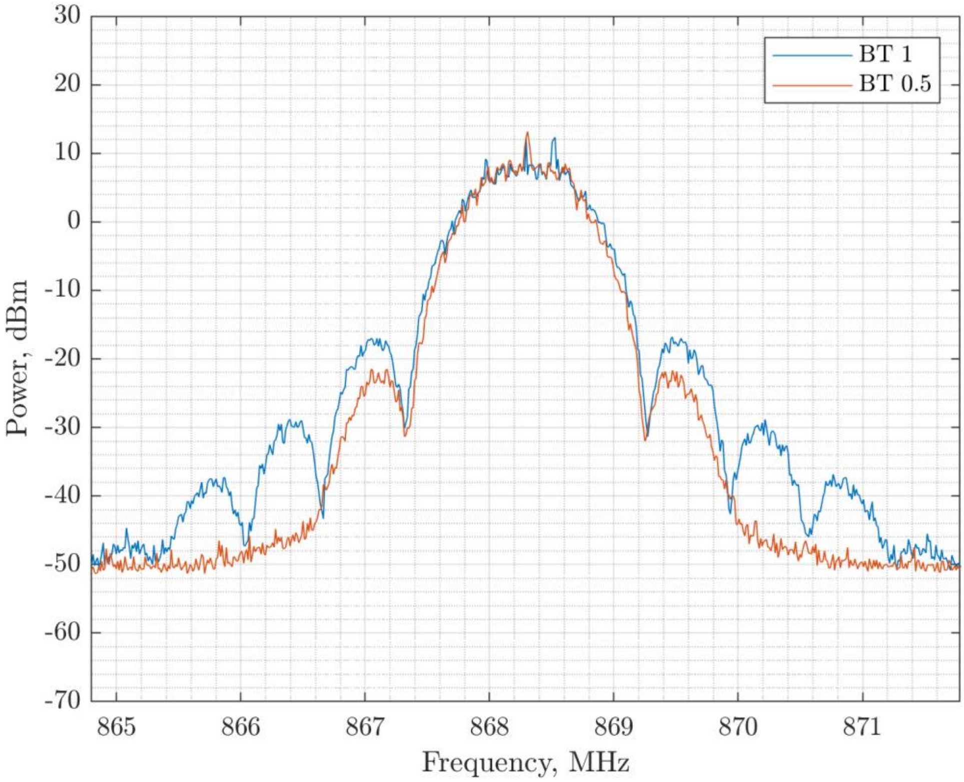


Figure 8. 1.3 Mbps: 1537 kHz OBW with BT = 1 and 1409 kHz OBW with BT = 0.5

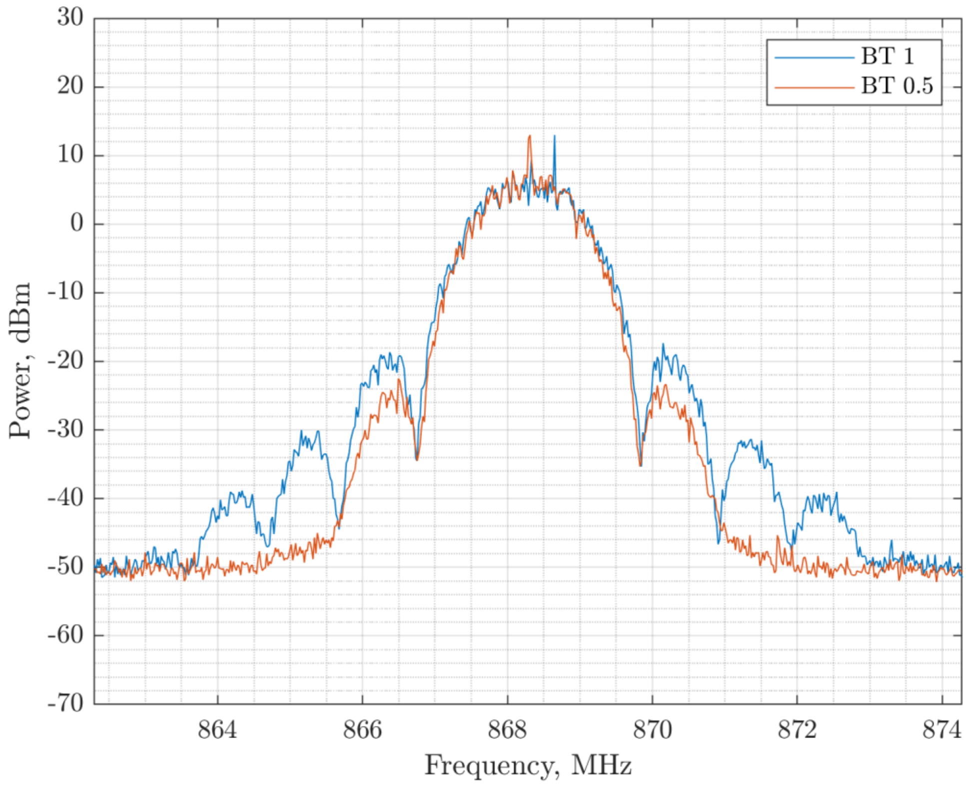


Figure 9. 2.08 Mbps: 2156 kHz OBW with BT = 1 and 2136 kHz OBW with BT = 0.5

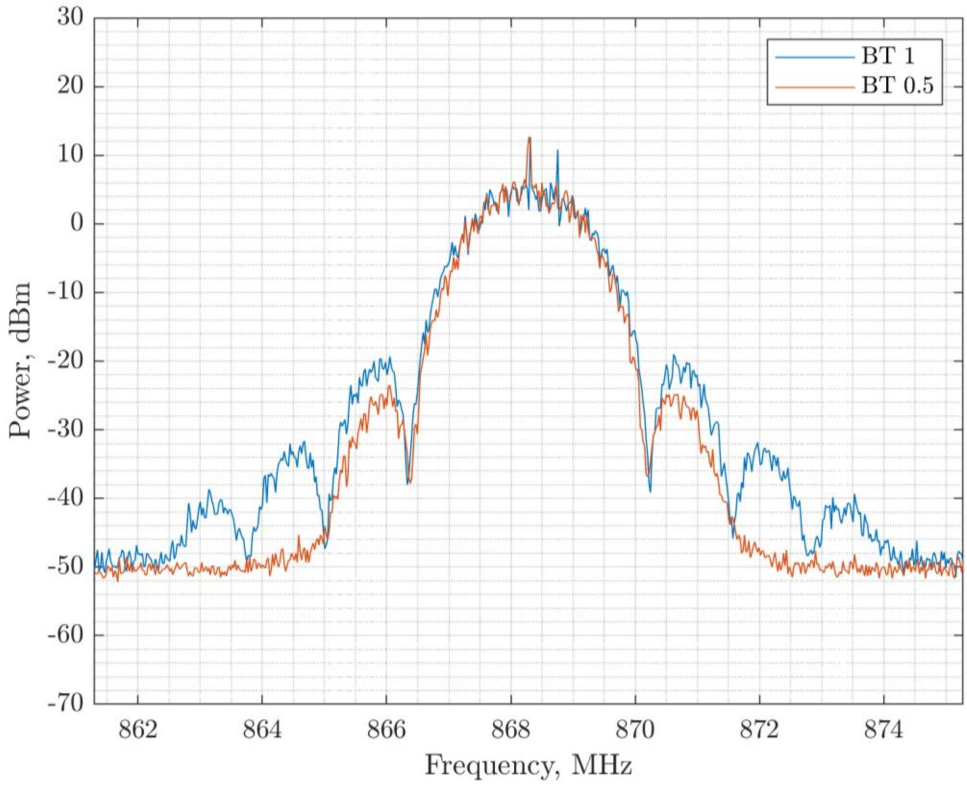


Figure 10. 2.6 Mbps: 2772 kHz OBW with BT = 1 and 2702 kHz OBW with BT = 0.5

2.7 Tx Out of Band Emissions Regulations

Section 5.8 of the EN 300 220 outlines the spectral mask each transmission *within* an ISM band must respect. Following the method outlined in the EN 300 220, compliance with this spectral mask is tested for the LR2021 FLRC modem using a centre frequency of 868.3 MHz and an output power of +14 dBm.

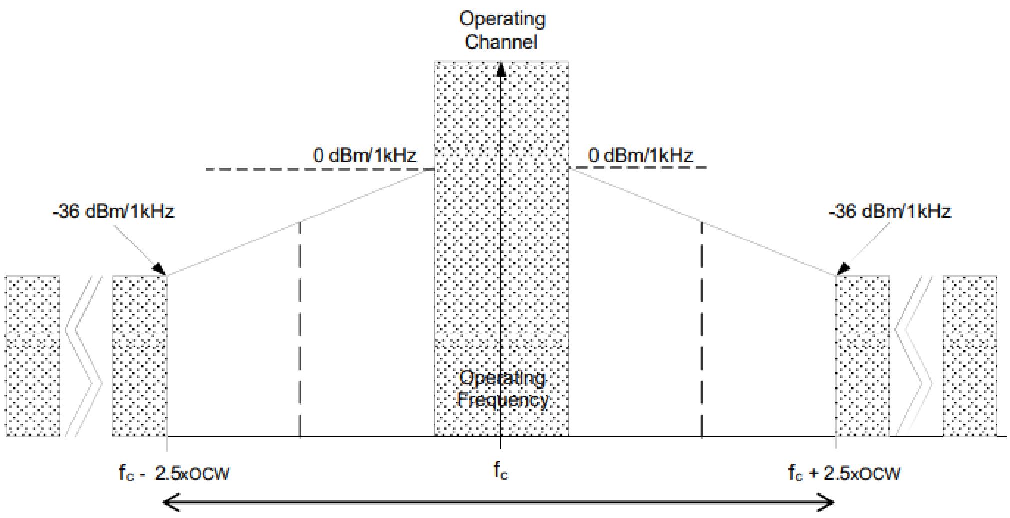


Figure 11. The Tx Out of Band Emission Mask for Signals within the ISM Band

2.8 Tx Out of Band Emissions Measurements

The results are presented in the following figures, note that the central operating channel-wide portion of the spectral mask will scale with the OCW.

For measurement bandwidths refer to previous figure. Please take note of the increasing frequency span in the following plots.

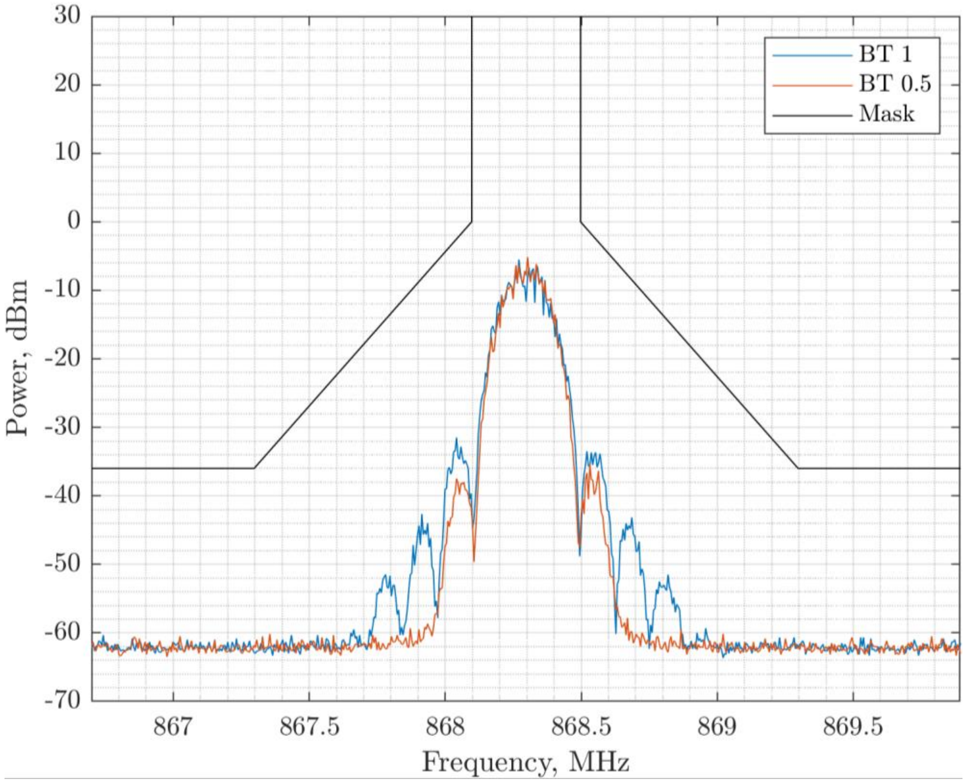


Figure 12. 260 kbps Out of Band Emissions

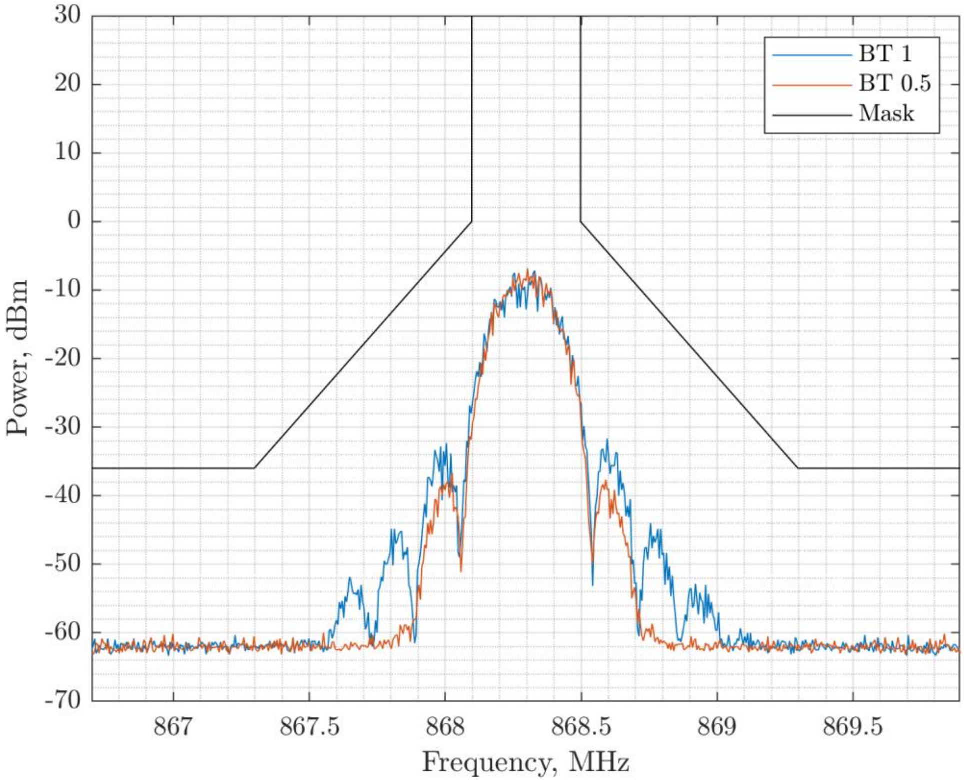


Figure 13. 325 kbps Out of Band Emissions

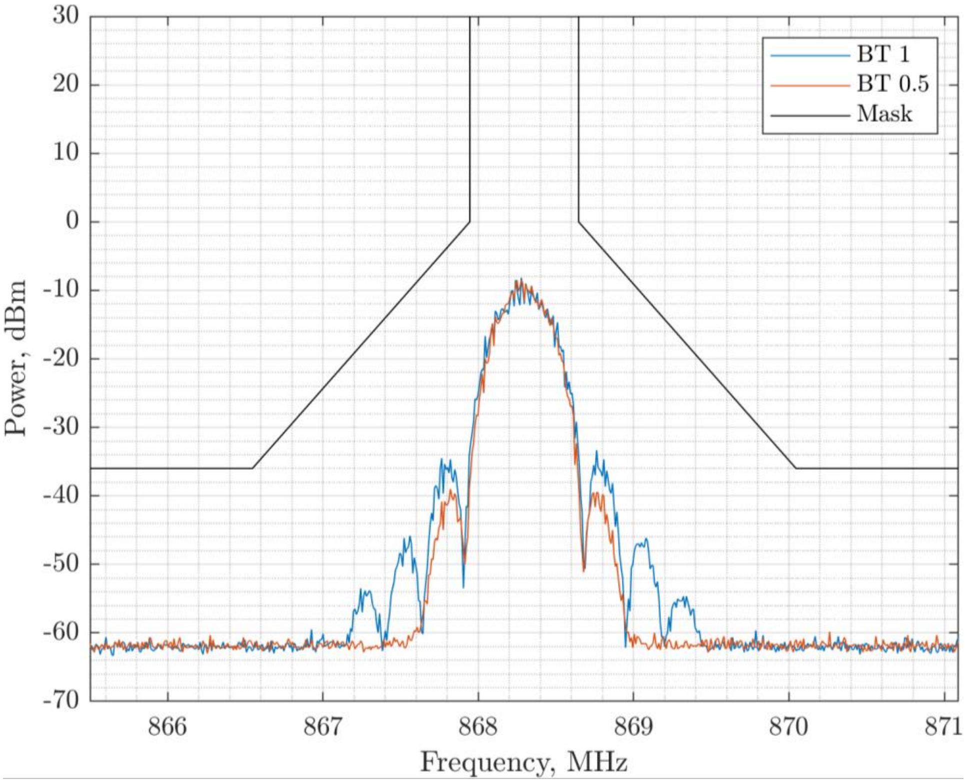


Figure 14. 520 kbps Out of Band Emissions

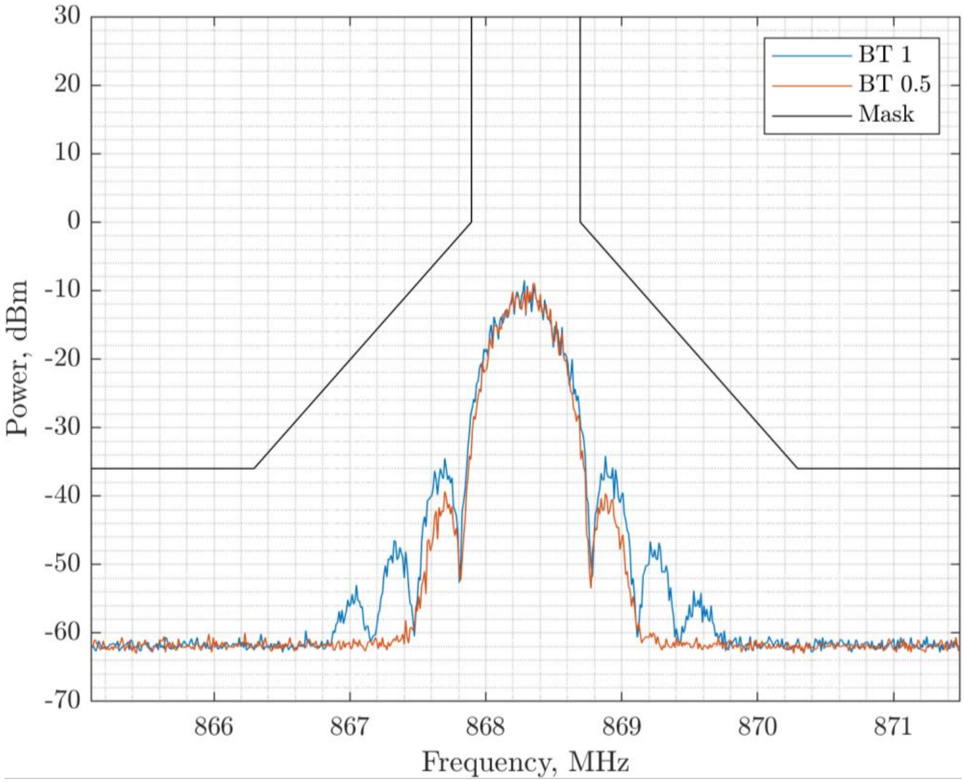


Figure 15. 650 kbps Out of Band Emissions

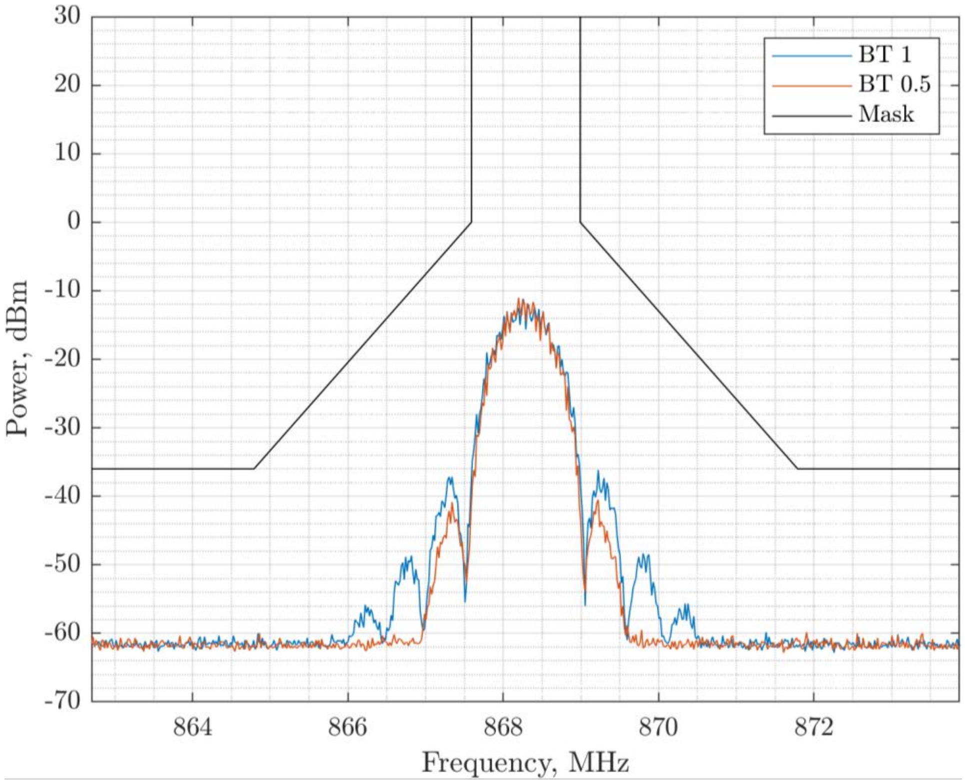


Figure 16. 1.040 Mbps Out of Band Emissions

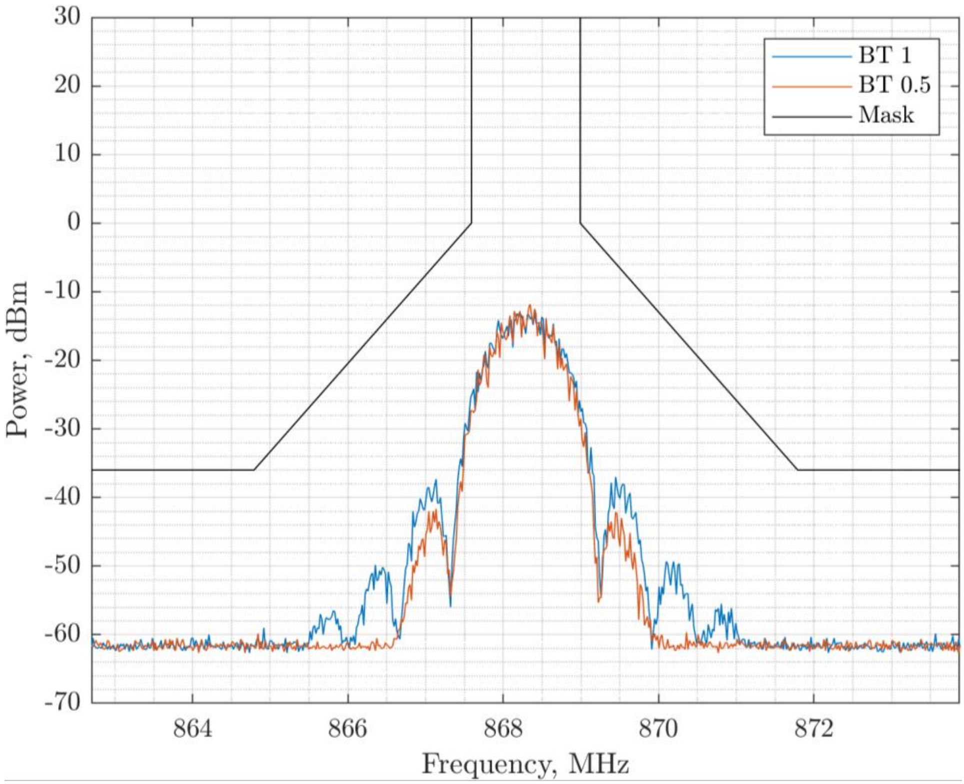


Figure 17. 1.300 Mbps Out of Band Emissions

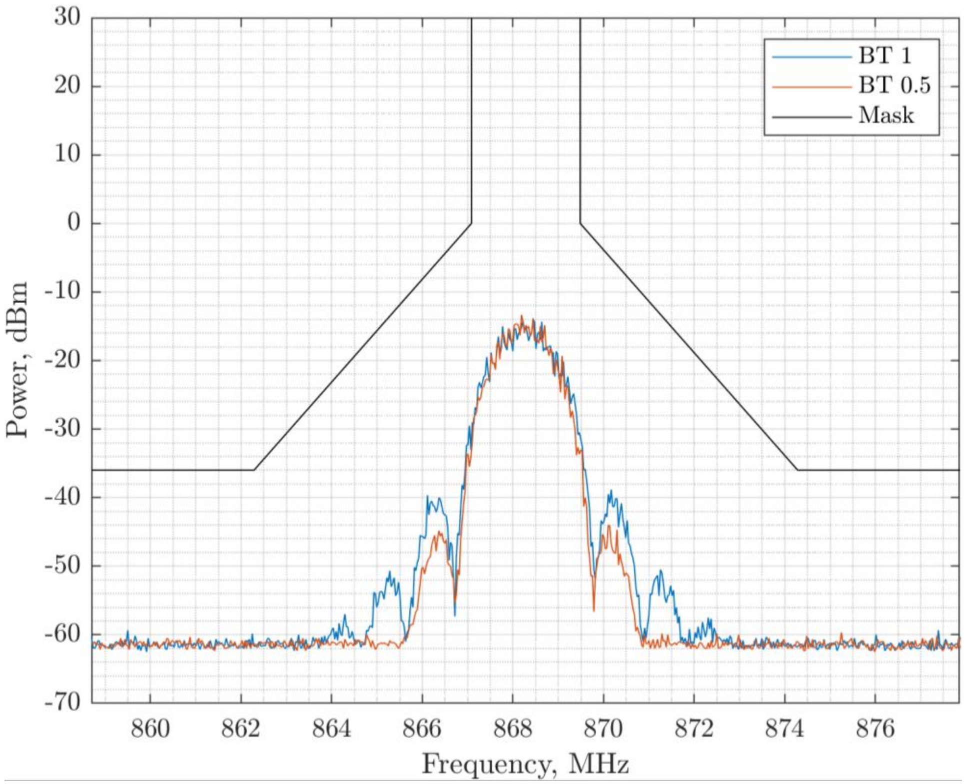


Figure 18. 2.080 Mbps Out of Band Emissions

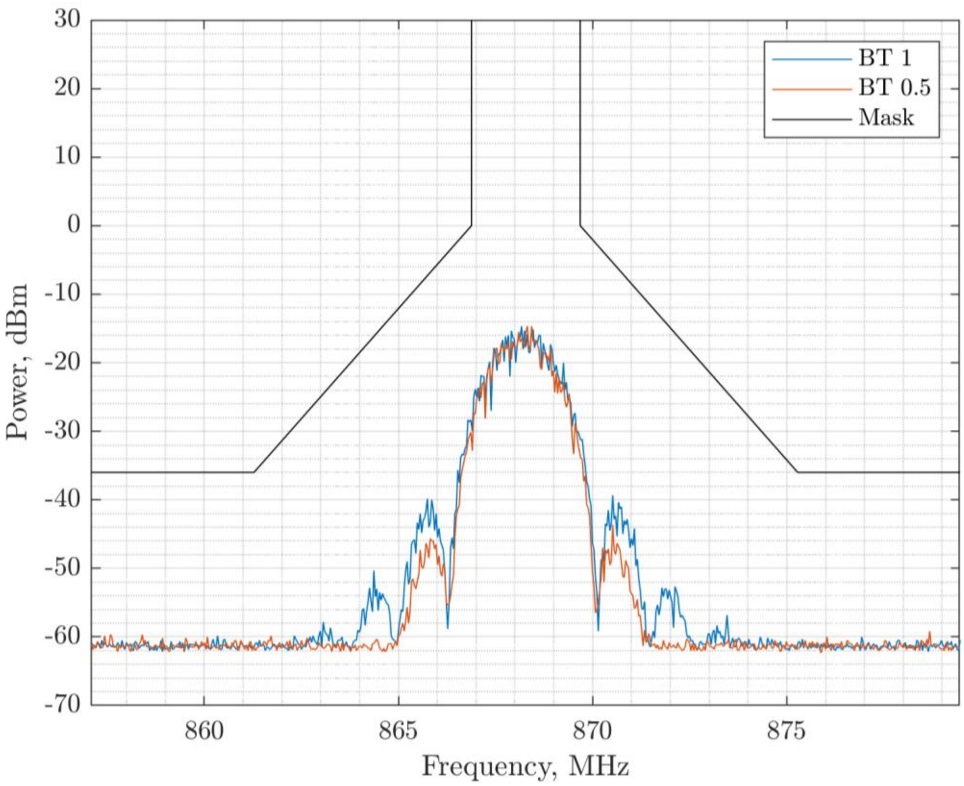


Figure 19. 2.6000 Mbps Out of Band Emissions

2.9 Unwanted Emissions in Spurious Domain Regulations

The bands identified in the ERC REC 70-03 that are applicable to FLRC are shown diagrammatically in the figure below. From this we can see that the band edge measurements must be performed across five bands. We note that the Annex 10 sub-band (see Table 2) is the same as the h1.3 band (see Table 1), but at a lower less demanding, output power, so is not retested.

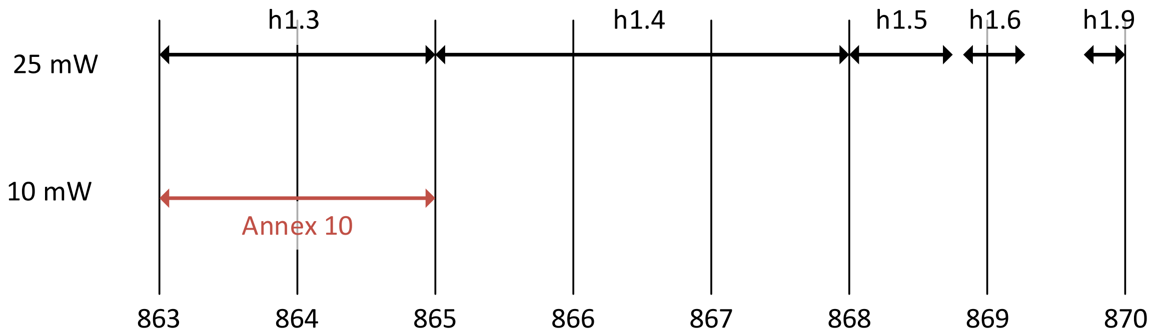


Figure 20. FLRC Compatible Sub-Band Edges and Output Power

Due to the bandwidth restrictions, not all combinations of FLRC bitrate are valid for each sub-band. The legal data rates for each of the 863-870 MHz sub-bands are indicated with a Y in the corresponding column. Note that Y indicates that the data rate can comply, but only a single channel can be accommodated.

Table 5. FLRC Compatible Data Rates versus Sub-Band

Data Rate (Mbps)	BW (MHz)	OCW (MHz)	h1.3	h1.4	h1.5	h1.6	h1.9	A10
			Bandwidth (MHz)					
			2.0	3.0	0.6	0.5	0.3	2
2.600	2.666	2.8	-	Y	-	-	-	-
2.080	2.222	2.4	-	Y	-	-	-	-
1.300	1.333	1.4	Y	Y	-	-	-	Y
1.040	1.333	1.4	Y	Y	-	-	-	Y
0.650	0.740	0.8	Y	Y	-	-	-	Y
0.520	0.571	0.7	Y	Y	-	-	-	Y
0.325	0.357	0.4	Y	Y	Y	Y	Y	Y
0.260	0.307	0.4	Y	Y	Y	Y	Y	Y

The configurations in Table 5 are tested against the spectral mask imposed for band edge emissions, shown below. All testing is performed at +14 dBm output power.

For such testing we must assume a channelisation: the wanted signal in all cases is placed 1 OCW away from the band edge. Where only a single channel can be accommodated in the sub-band (see previous table) the channel is instead centred on the sub-band.

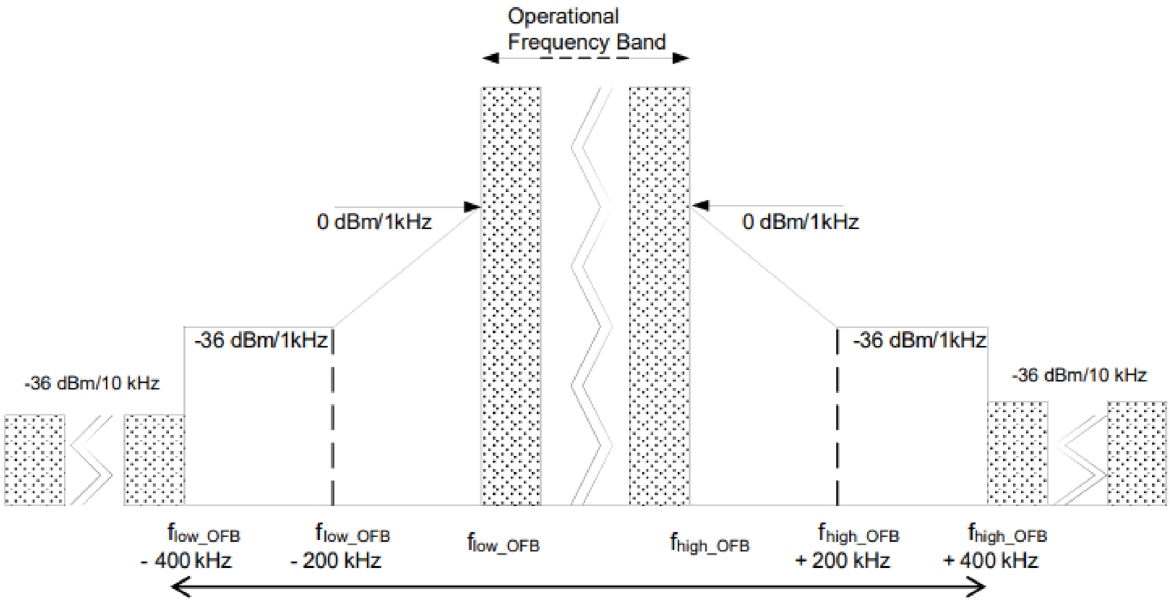


Figure 21. The Tx Out of Band Emissions Mask for the Band Edge

Measurements on the LR2021 Evaluation Kit (with 32MHz crystal oscillator) for each *compatible* data rate, Gaussian filtering setting and data rate are shown. Compatibility is evaluated on the test signal exceeding the extent of the mask. Note that in real applications, the frequency tolerance of the reference oscillator may also need to be considered.

2.10 Unwanted Emissions in Spurious Domain Measurements

2.10.1 Sub-Band h1.3

For measurement bandwidths refer to previous figure.

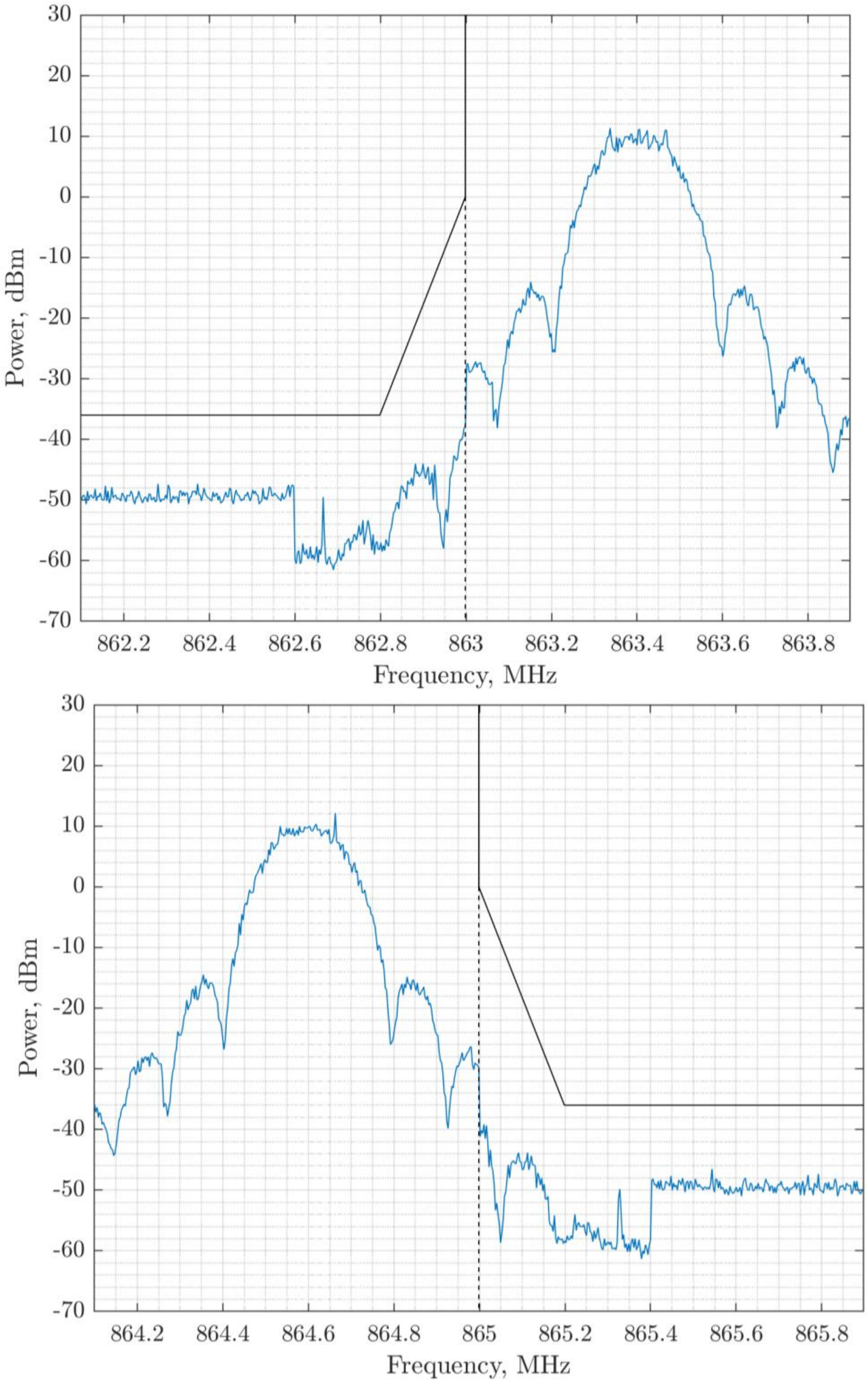


Figure 22. h1.3 Band Edge Compliance at 260 kbps BT = 1

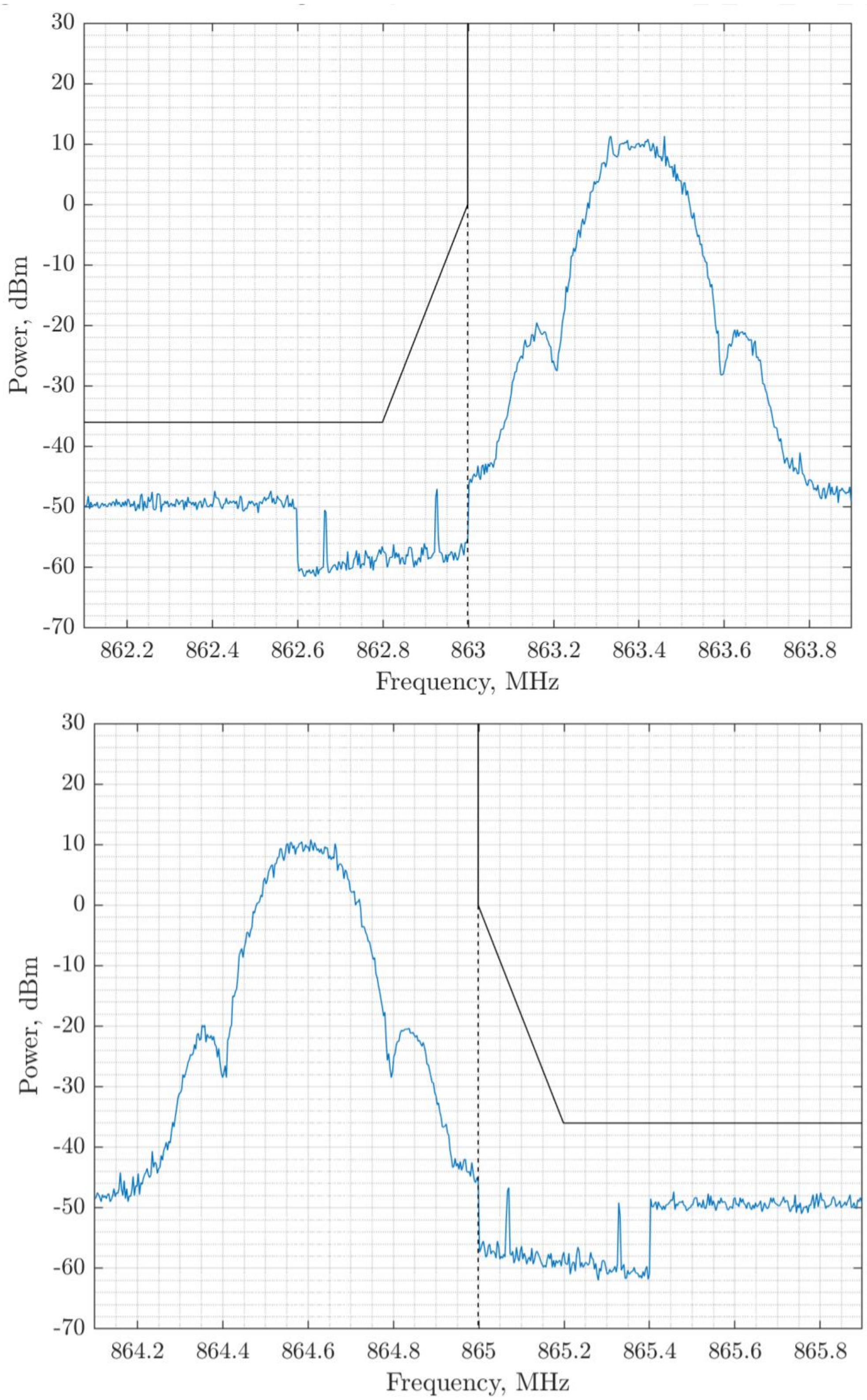


Figure 23. h1.3 Band Edge Compliance at 260 kbps BT = 0.5

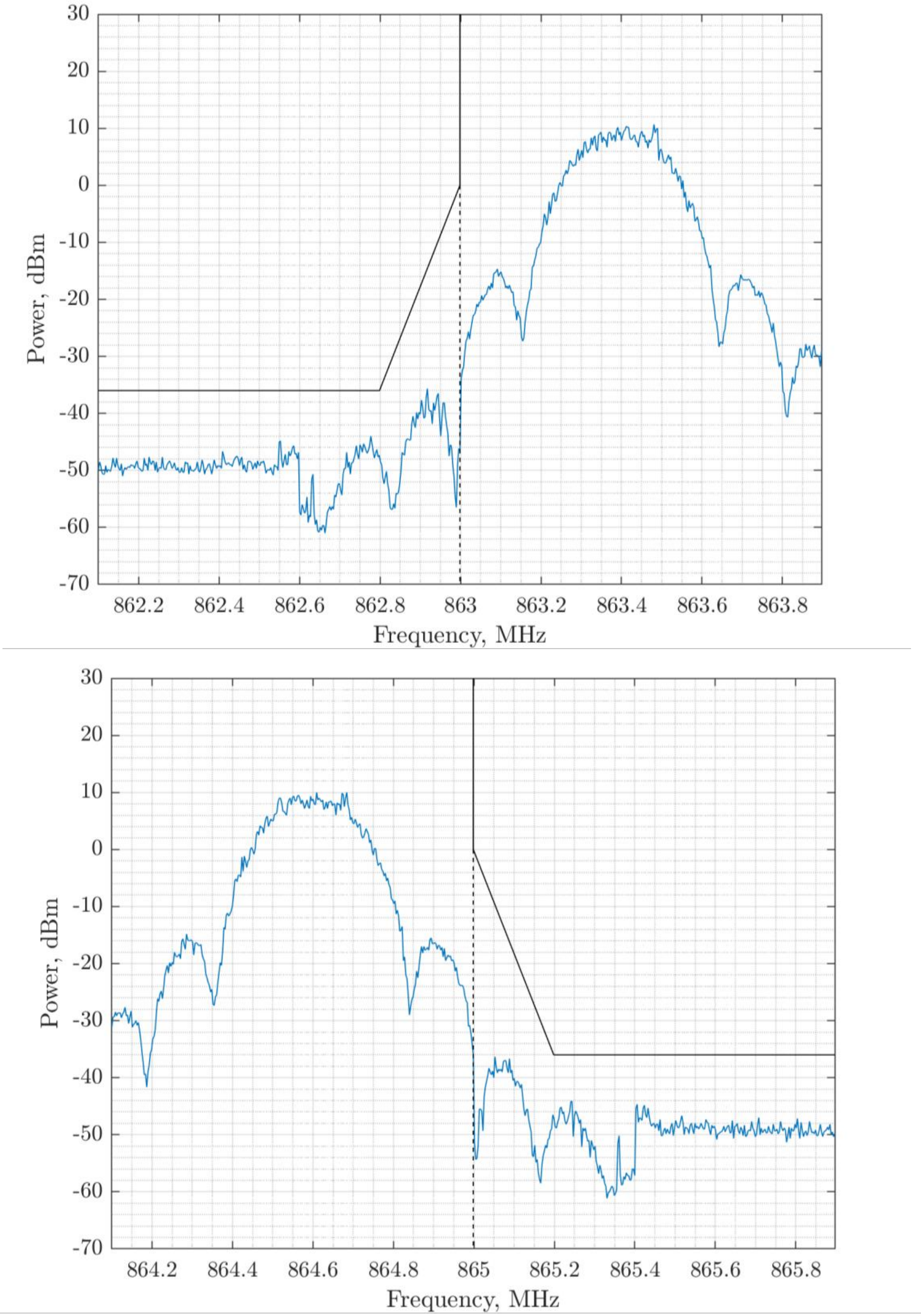


Figure 24. h1.3 Band Edge Compliance at 325 kbps BT = 1

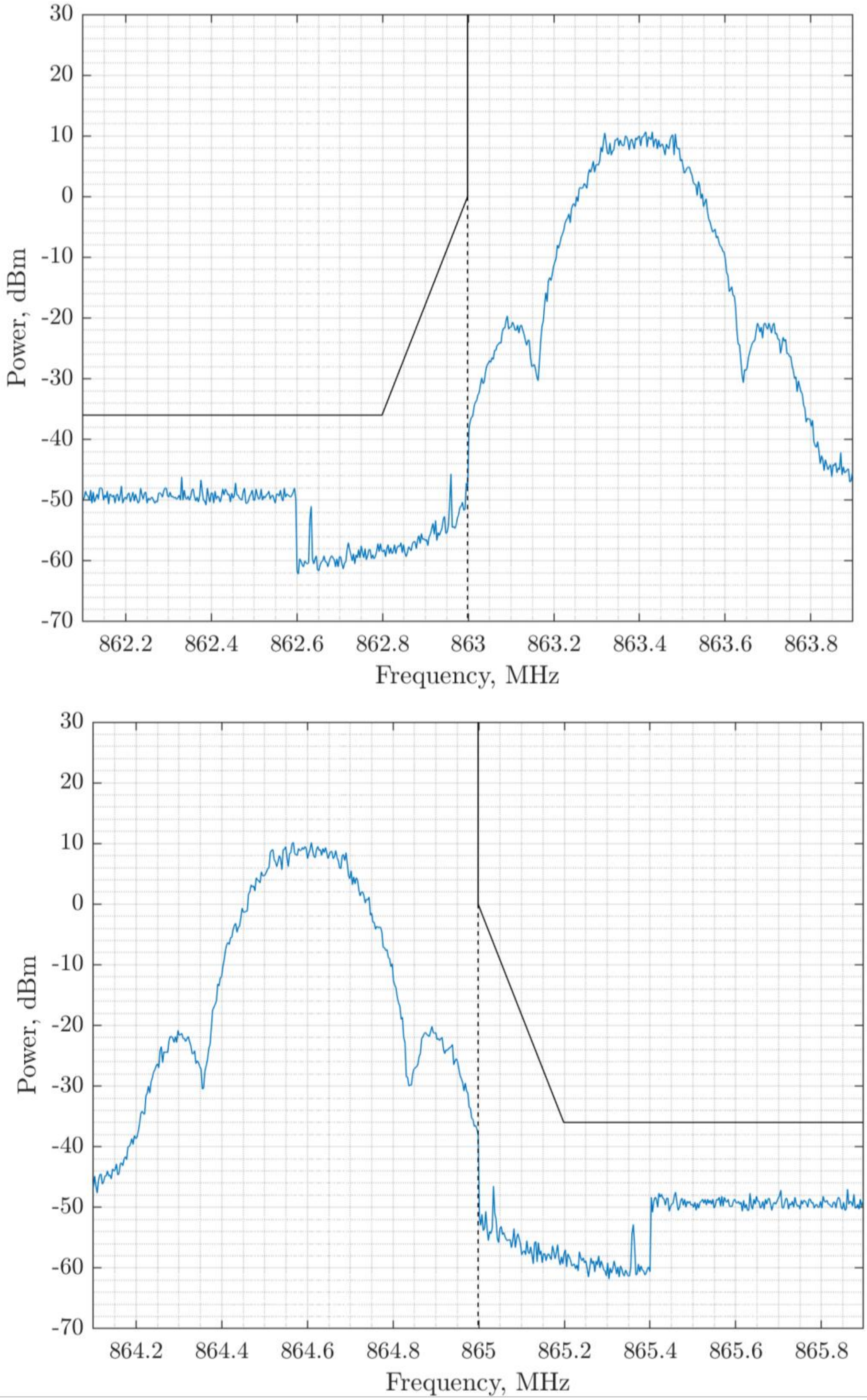


Figure 25. h1.3 Band Edge Compliance at 325 kbps BT = 0.5

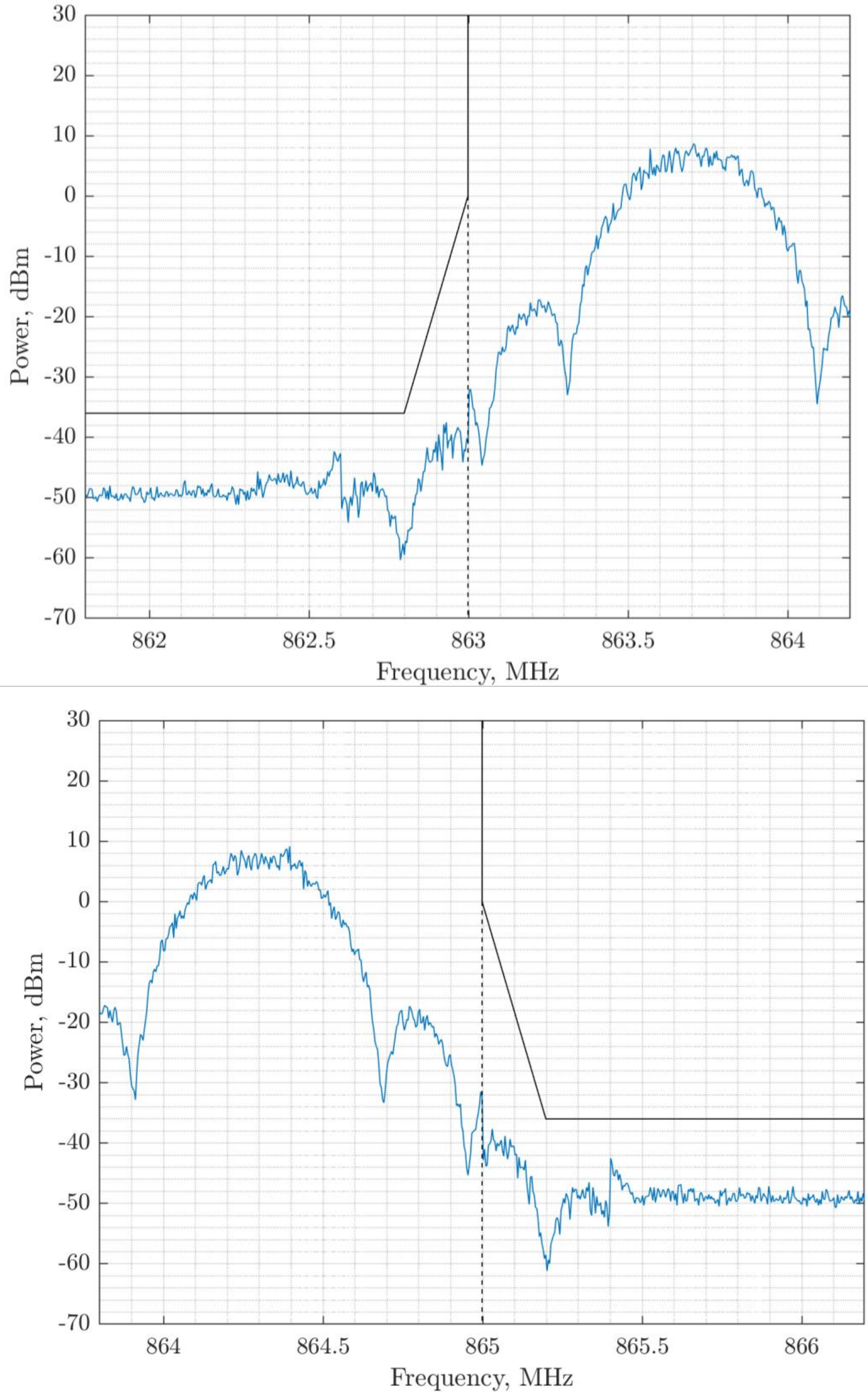


Figure 26. h1.3 Band Edge Compliance at 520 kbps BT = 1

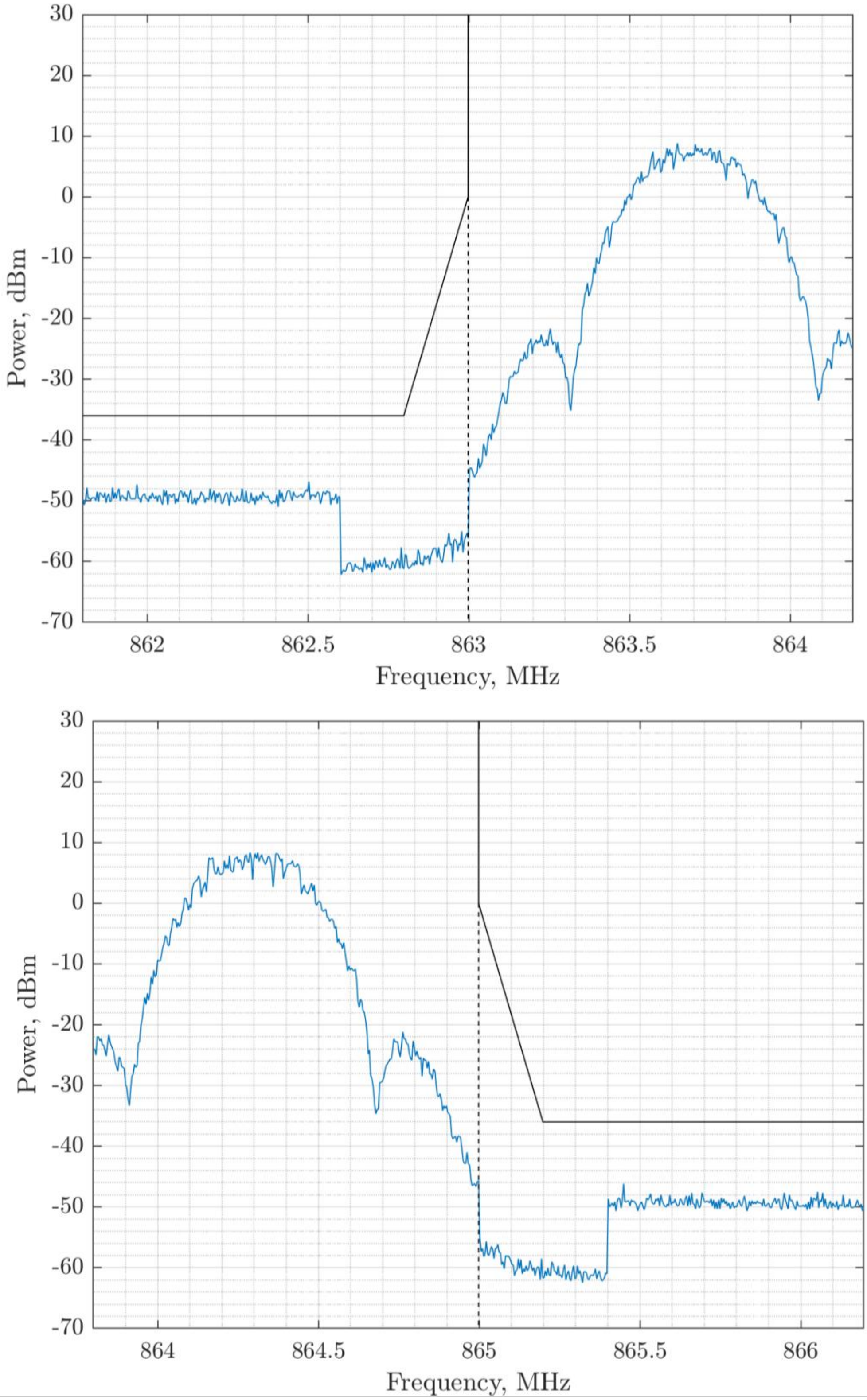


Figure 27. h1.3 Band Edge Compliance at 520 kbps BT = 0.5

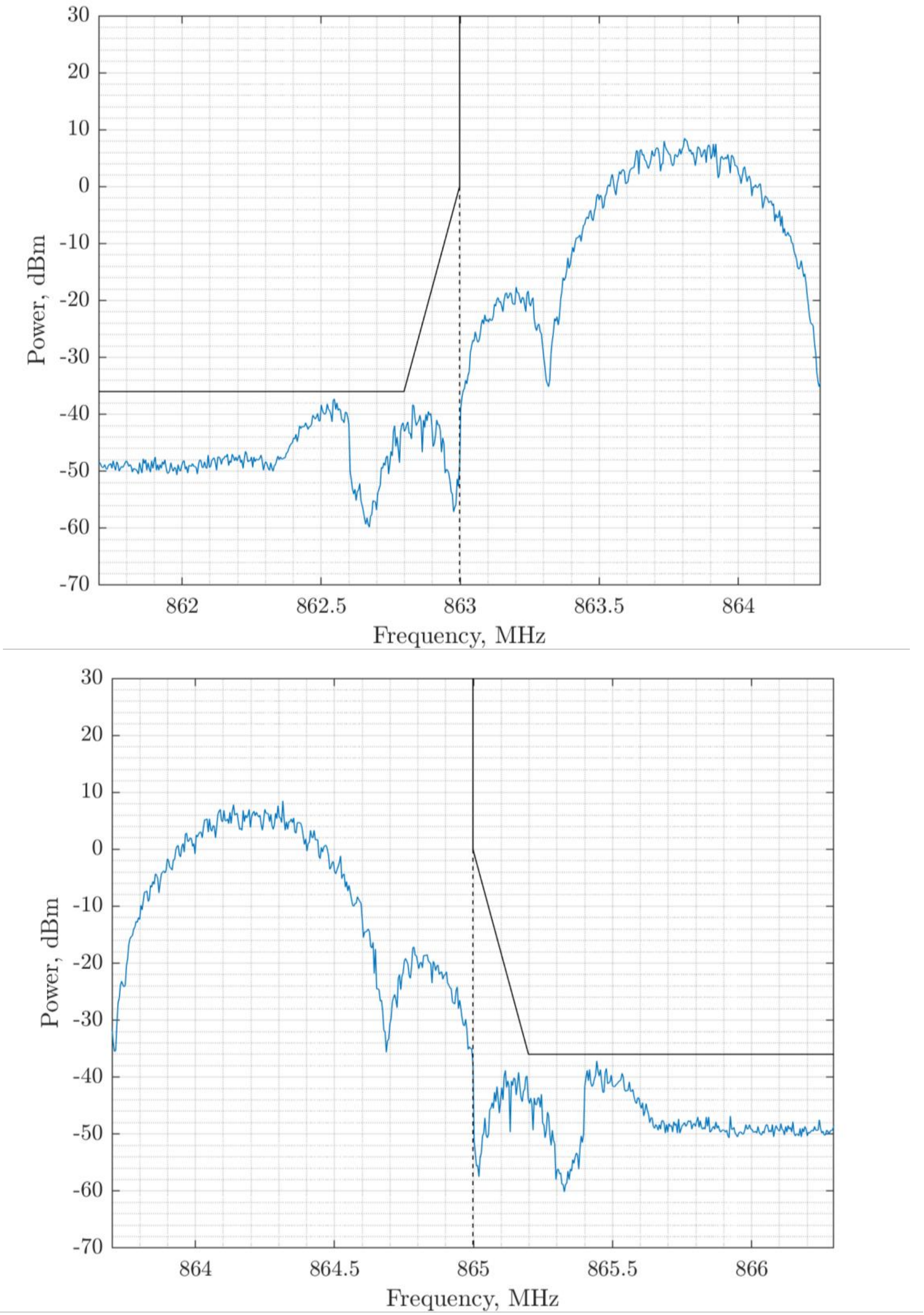


Figure 28. h1.3 Band Edge Compliance at 650 kbps BT = 1

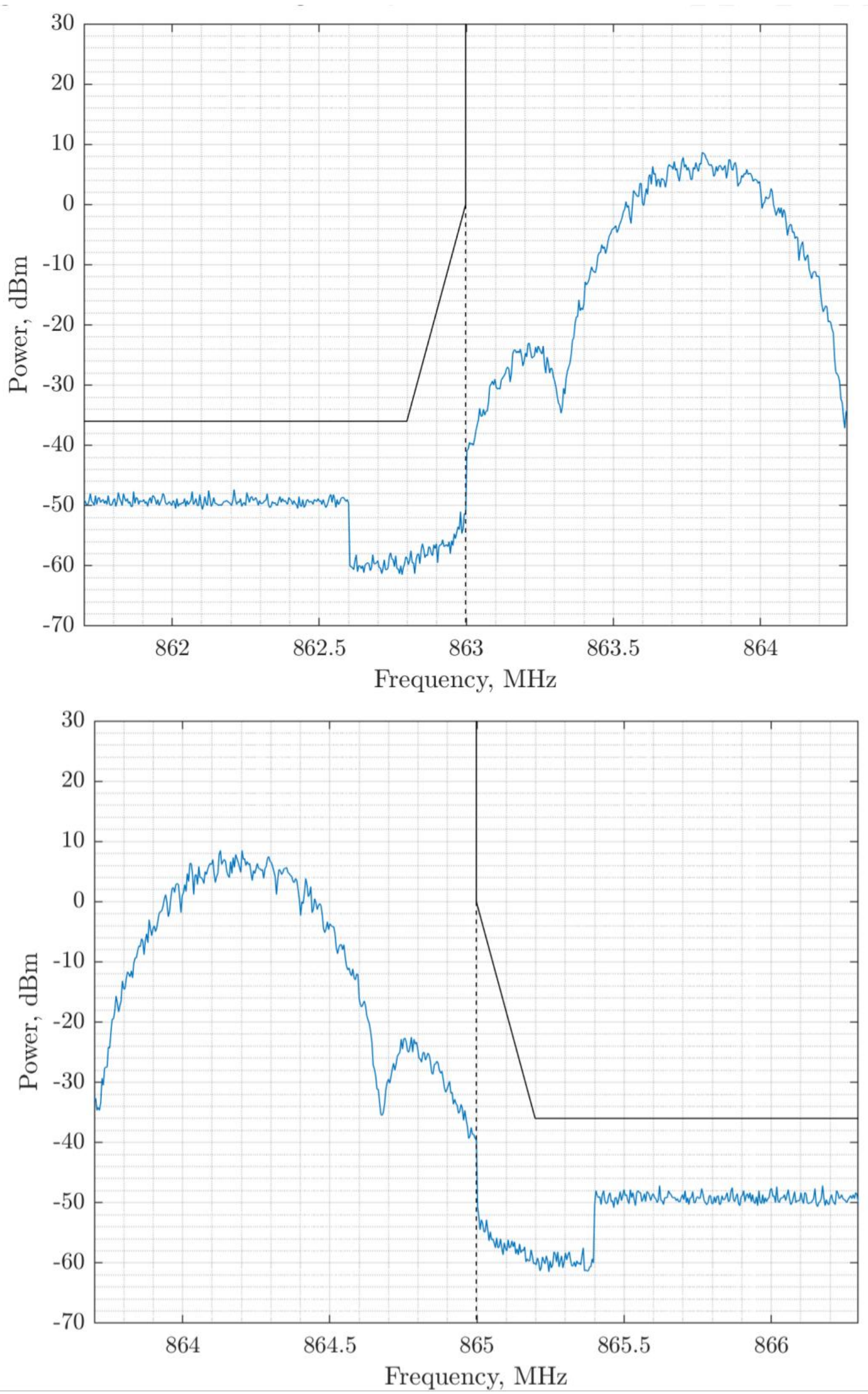


Figure 29. h1.3 Band Edge Compliance at 650 kbps BT = 0.5

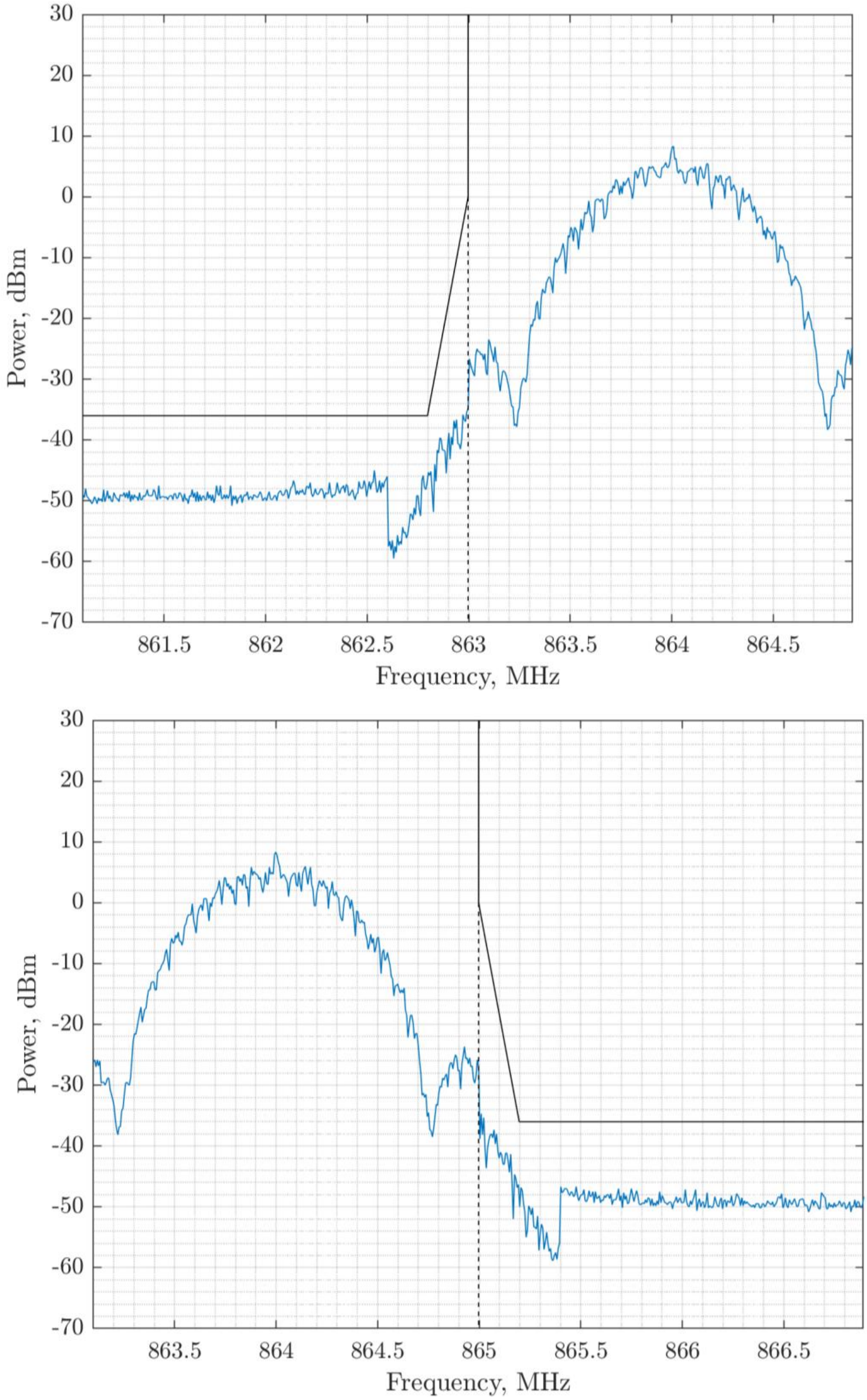
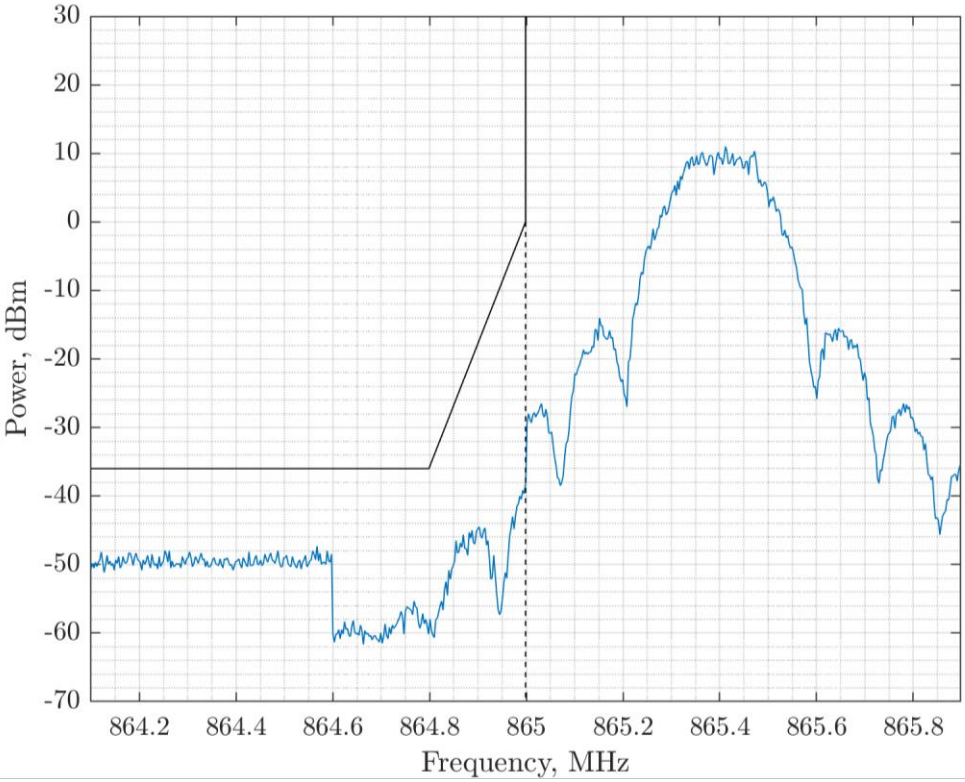


Figure 30. h1.3 Band Edge Compliance at 1.040 Mbps BT = 0.5

2.10.2 Sub-Band h1.4



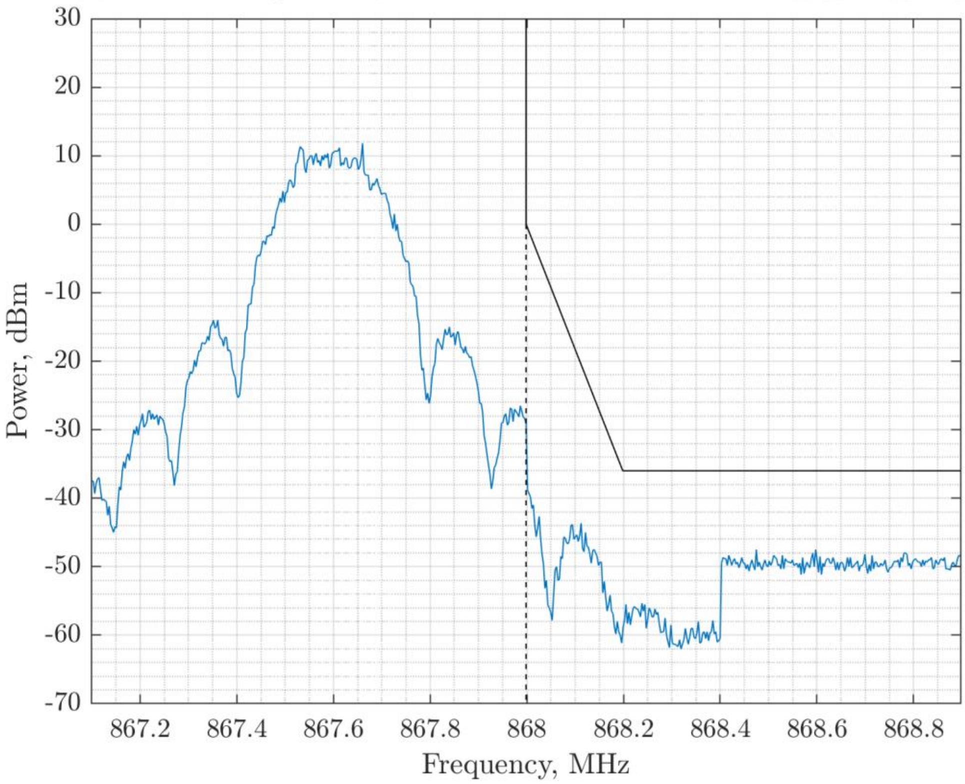


Figure 31. h1.4 Band Edge Compliance at 260 kbps BT = 1

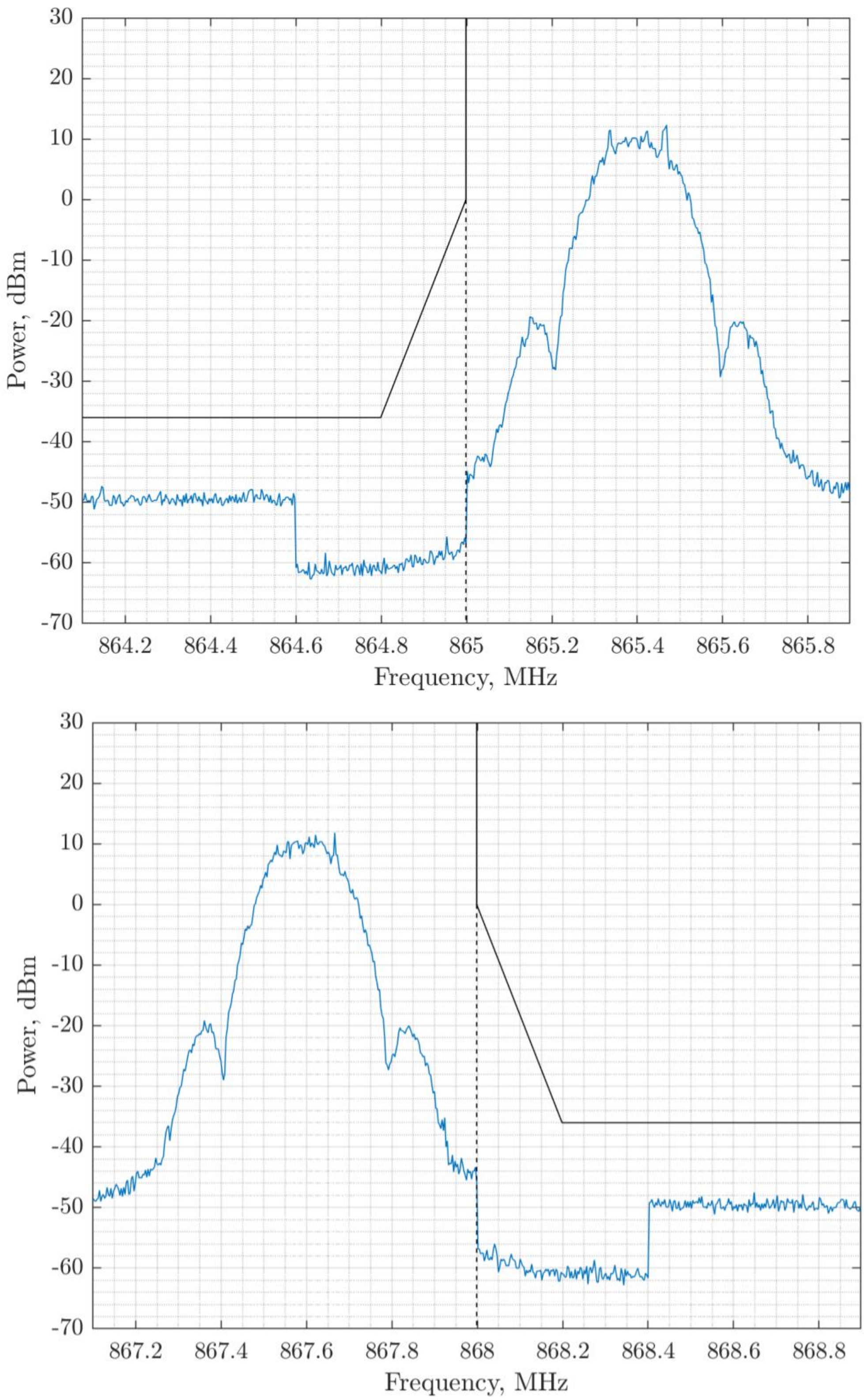


Figure 32. h1.4 Band Edge Compliance at 260 kbps BT = 0.5

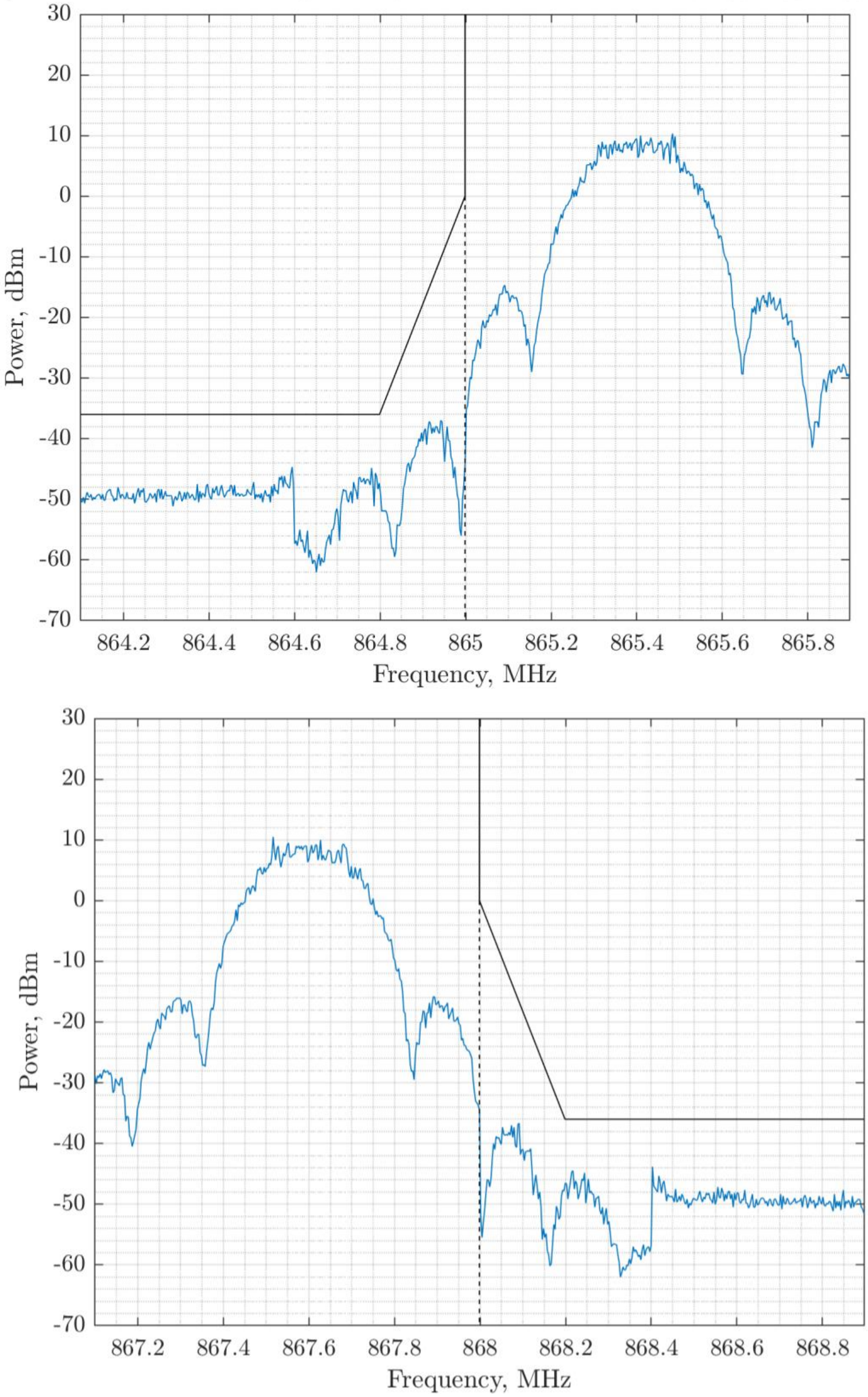


Figure 33. h1.4 Band Edge Compliance at 325 kbps BT = 1

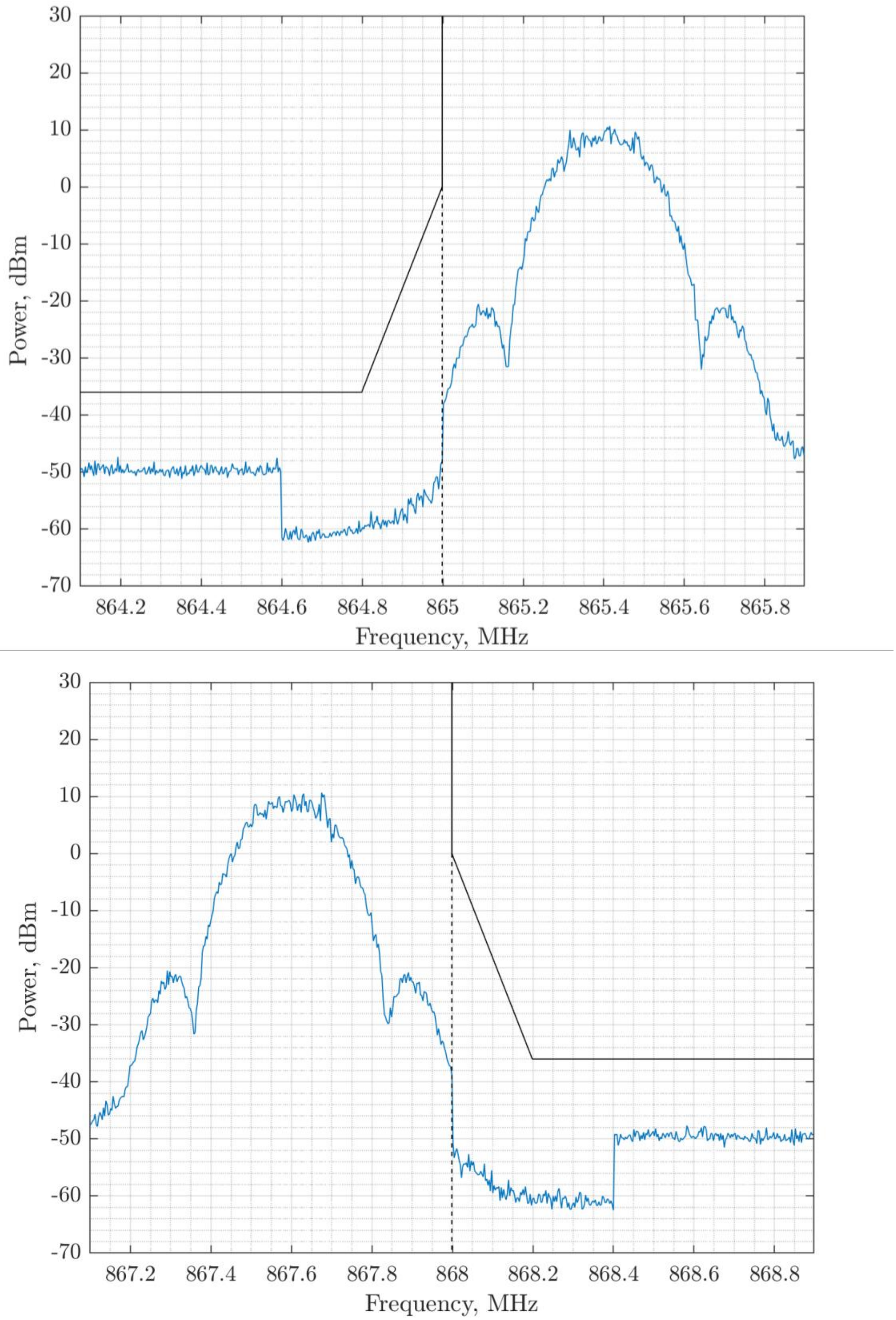


Figure 34. h1.4 Band Edge Compliance at 325 kbps BT = 0.5

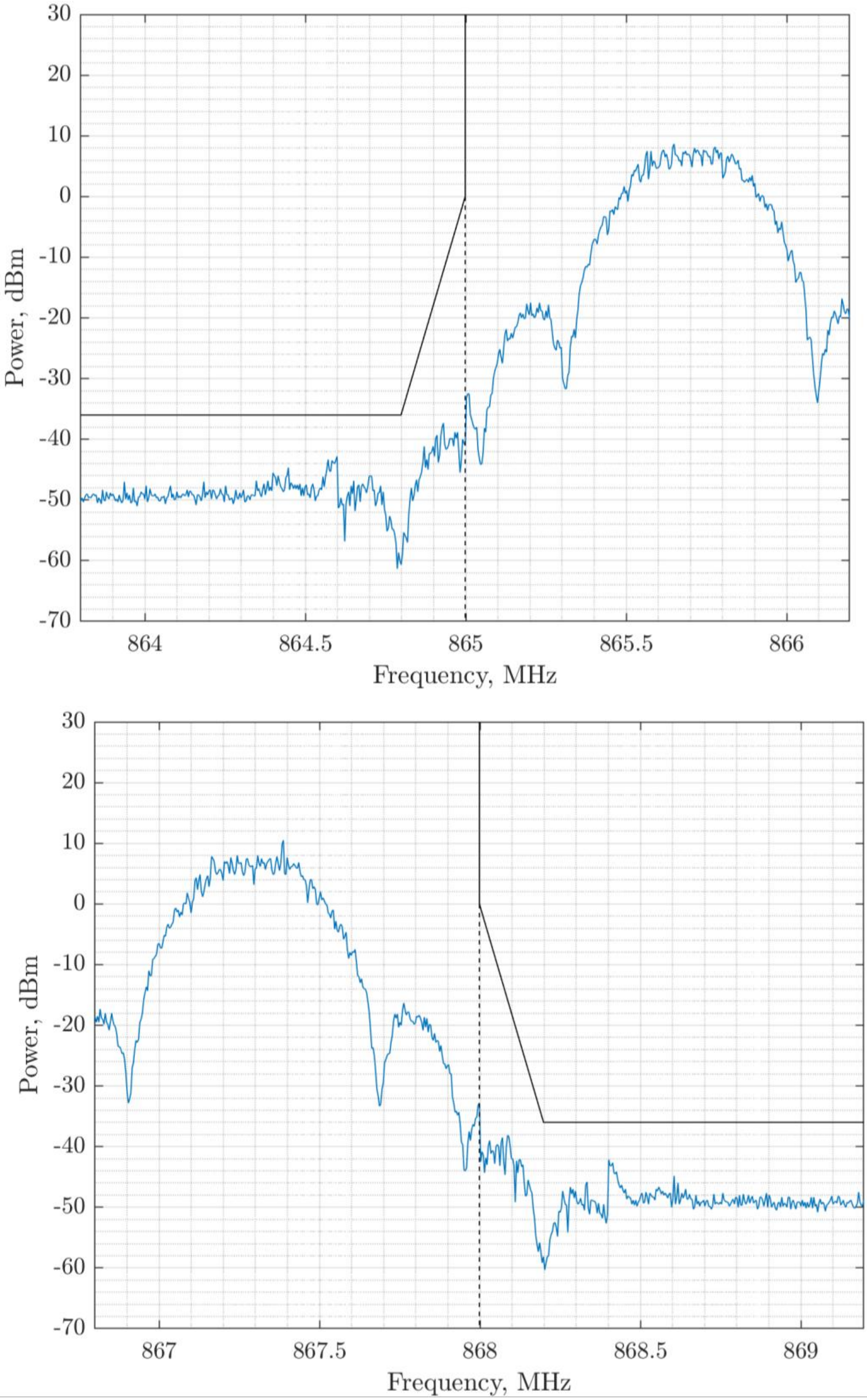


Figure 35. h1.4 Band Edge Compliance at 520 kbps BT = 1

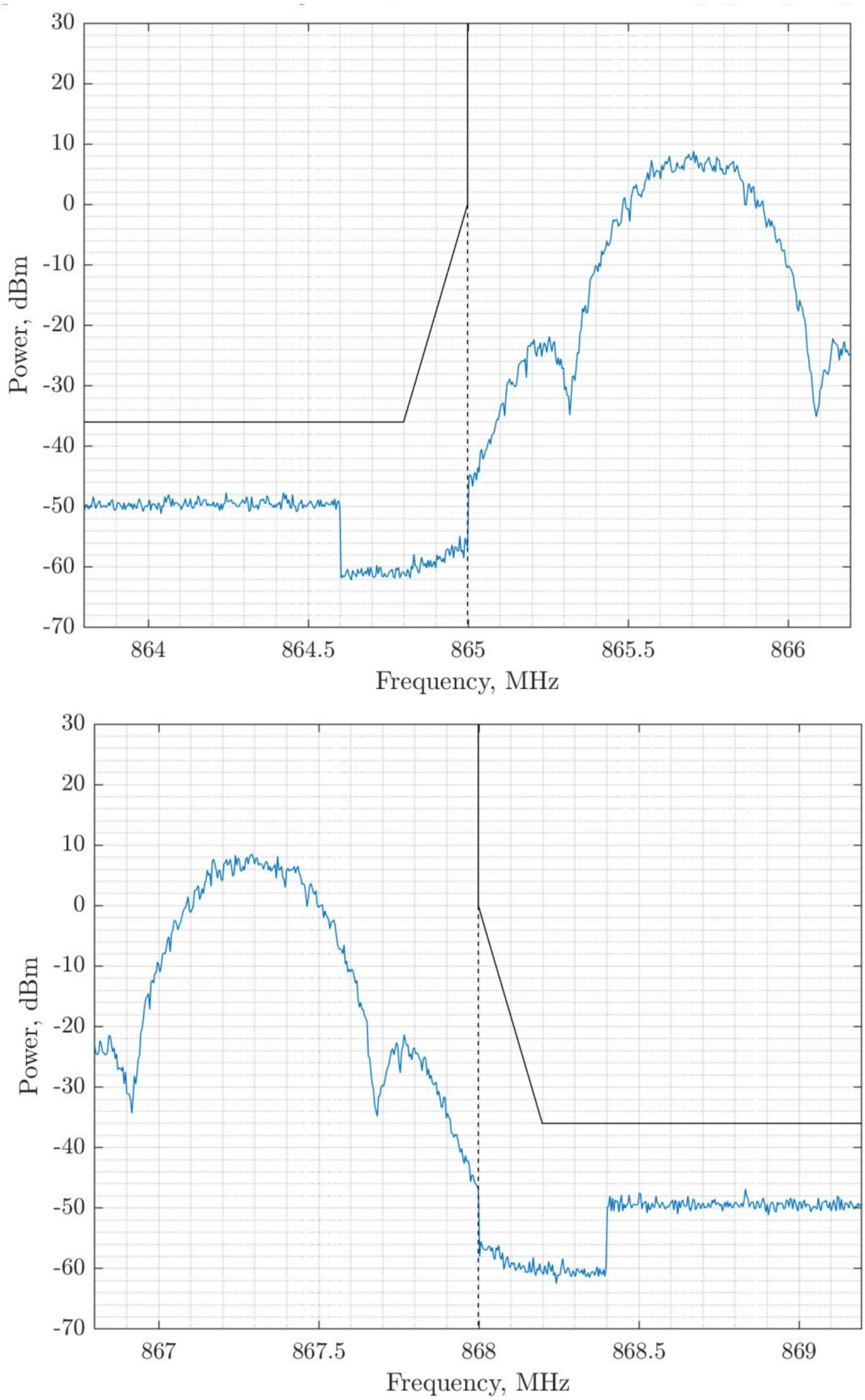


Figure 36. h1.4 Band Edge Compliance at 520 kbps BT = 0.5

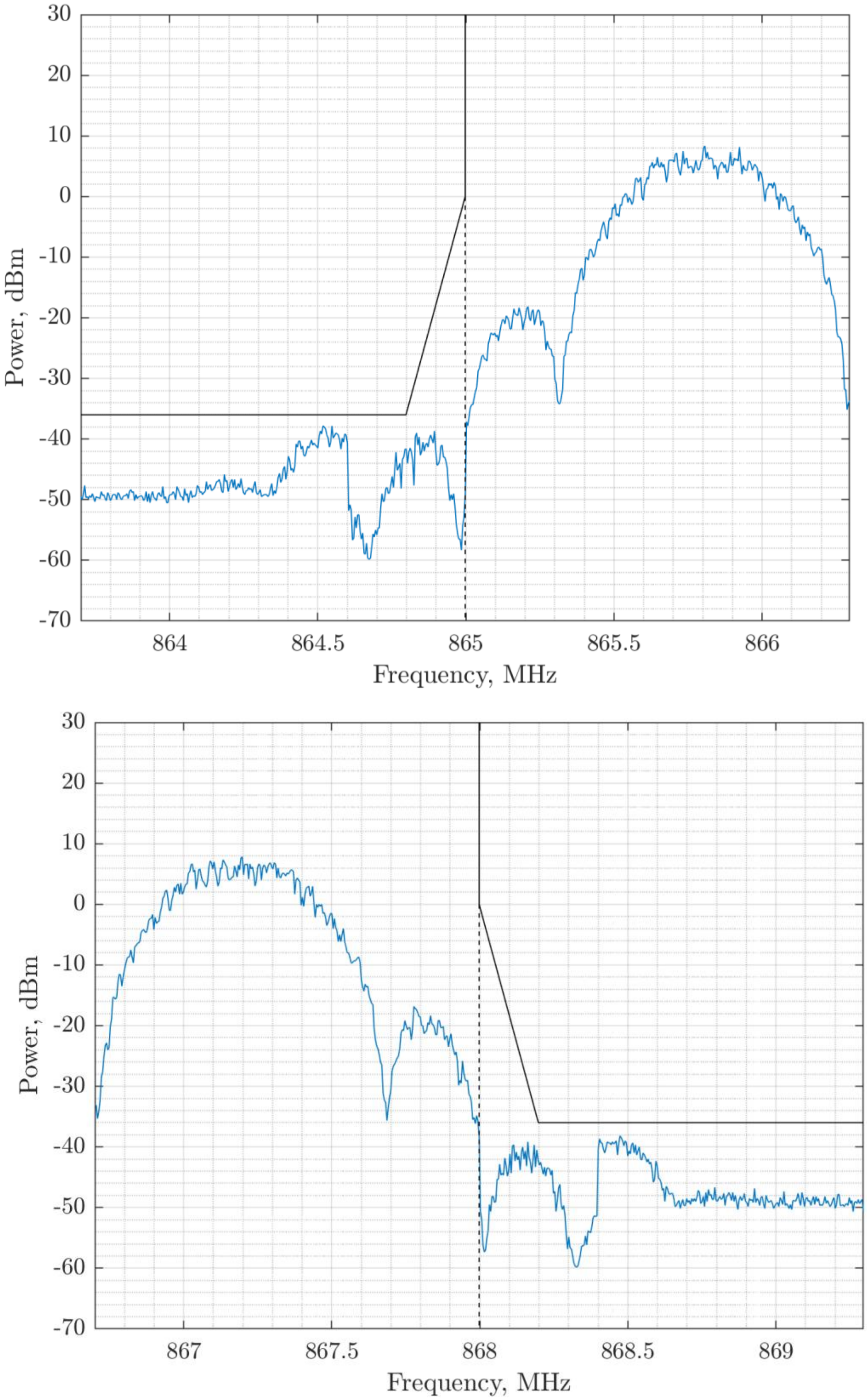


Figure 37. h1.4 Band Edge Compliance at 650 kbps BT = 1

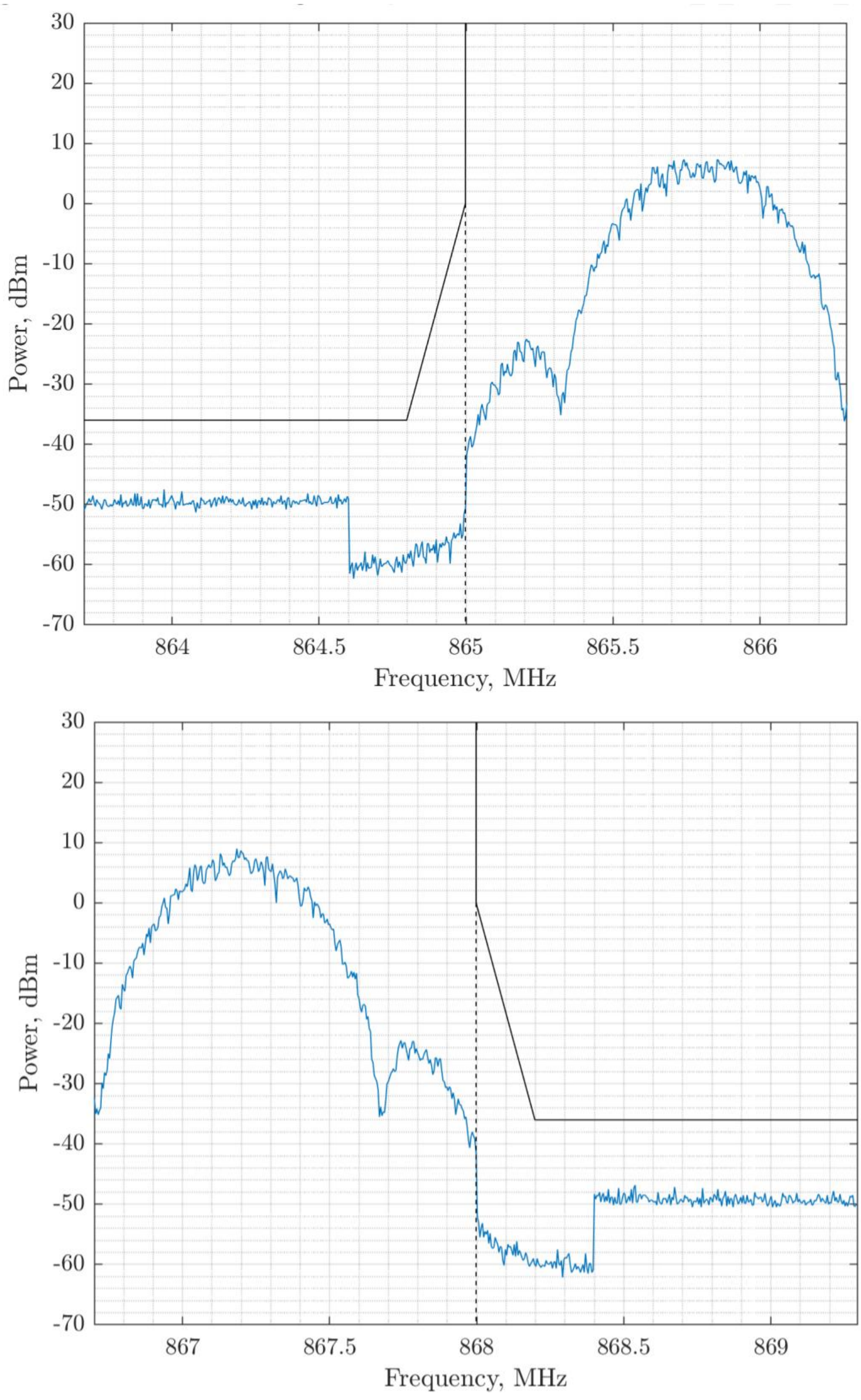


Figure 38. h1.4 Band Edge Compliance at 650 kbps BT = 0.5

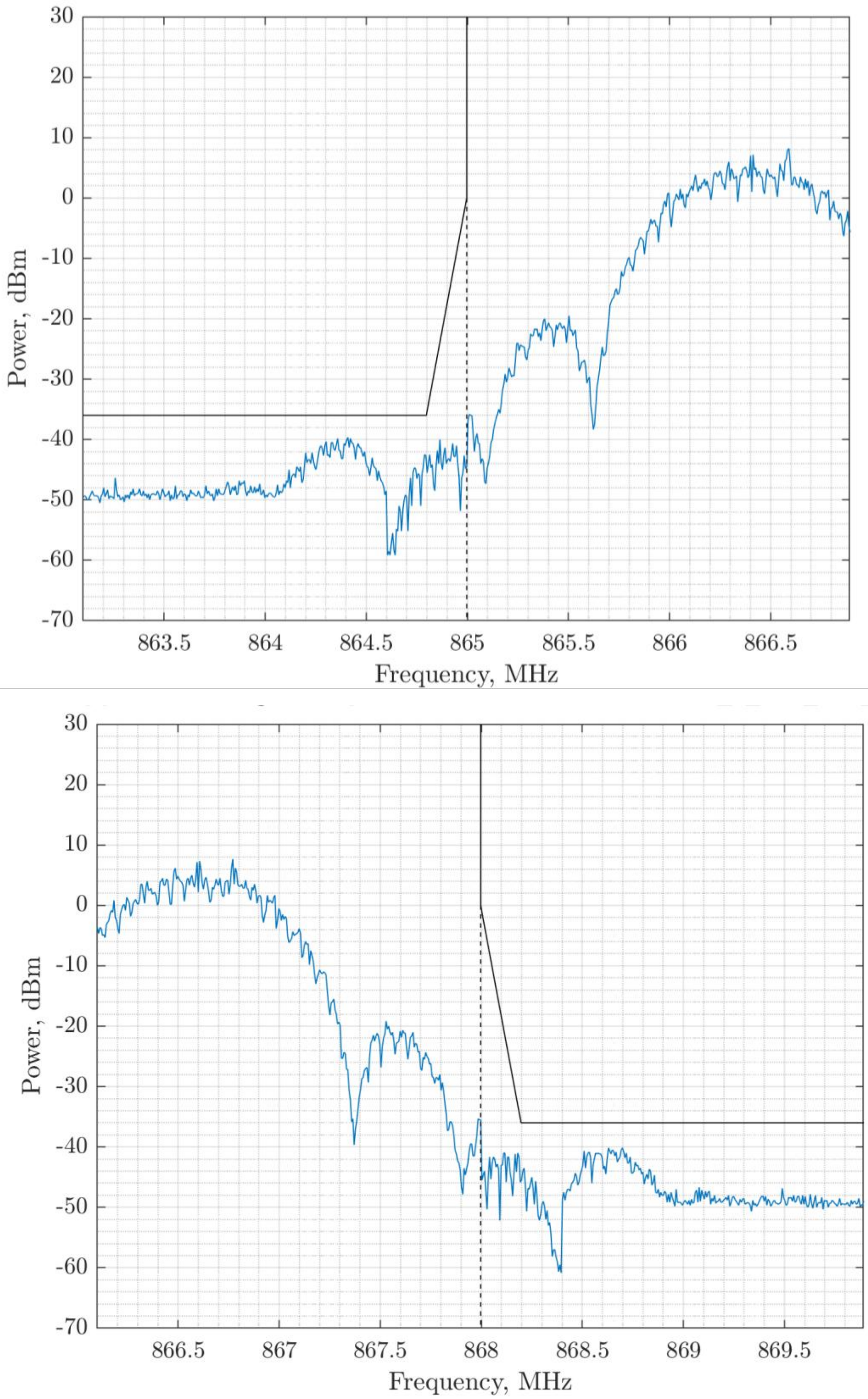


Figure 39. h1.4 Band Edge Compliance at 1.040 Mbps BT = 1

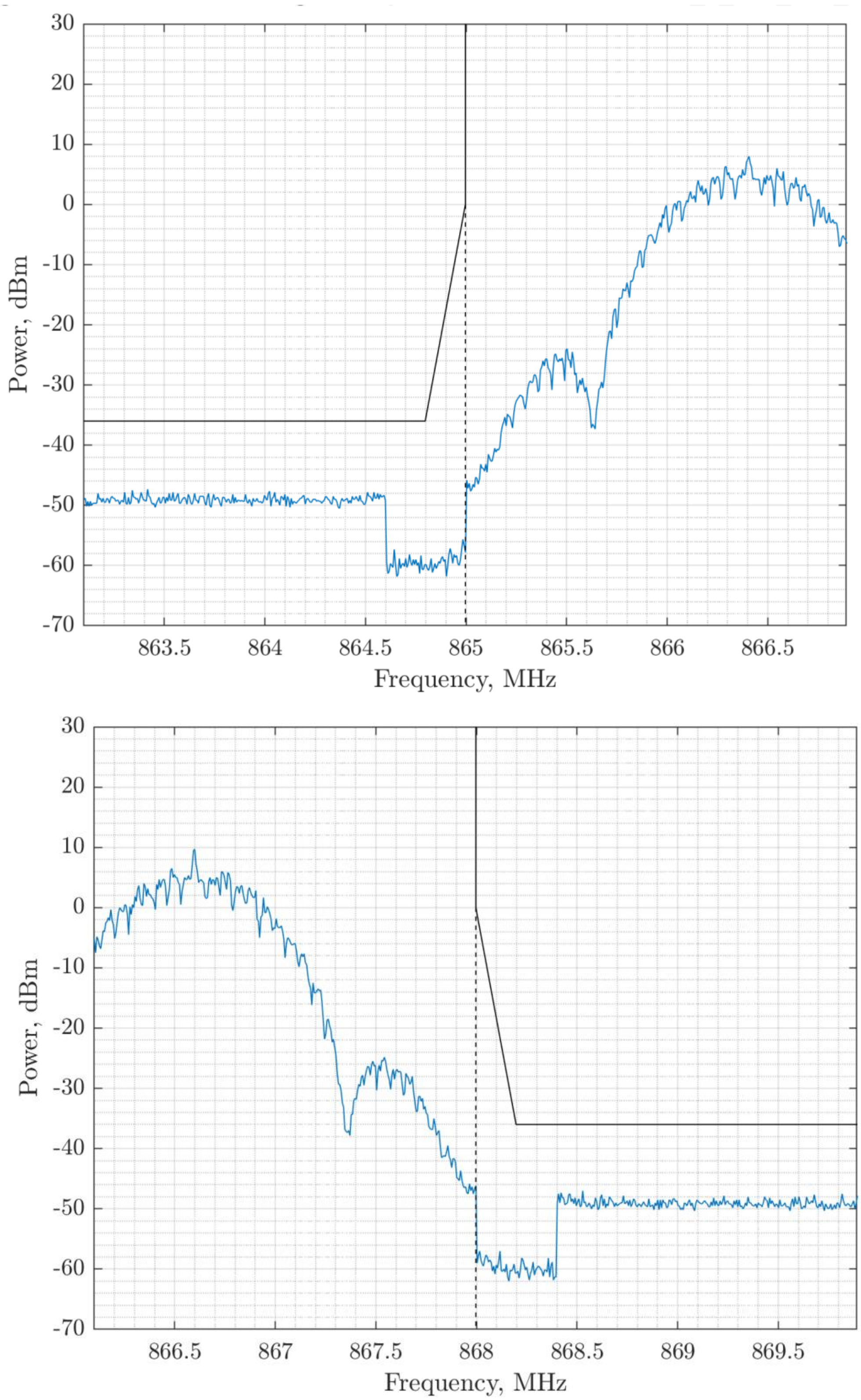


Figure 40. h1.4 Band Edge Compliance at 1.040 Mbps BT = 0.5

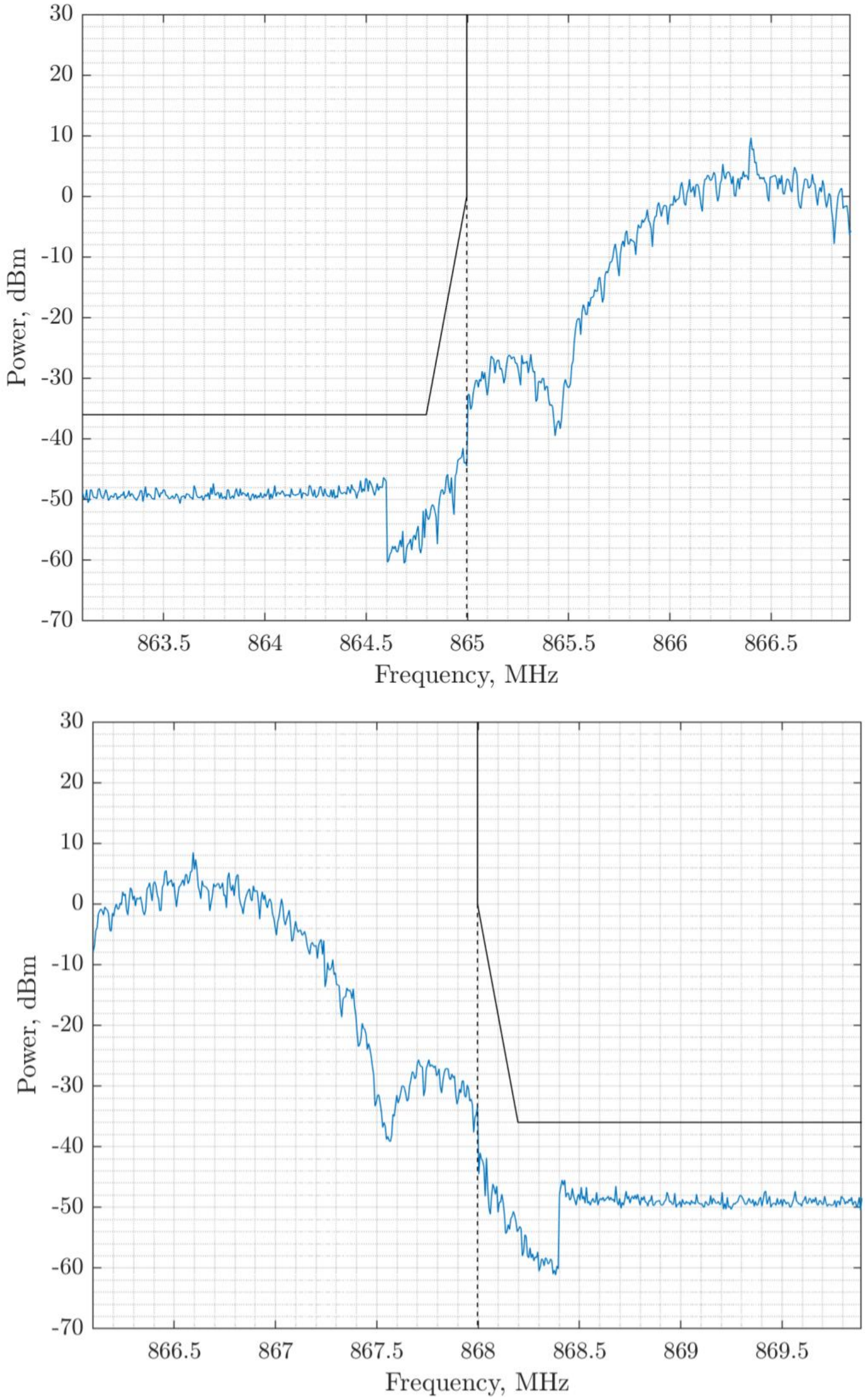


Figure 41. h1.4 Band Edge Compliance at 1.30 Mbps BT = 0.5

2.10.3 Sub-Band h1.5

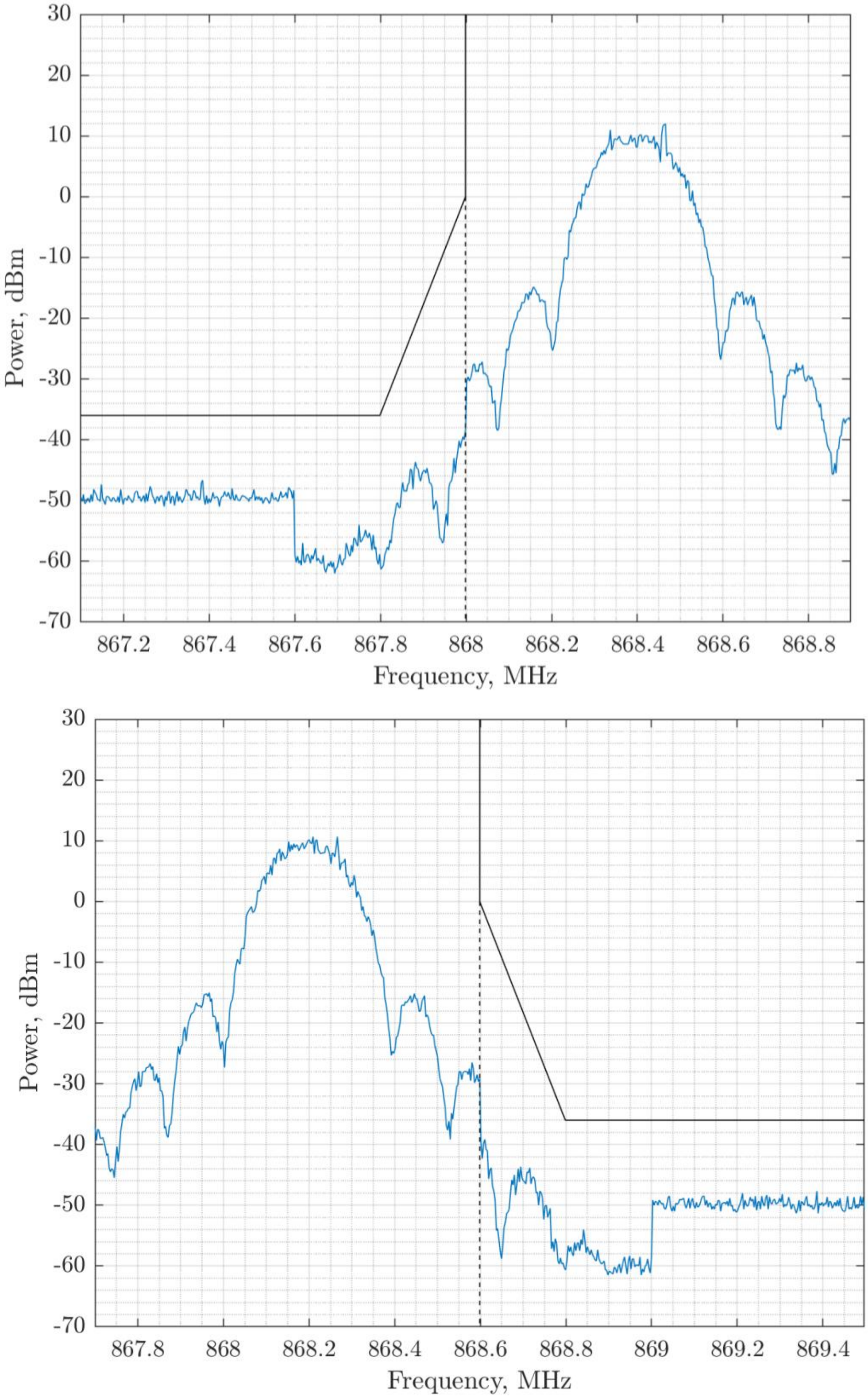


Figure 42. h1.5 Band Edge Compliance at 260 kbps BT = 1

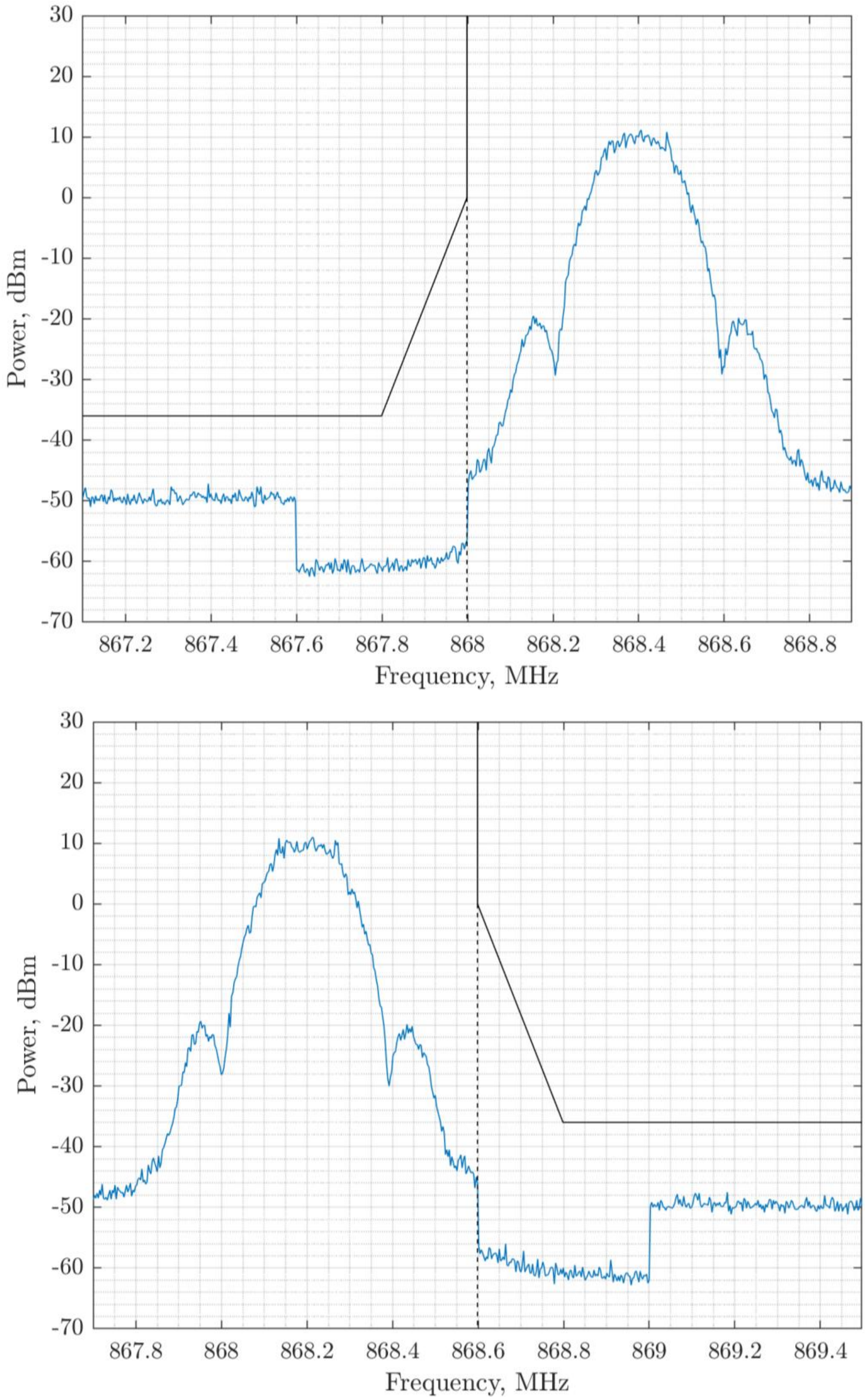


Figure 43. h1.5 Band Edge Compliance at 260 kbps BT = 0.5

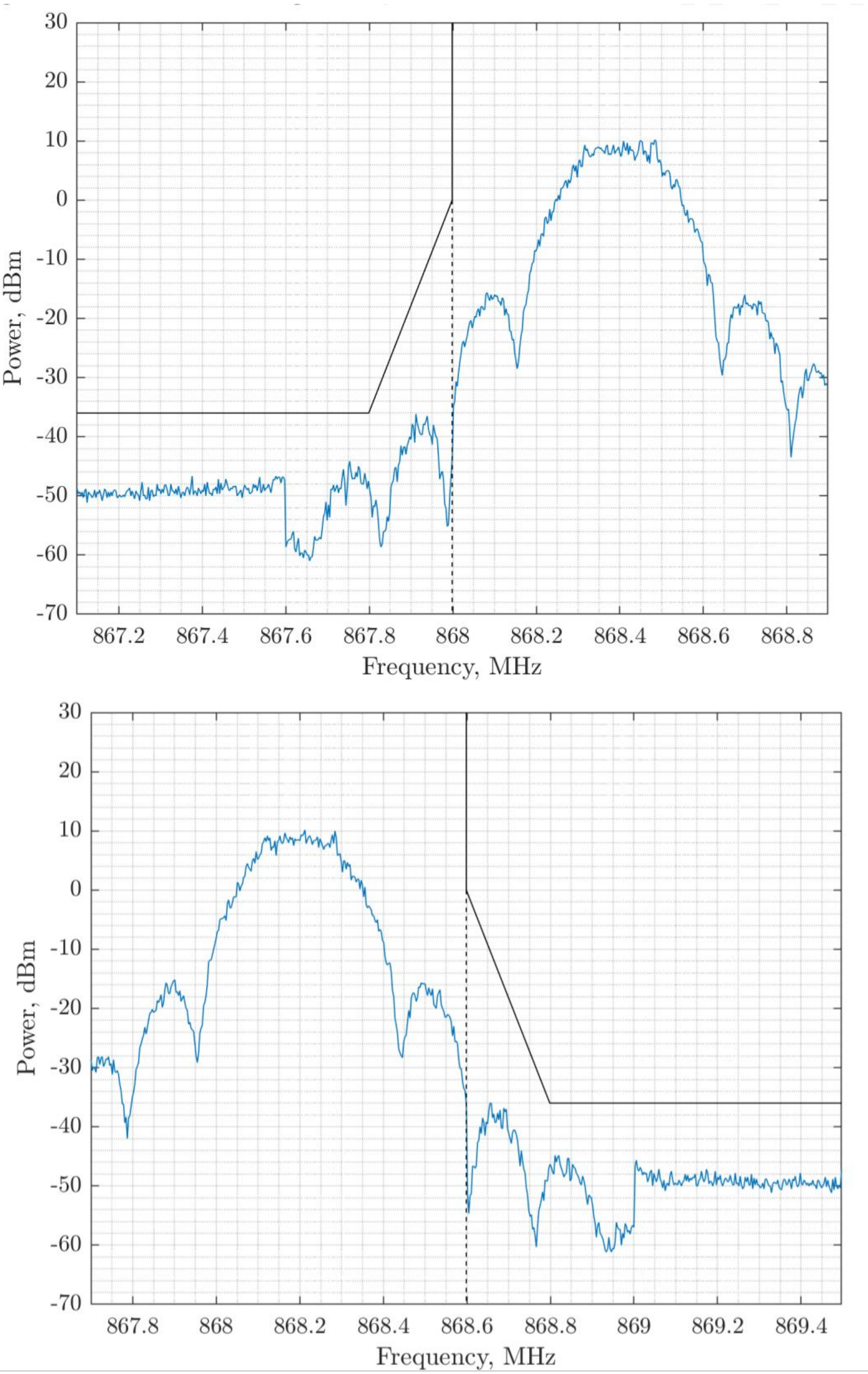


Figure 44. h1.5 Band Edge Compliance at 325 kbps BT = 1

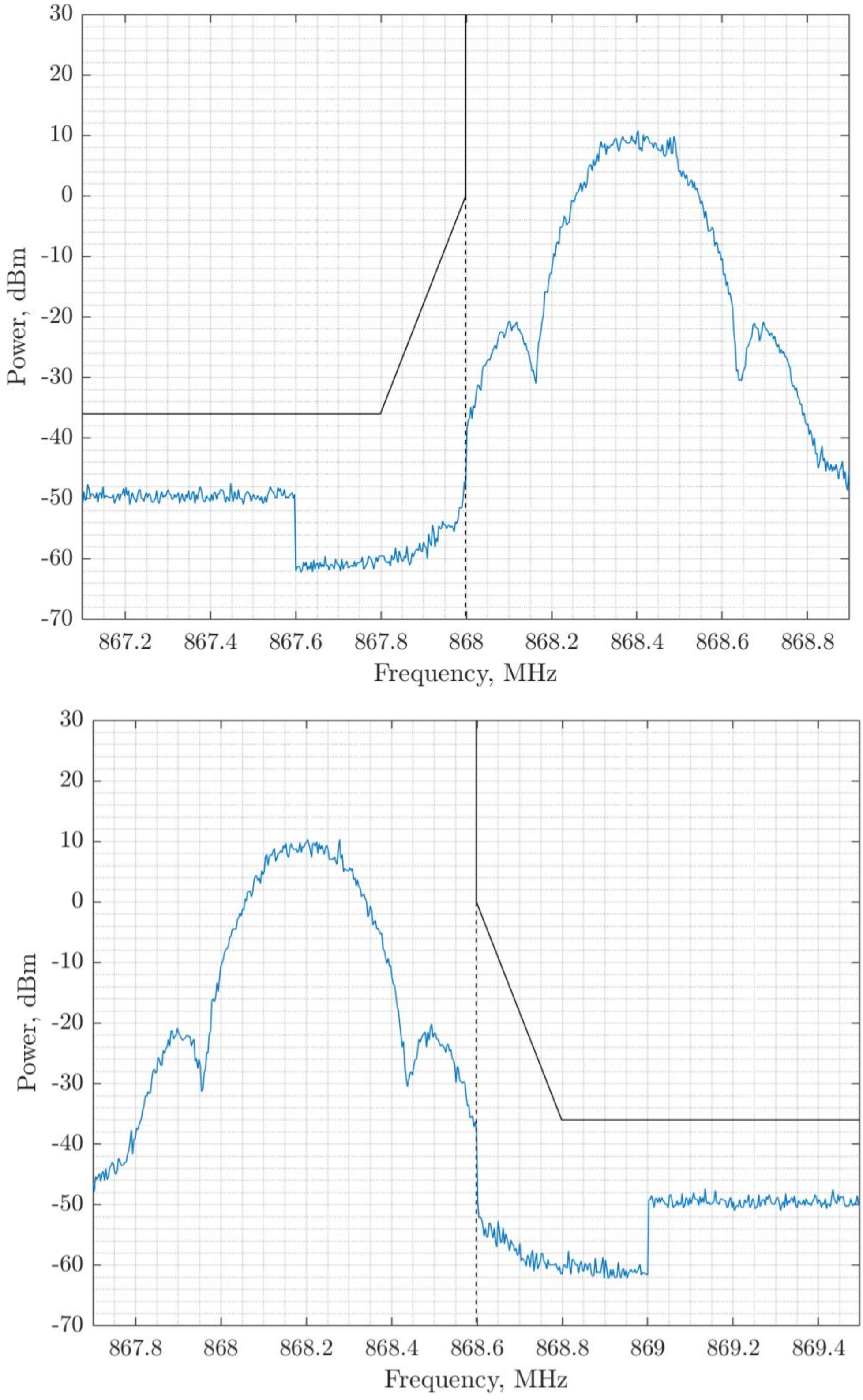


Figure 45. h1.5 Band Edge Compliance at 325 kbps BT = 0.5

2.10.4 Sub-Band h1.6

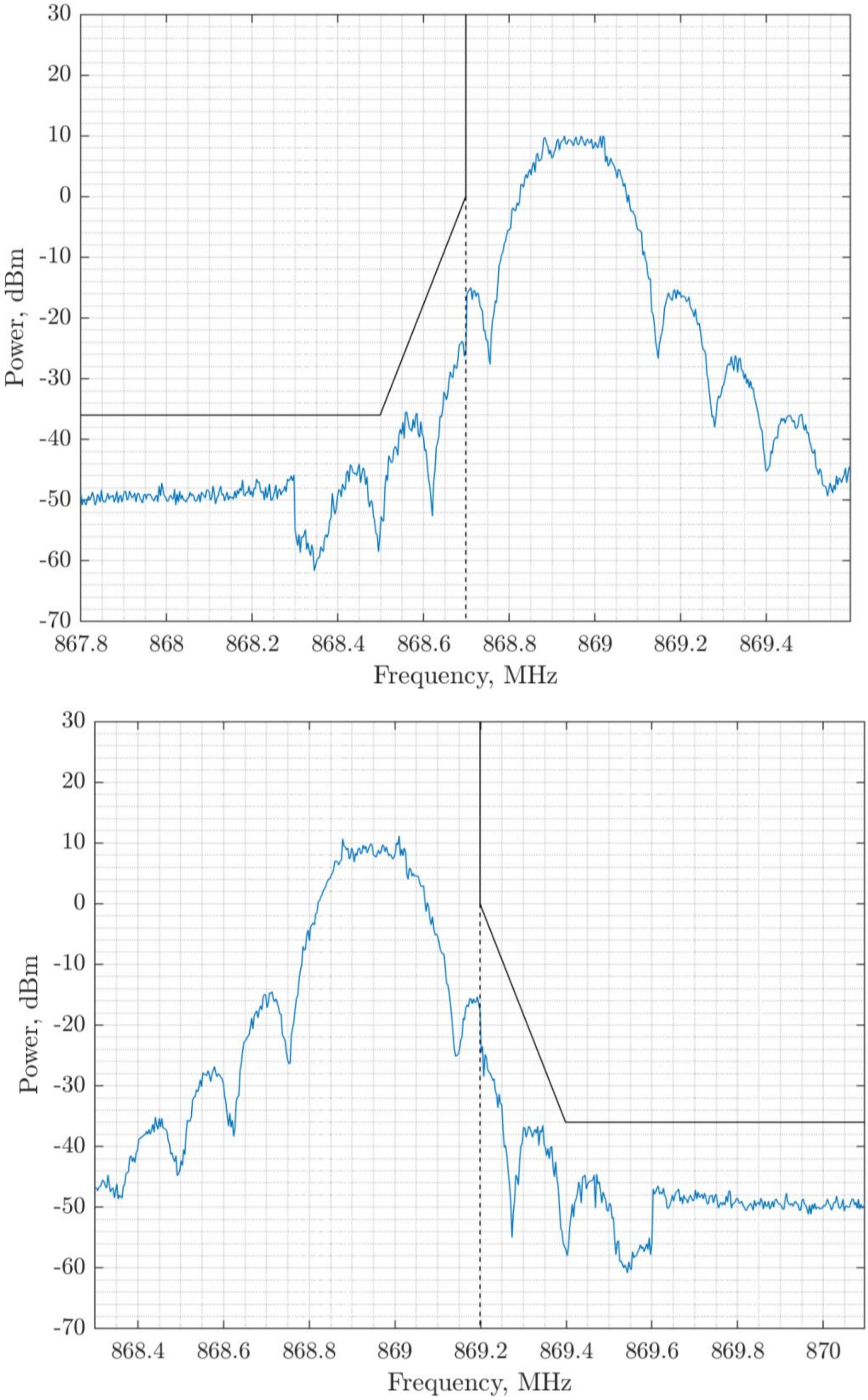


Figure 46. h1.5 Band Edge Compliance at 260 kbps BT = 1

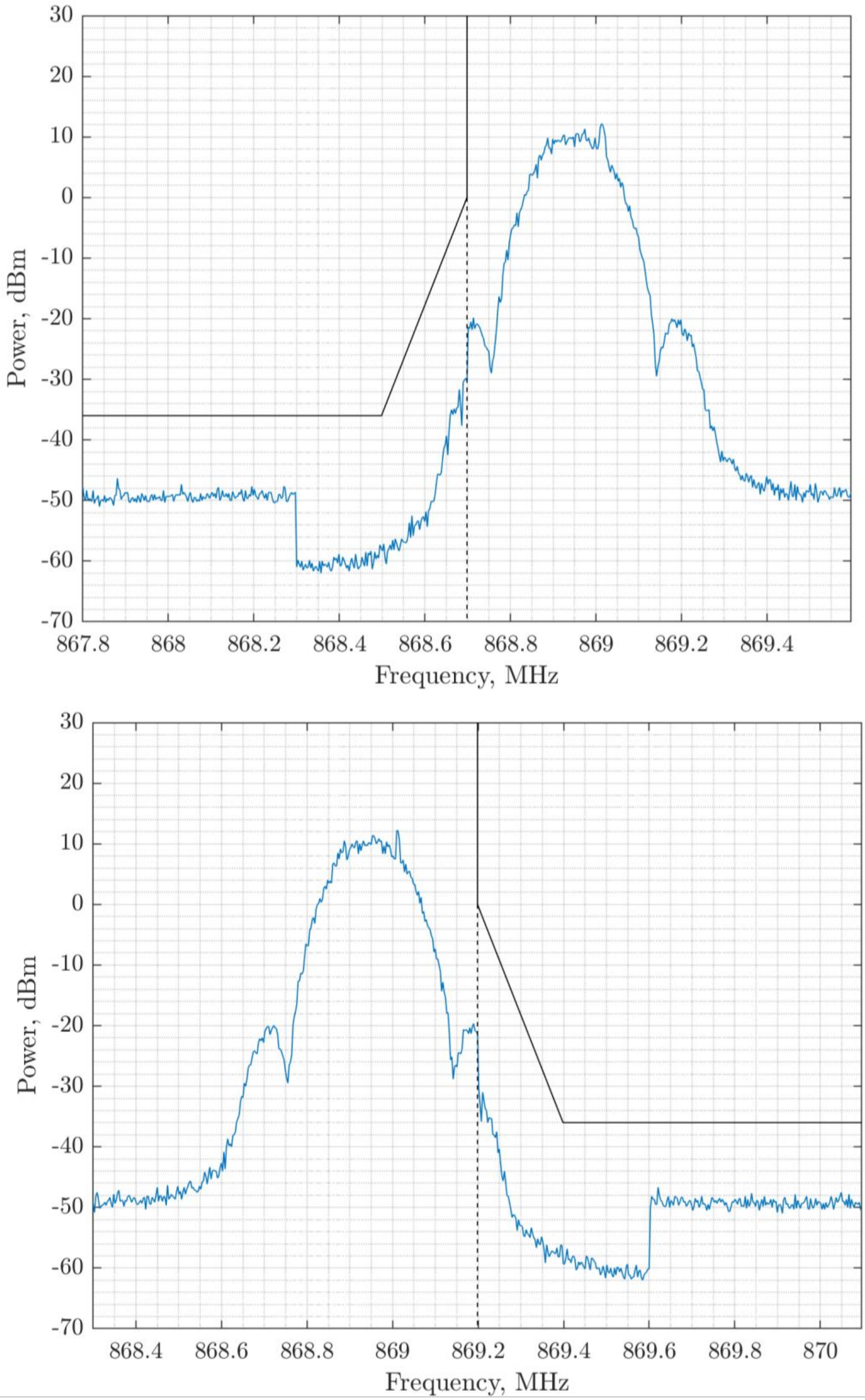


Figure 47. h1.6 Band Edge Compliance at 260 kbps BT = 0.5

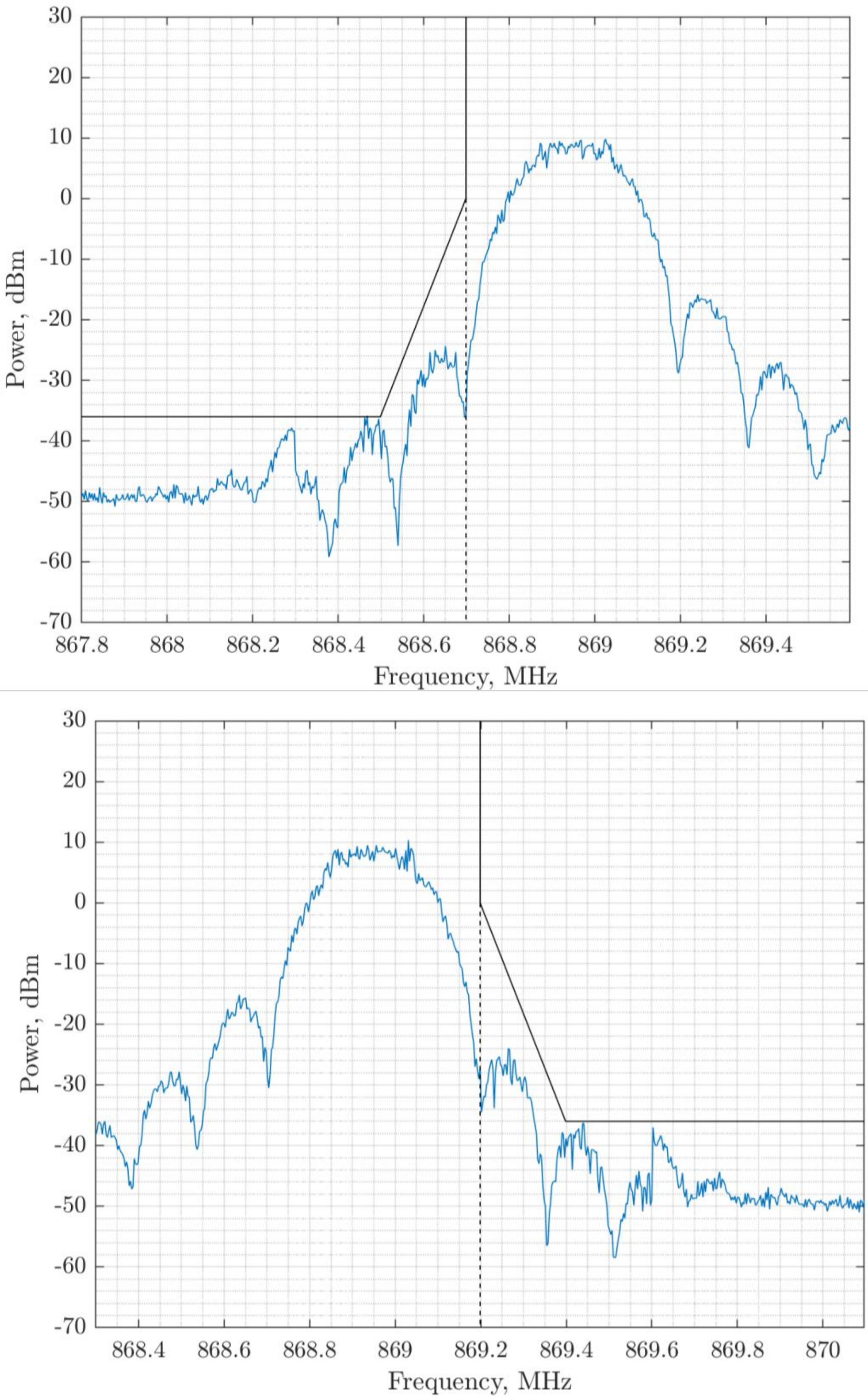


Figure 48. h1.6 Band Edge Compliance at 325 kbps BT = 1

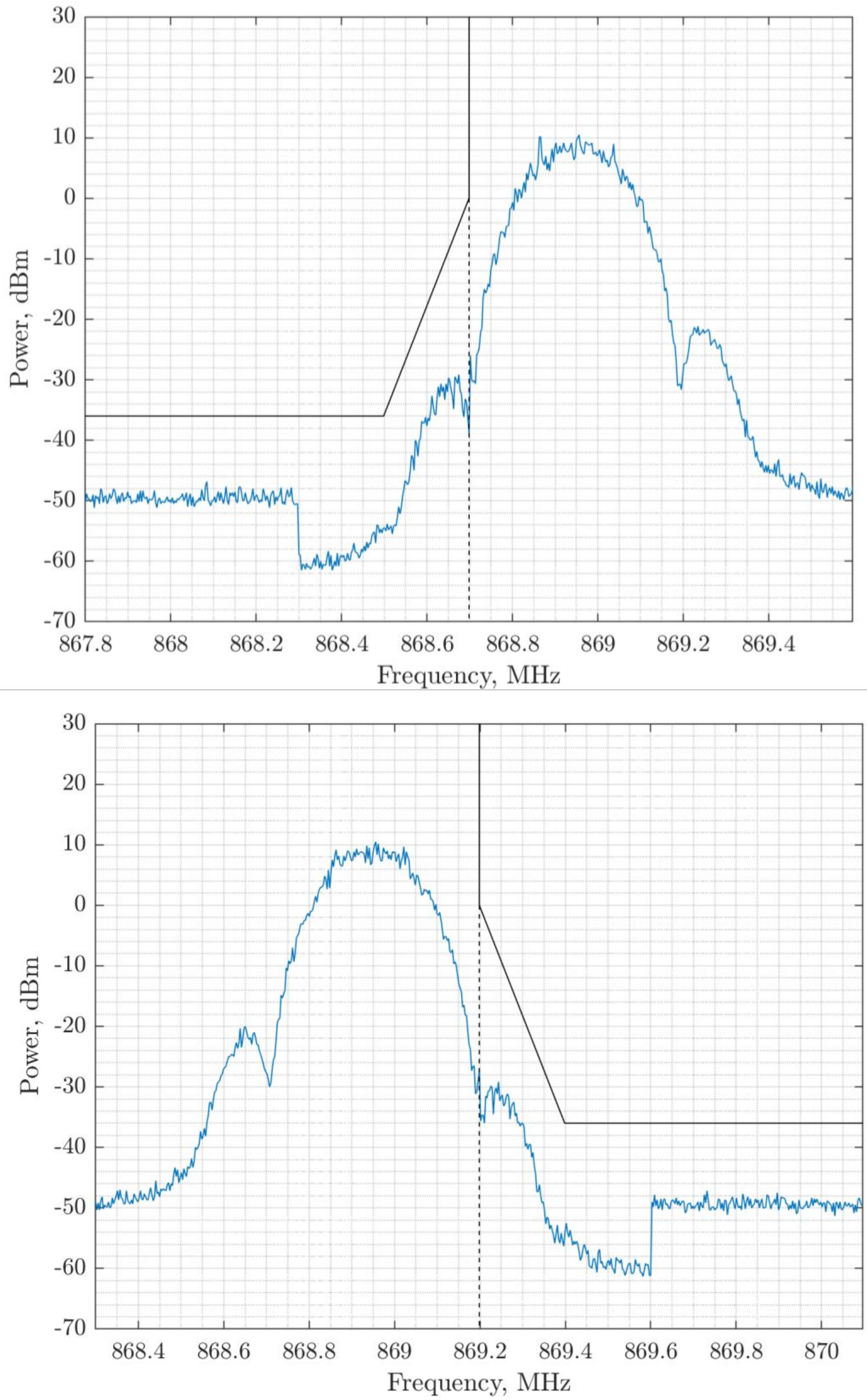


Figure 49. h1.6 Band Edge Compliance at 325 kbps BT = 0.5

2.10.5 Sub-Band h1.9

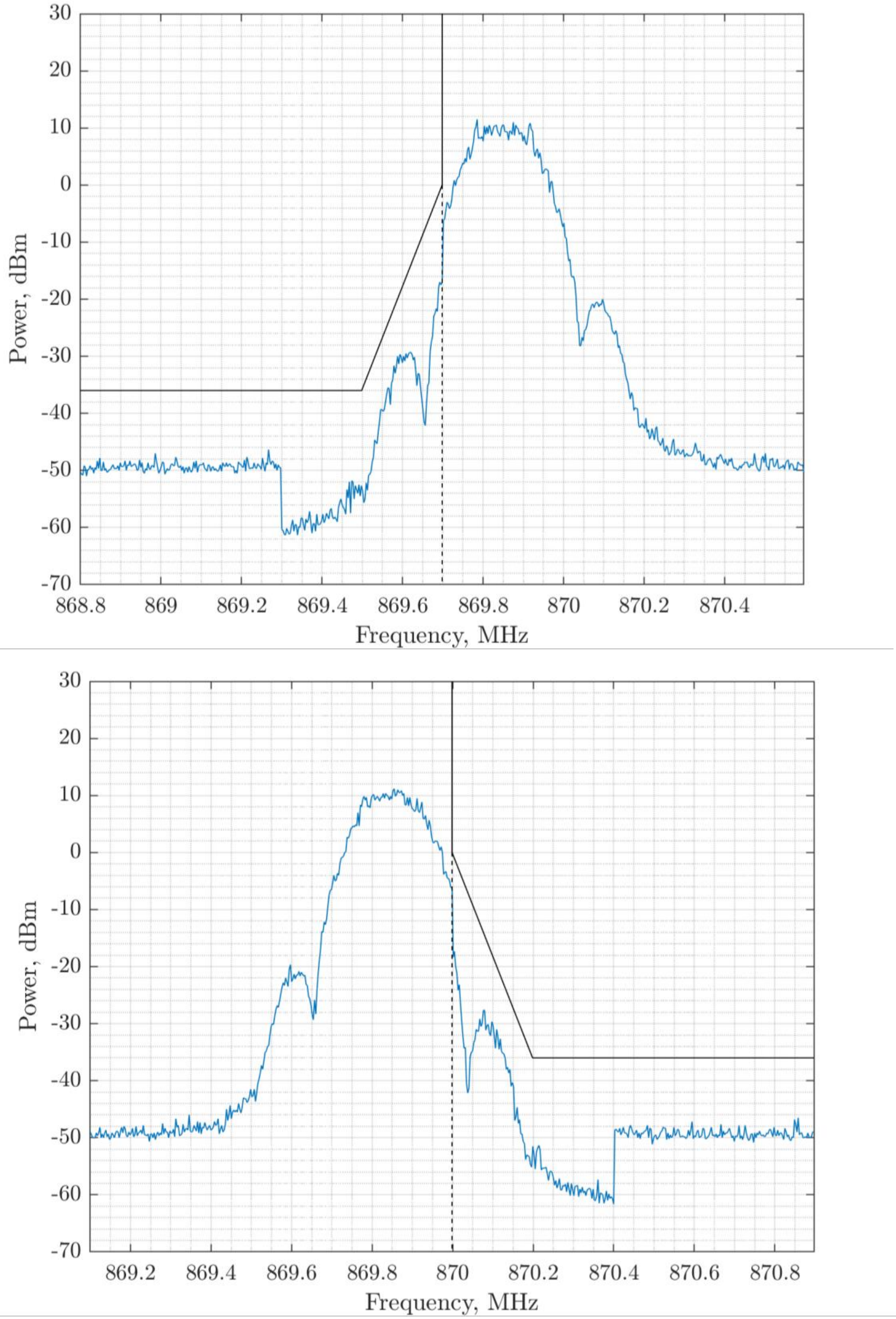


Figure 50. h1.9 Band Edge Compliance at 260 kbps BT = 0.5

3 Part II: US Regulatory Compliance

3.1 Introduction

This Part of the Application Note focuses on the application of FLRC to the FCC Part 15.247 [3] and the US 902 - 928 MHz ISM license-exempt frequency band. The test methodology also refers to the FCC OET guidance for the test methodology within this band [4].

For FLRC there are three possible modes of operation:

- Systems employing frequency hopping spread-spectrum (**FHSS**)
- Systems employing digital modulation techniques (**DTS**)
- Systems employing a combination of these modes, known as **Hybrid** mode operation

We examine each mode of operation in turn and determine the relevant subset of regulatory requirements relevant to each regime of operation and evaluate them. Although each mode of regulatory employment is different, each imposes constraints in three key regards:

1. **Modulation Bandwidth:** Some applications are constrained to certain measured transmission bandwidths.
2. **Output Power**: All ISM band devices are constrained to low power operation, the specific constraints on each mode of operation will be evaluated.
3. **Spurious Emissions**: In this FLRC-centric Application Note only emissions close to RF operating frequency are considered. General radiated emissions are covered in AN1200.108 [5].

3.2 FHSS Operation

3.2.1 FHSS Modulation Bandwidth Regulations

15.247 a) States that "if the 20 dB bandwidth of the hopping channel is 250 kHz or greater, the system shall use at least 25 hopping frequencies … the maximum allowed 20 dB bandwidth of the hopping channel is 500 kHz."

For this reason, only the two lowest FLRC data rates (with modulation bandwidths in the permitted range) are relevant. The measurement procedure is as detailed below:

1. Set the frequency span of the spectrum analyzer to approximately 2 to 3 times the 20 dB BW.
2. Set the RBW to approximately 1% of the expected 20 dB BW and Video Bandwidth (VBW) is set to 3*RBW.
3. Measure using the analyzer's detector function to peak and use the max hold when displaying the trace.

3.2.2 FHSS Modulation Bandwidth Measurements

Measured modulation bandwidths *less* than 500 kHz are recorded in the following sequence of plots.

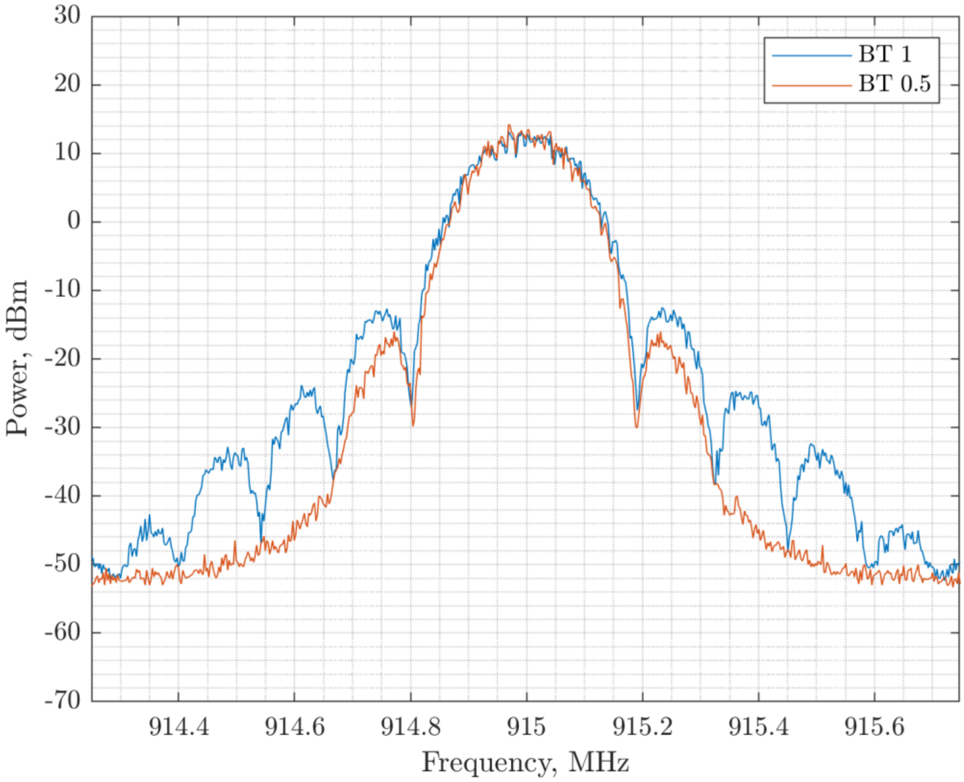


Figure 51. 260 kbps: 329 kHz Modulation Bandwidth with BT = 1 and 304 kHz with BT = 0.5

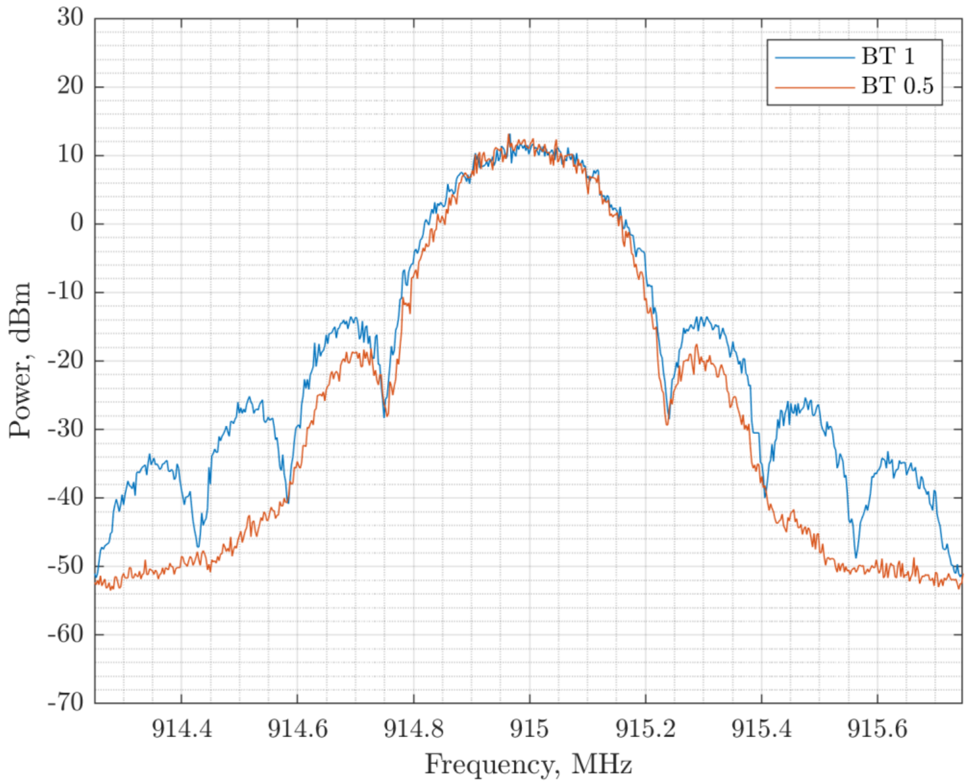


Figure 52. 325 kbps: 419 kHz Modulation Bandwidth with BT = 1 and 386 kHz with BT = 0.5

3.2.3 FHSS Output Power Regulations

15.247 b) States that "for frequency hopping systems operating in the 902-928 MHz band: … 0.25 watts for systems employing less than 50 hopping channels, but at least 25 hopping channels".

3.2.4 FHSS Output Power Measurements

The LR2021 can generate output power up to +22 dBm, so compliance is guaranteed with the 24 dBm limit. Nonetheless a CW conducted output power measurement is included below, conducted at 915 MHz.

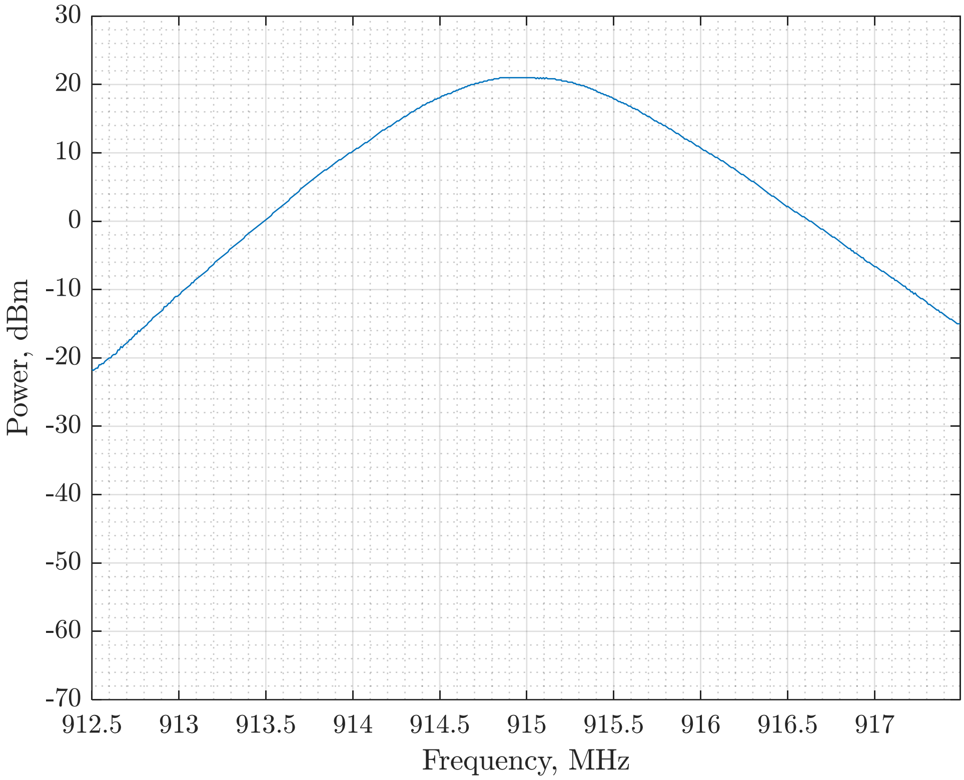


Figure 53. FCC Conducted Output Power Measured from the LR2021 at +21.1 dBm

3.2.5 FHSS Spurious Emissions Regulations

15.247 d) States "In any 100 kHz bandwidth outside the frequency band in which the spread spectrum or digitally modulated intentional radiator is operating, the radio frequency power that is produced by the intentional radiator shall be at least 20 dB below that in the 100 kHz bandwidth within the band that contains the highest level of the desired power, based on either an RF conducted or a radiated measurement, provided the transmitter demonstrates compliance with the peak conducted power limits. If the transmitter complies with the conducted power limits based on the use of RMS averaging over a time interval, as permitted under paragraph (b)(3) of this section, the attenuation required under this paragraph shall be 30 dB instead of 20 dB. Attenuation below the general limits specified in§ 15.209(a) is not required. In addition, radiated emissions which fall in the restricted bands, as defined in§ 15.205(a), must also comply with the radiated emission limits specified in§ 15.209(a) (see § 15.205(c))."

These limits are specified as radiated limits, from which we can derive the following conducted equivalent limits.

Table 6. Conducted Equivalent Spurious Emission Limits from§ 15.209(a)

Frequency Range [MHz]	Limit Field Strength [uV/m]	Equivalent Conducted Power [dBm]
30-88	100	-55.2
88-216	150	-51.7
216-960	200	-49.2
>960	500	-41.2

3.2.6 FHSS Spurious Emissions Measurements

The conducted measurement of a limited frequency range from 30 MHz to 1 GHz was performed to test for any modulation specific spurious emissions. The test was performed at a centre frequency of 915 MHz.

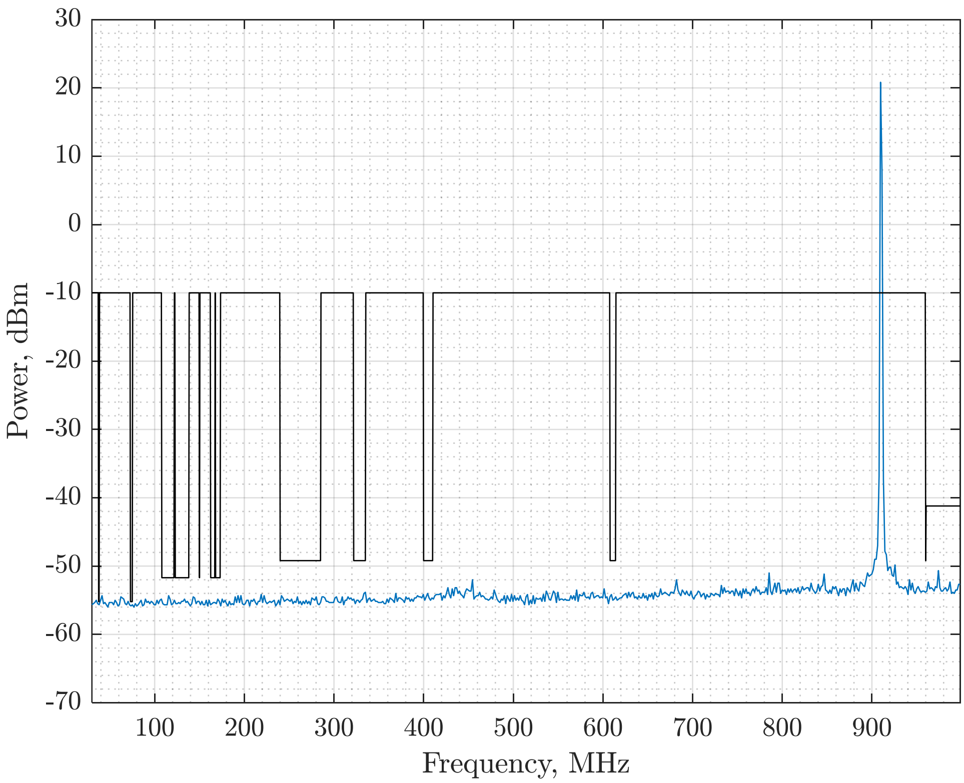


Figure 54. Spurious Emission Testing Results.

The resulting spectrum shows no observable spurious emissions above the measurement receiver's noise floor.

3.3 DTS Operation

3.3.1 DTS Modulation Bandwidth Regulations

15.247 e) States that "the minimum 6 dB bandwidth shall be at least 500 kHz". The bandwidth measurement process is as follows:

1. Set the Resolution Bandwidth (RBW) of the spectrum analyzer to 100 kHz and the video bandwidth (VBW) to ≥ 3 * RBW.

2. Using the spectrum analyzer's peak detector and with the trace mode set to max. hold, allow the trace to stabilize.

3. Measure the maximum width of the emission between upper and lower frequency points that are attenuated by 6 dB, relative to the maximum level measured in the fundamental emission.

3.3.2 DTS Modulation Bandwidth Measurements

The measured modulation bandwidths in excess of 500 kHz are plotted in the following sequence of graphs. Note that the first setting 520 kbps, BT=0.5 is non-compliant as it provides a bandwidth below 500 kHz. All other settings plotted are compliant.

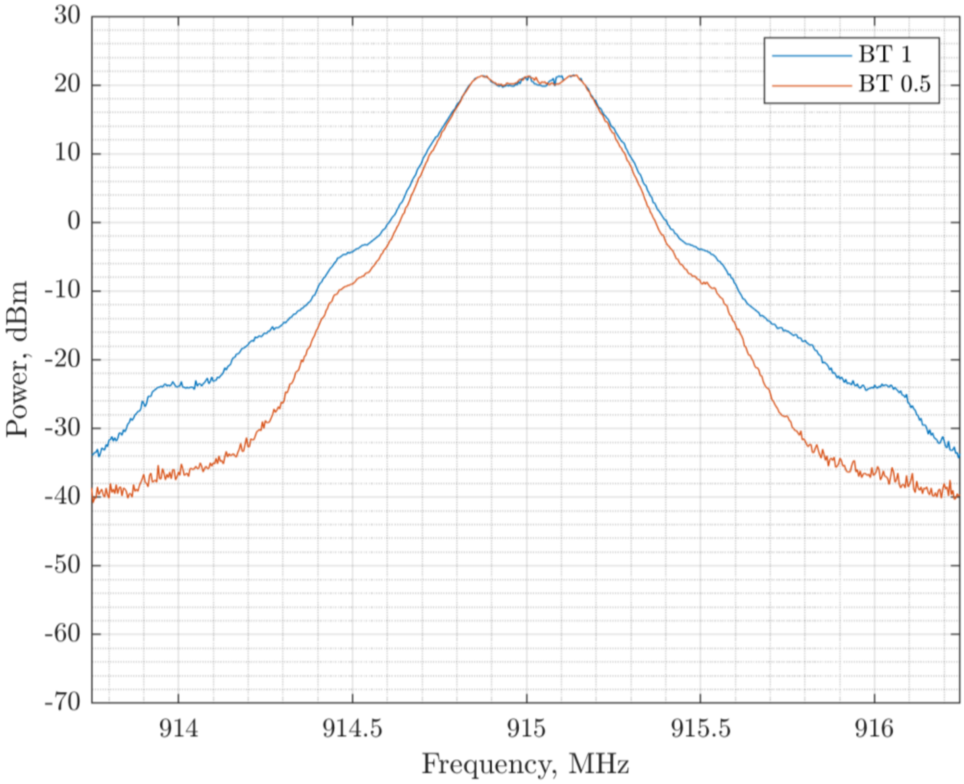


Figure 55. 520 kbps: 449 kHz Modulation Bandwidth with BT = 1 and 432 kHz with BT = 0.5

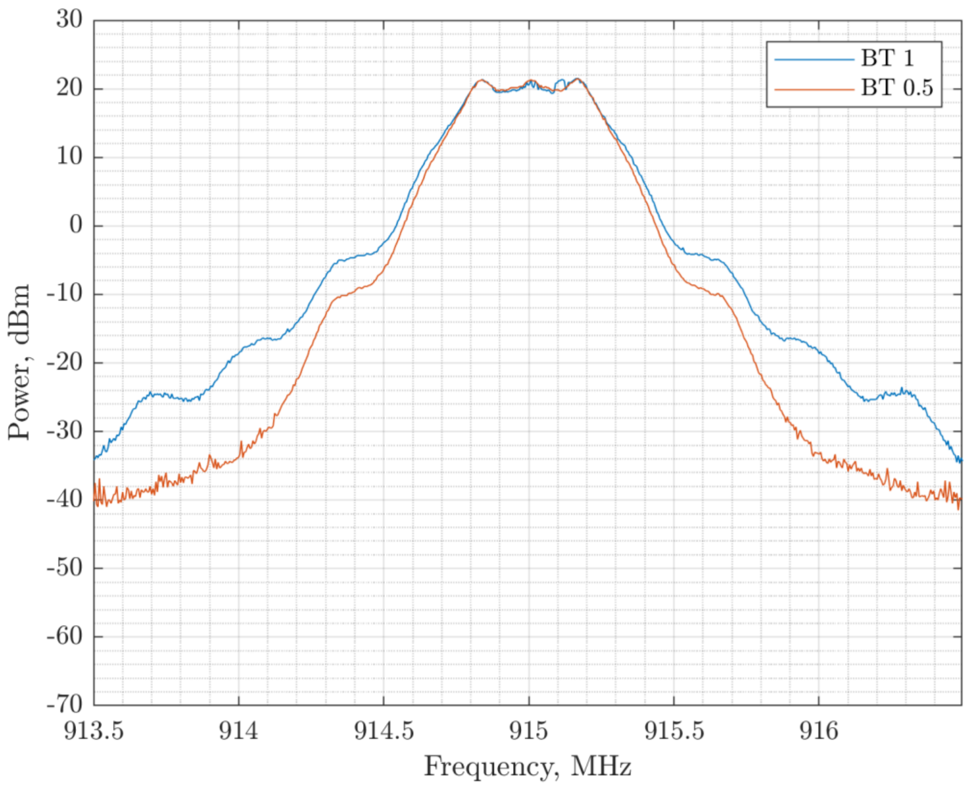


Figure 56. 650 kbps: 529 kHz Modulation Bandwidth with BT = 1 and 509 kHz with BT = 0.5

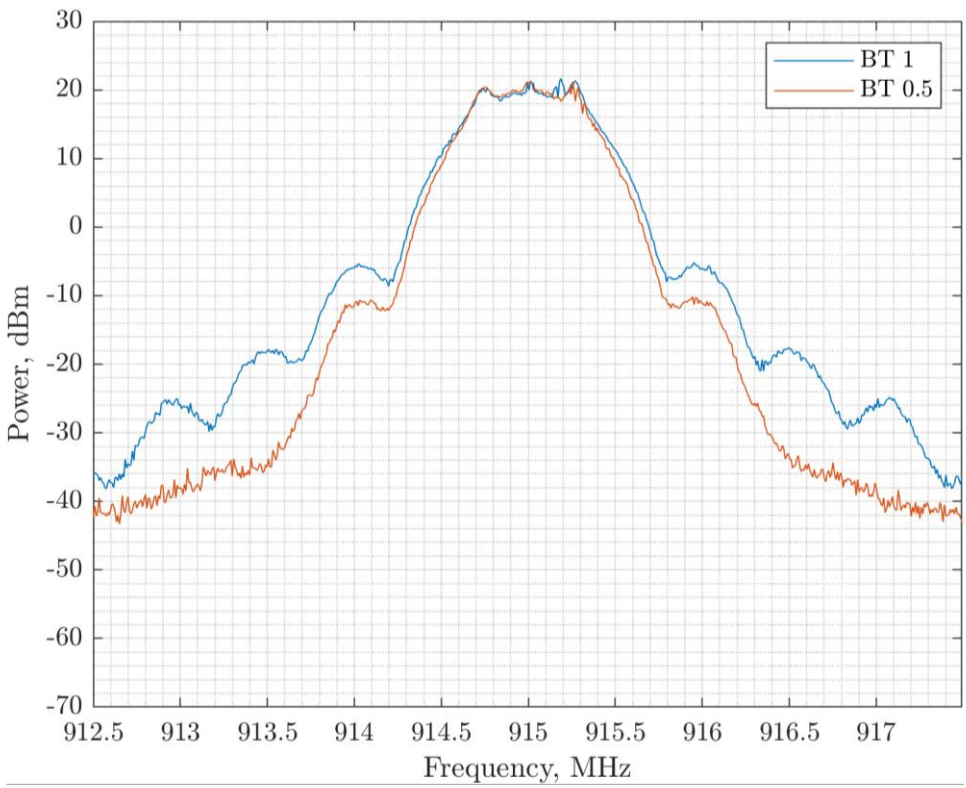


Figure 57. 1.040 Mbps: 748 kHz Modulation Bandwidth with BT = 1 and 748 kHz with BT = 0.5

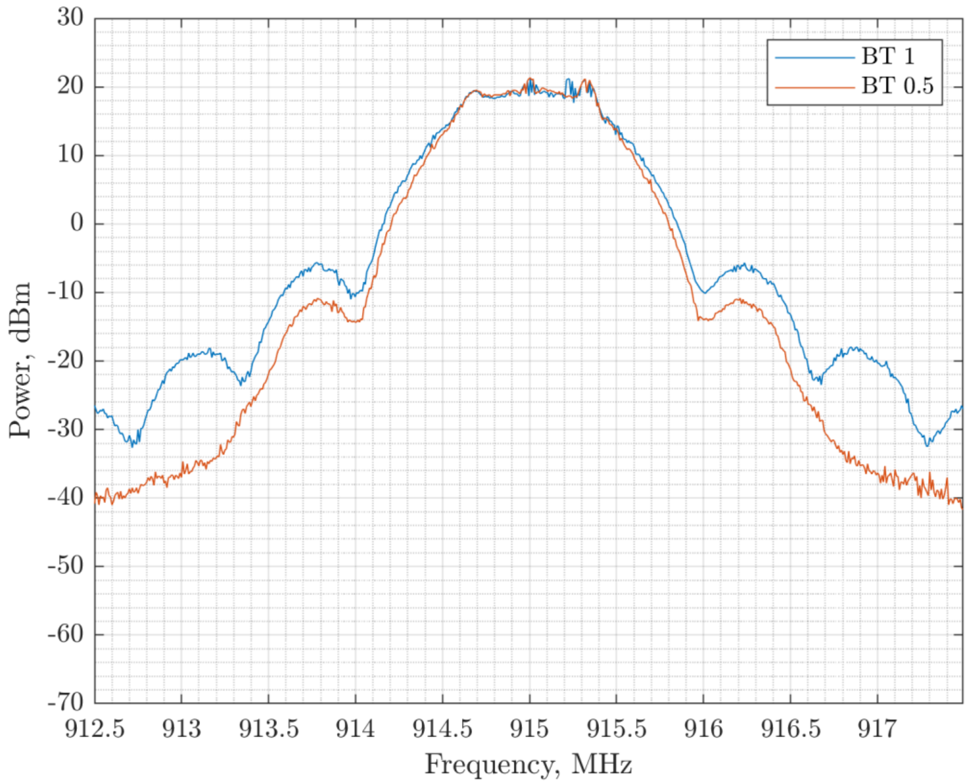


Figure 58. 1.30 Mbps: 873 kHz Modulation Bandwidth with BT = 1 and 881 kHz with BT = 0.5

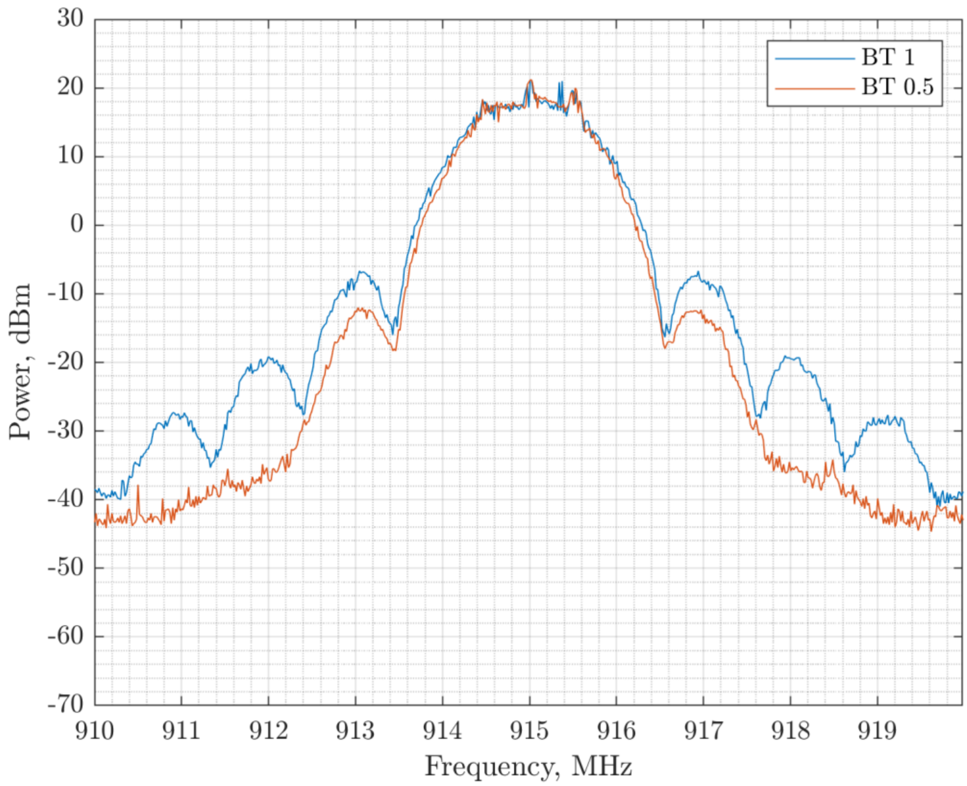


Figure 59. 2.08 Mbps: 1048 kHz Modulation Bandwidth with BT = 1 and 1181 kHz with BT = 0.5

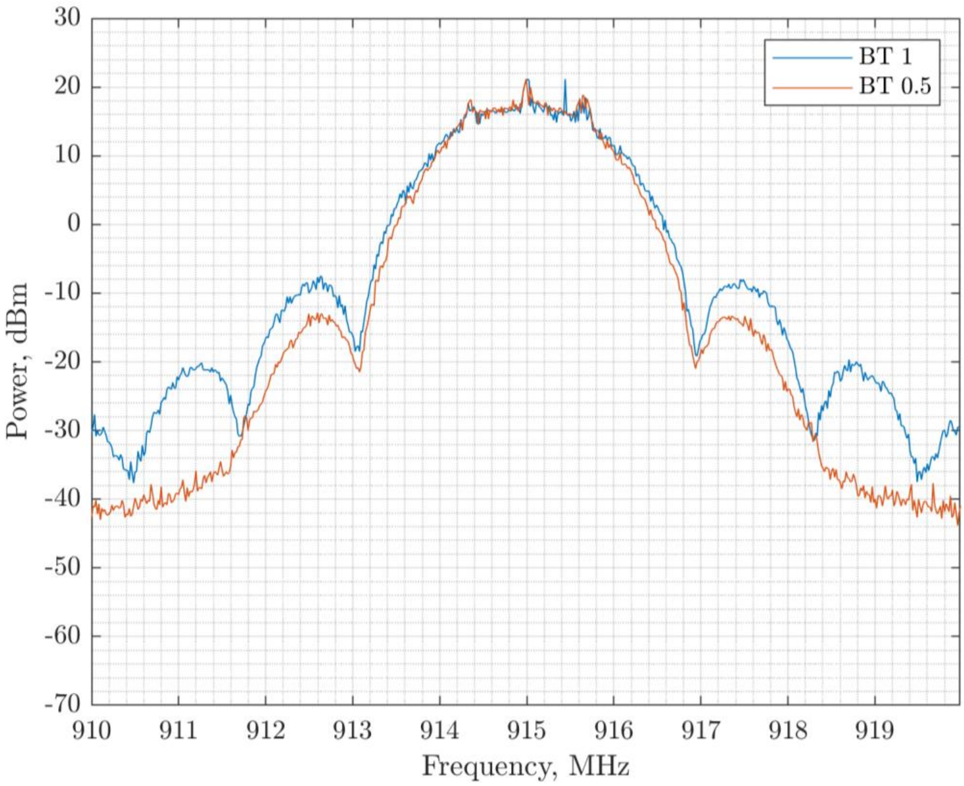


Figure 60. 2.6 Mbps: 1064 kHz Modulation Bandwidth with BT = 1 and 1214 kHz with BT = 0.5

3.3.3 DTS Output Power Regulations

15.247 e) States: "the power spectral density conducted from the intentional radiator to the antenna shall not be greater than 8 dBm in any 3 kHz band".

3.3.4 DTS Output Power Measurements

Following the measurement method outlined by the OET, the measurement of the PSD as a function of the data rate is given in the following measurement plots.

Note that data rates of 260 kbps and 325 kbps are presented here for Hybrid Mode applications (see next Section).

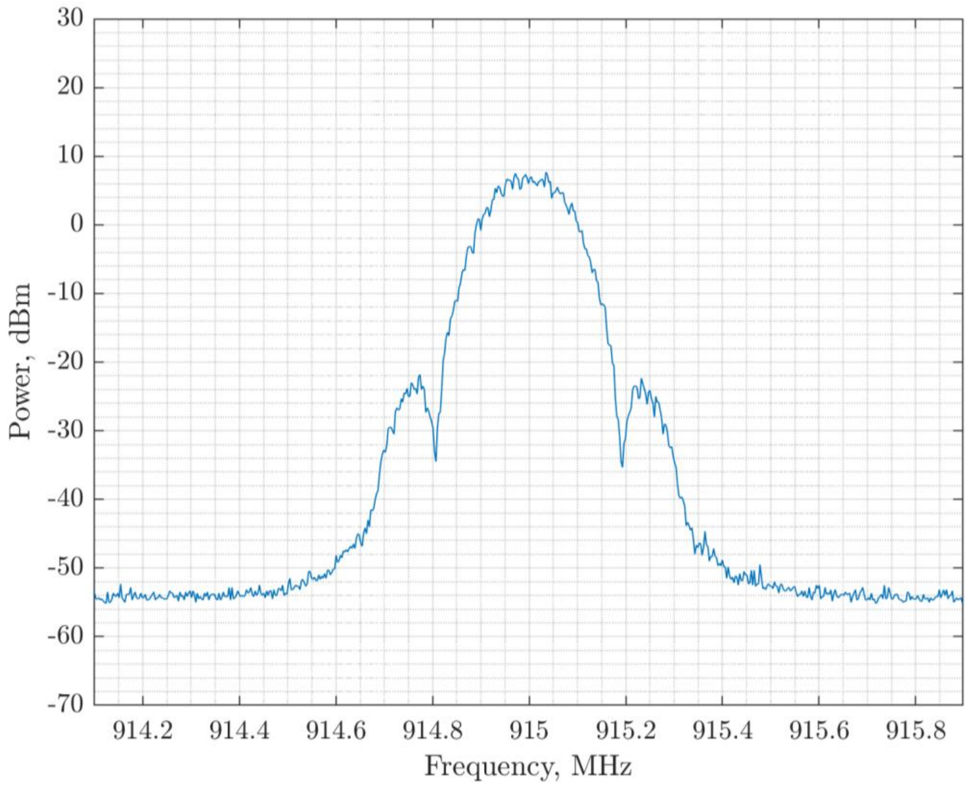


Figure 61. 260 kbps: PSD Measurement at 7.6 dBm

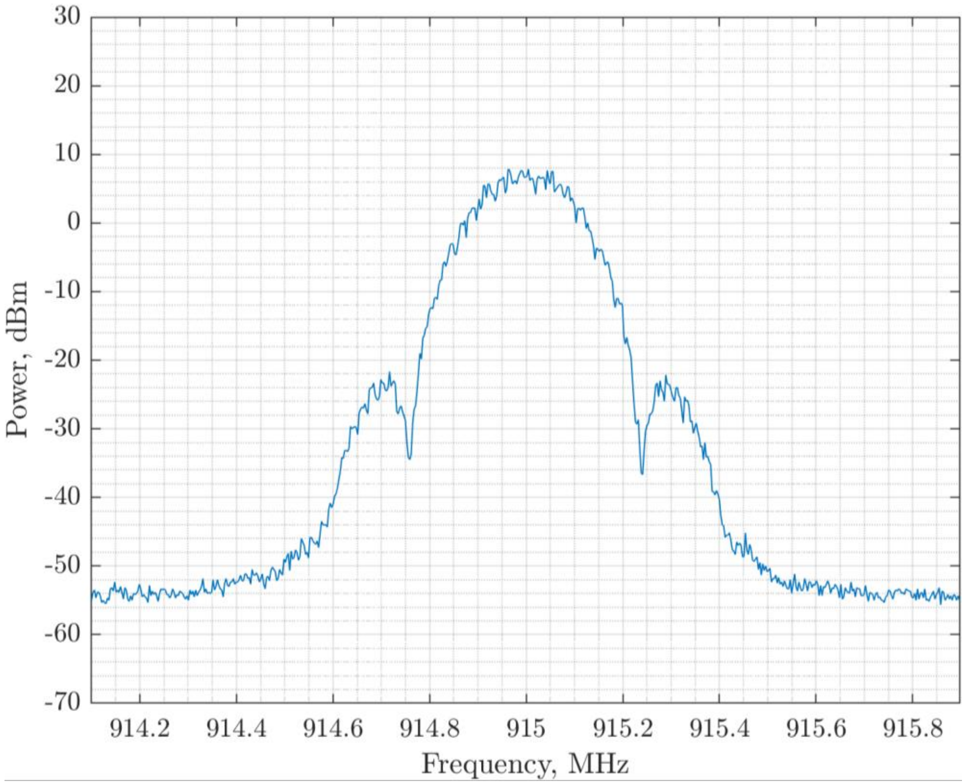


Figure 62. 325 kbps PSD Measurement at 7.8 dBm

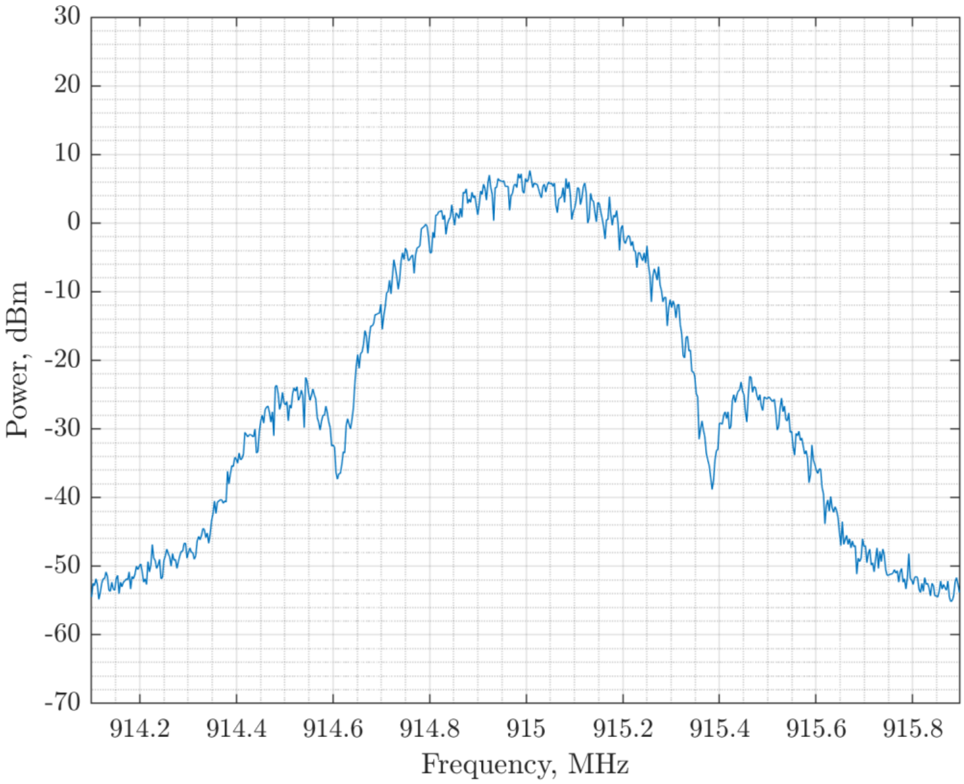


Figure 63. 520 kbps: PSD Measurement at 7.6 dBm

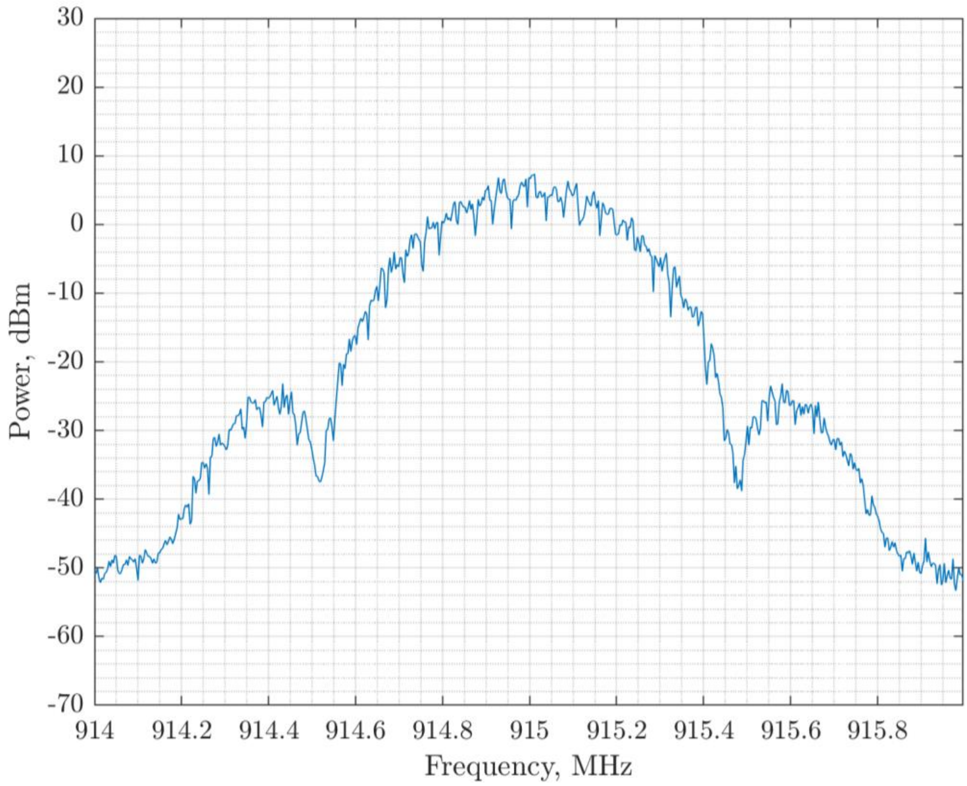


Figure 64. 650 kbps: PSD Measurement at 7.3 dBm

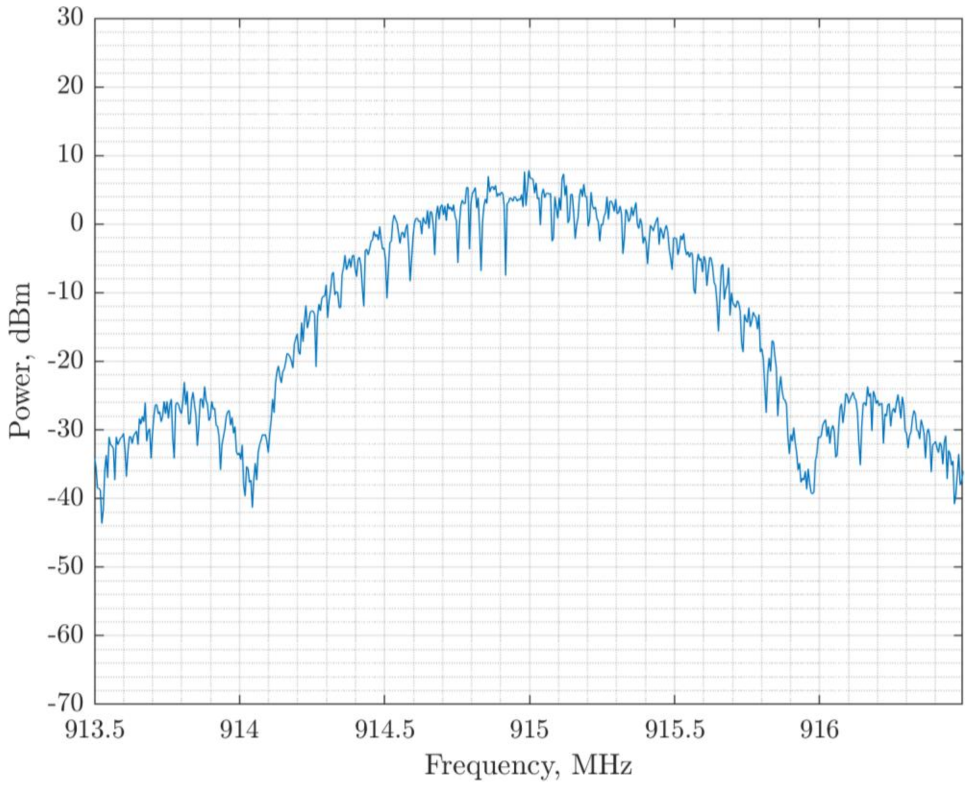


Figure 65. 1.04 Mbps: PSD Measurement at 7.9 dBm

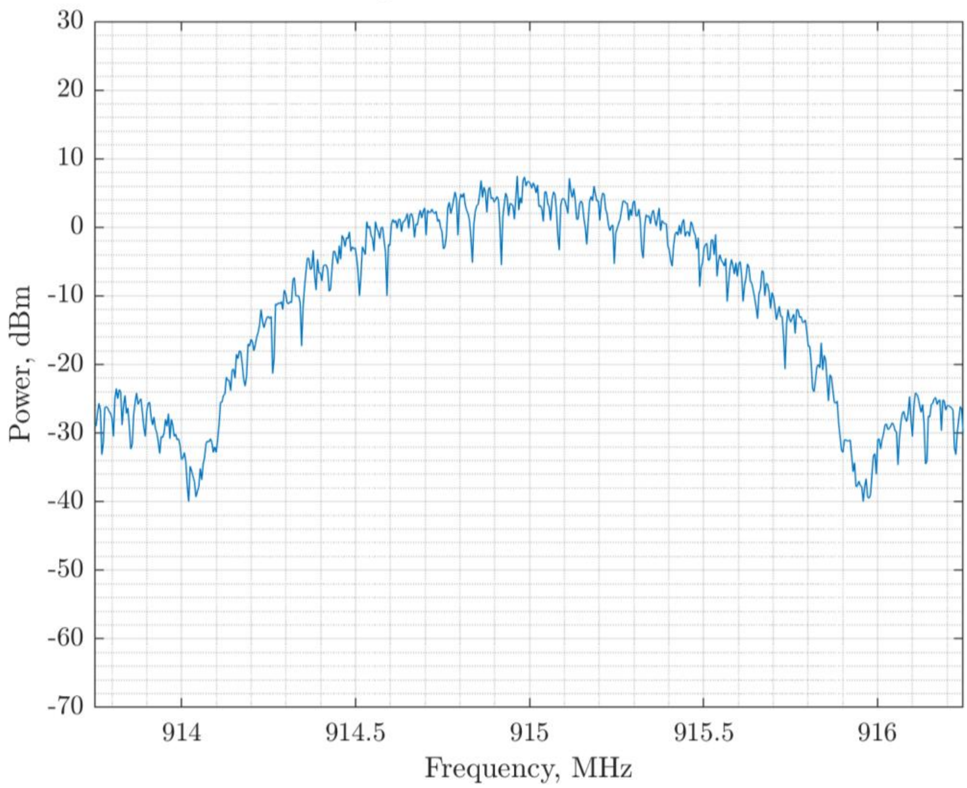


Figure 66. 1.3 Mbps: PSD Measurement at 7.5 dBm

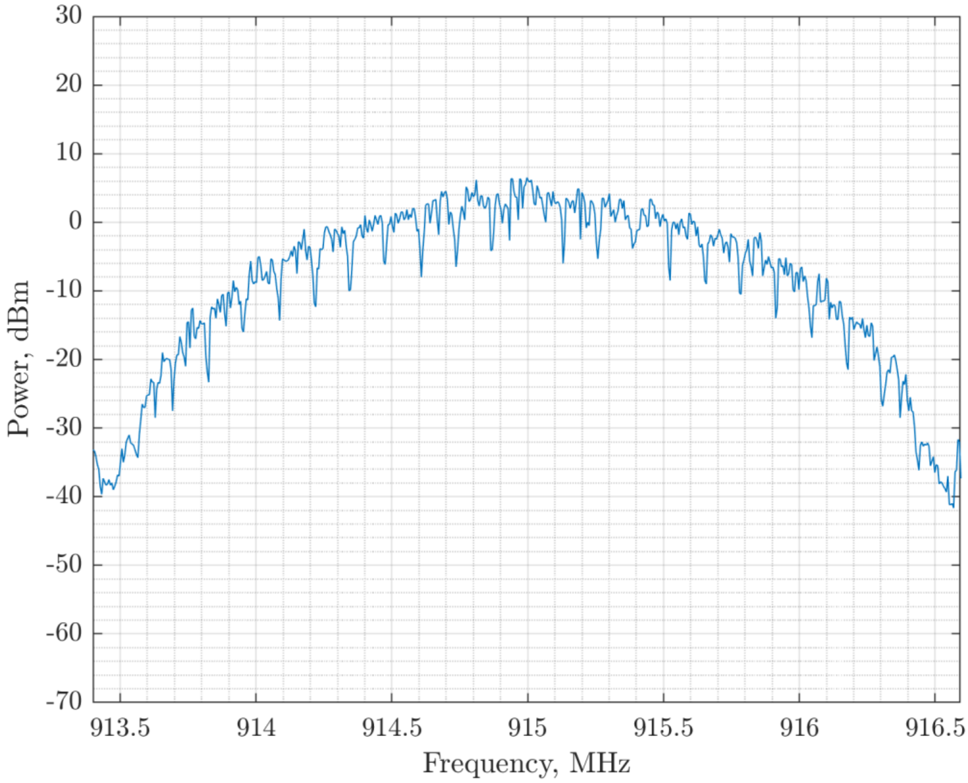


Figure 67. 2.08 Mbps: PSD Measurement at 6.5 dBm

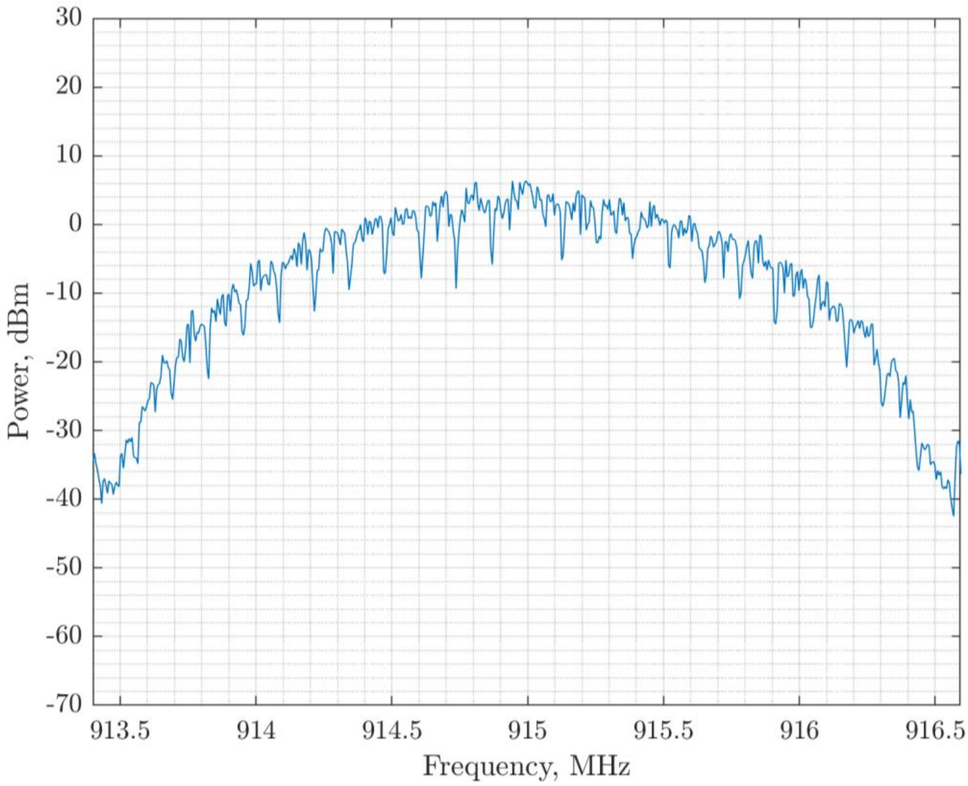


Figure 68. 2.6 Mbps: PSD Measurement at 6.4 dBm

The final measured PSD versus the programmed output power is summarised in the following table.

Table 7. Final Measured PSD versus Programmed Output Power

Data Rate (Mbps)	BW (MHz)	Programmed Power	PSD (dBm/3 kHz)
2.600	2.666	22	6.4
2.080	2.222	22	6.5
1.300	1.333	21	7.5
1.040	1.333	20	7.9
0.650	0.740	19	7.3
0.520	0.571	19	7.6
0.325	0.357	17	7.8
0.260	0.307	16	7.6

As the data rate increases, so the whole output power of the LR2021 may be exploited as the power spectral density decreases.

3.3.5 DTS Spurious Emissions

DTS systems follow the same methodology and results as for FHSS.

3.4 Hybrid Mode

3.4.1 Hybrid Mode Modulation Bandwidth Regulations

No modulation bandwidth constraint is enforced.

3.4.2 Hybrid Mode Output Power Measurements

The output power limitations are the same as for DTS mode, except no bandwidth limitation is in force: on this basis, all FLRC data rates can be employed.

3.4.3 Hybrid Mode Spurious Emissions

Follow the same methodology and results as for FHSS and DTS operation.

4 Conclusion

These results aim to clarify the range of options available for FLRC PHY layer regulatory trade-off choices. These results do not replace the regulatory certification process.

4.1 EU Regulatory Compliance Conclusion

In Part I of this Application Note we have demonstrated compliance of FLRC modulation in multiple configurations and sub bands within the 863-870 MHz EU ISM band. Care must still be taken to ensure compliance of the wider radio system with the relevant Duty Cycle, frequency tolerance and general emission and receiver limits. Please see [1] and [2] for more details.

The principal limitation on FLRC operation in Europe is the band-edge transmission mask compliance. Compliant combinations of data rate and sub-band are tabulated below for 14 dBm output power.

Table 8. ETSI Compatibility of the FLRC Modem in the 868 MHz ISM Band

Data Rate (Mbps)	BW (MHz)	OCW (MHz)	h1.3	h1.4	h1.5	h1.6	h1.9	A10
			Bandwidth (MHz)					
			2.0	3.0	0.6	0.5	0.3	2
2.600	2.666	2.8	-	N	-	-	-	-
2.080	2.222	2.4	-	N	-	-	-	-
1.300	1.333	1.4	N	0.5	-	-	-	Tbc
1.040	1.333	1.4	0.5	Y	-	-	-	Tbc
0.650	0.740	0.8	Y	Y	-	-	-	Y
0.520	0.571	0.7	Y	Y	-	-	-	Y
0.325	0.357	0.4	Y	Y	Y	Y	N	Y
0.260	0.307	0.4	Y	Y	Y	Y	0.5	Y

Note that 0.5 denotes BT = 0.5 compliant only and Tbc is untested (at +10 dBm). Note also that, whilst effective operation up to the regulatory limit was shown for 1.3 Mbps usage of the FLRC modem at higher data rates may still be possible at lower output powers. Future work will focus on examining the feasibility of lower power operation at higher data rates.

4.2 US Regulatory Compliance Conclusion

In Part II of this Application Note we have demonstrated that US operation of the LR2021 FLRC modem is very straightforward in all three modes of regulatory operation, with the only noteworthy limitation being the sub-500 kHz bandwidth of the 520 kbps FLRC modulation for DTS operation.

Compliant settings for all other modes of employment of the FLRC modem have been presented for all three regulatory regimes.

5 References

[1] CEPT ERC REC 70-03

[2] ETSI EN 300 220

[3] FCC CFR Title 47, Chapter 1, Subchapter A, Part 15

[4] FCC OET, “Guidance for Compliance Measurements on Digital Transmission System, Frequency Hopping Spread Spectrum System, and Hybrid System Devices Operating Under Section 15.247 of the FCC Rules”, April 2, 2019

[5] AN1200.108 “FCC and ETSI Certification of the LR2021”

[6] AN1200.101 “LR2021 FLRC Performance”

6 Revision History

Version	ECO	Date	Changes and/or Modifications
1.0	ECO-076647	October 2025	Initial Version

Important Notice